

A MISSING CHAPTER FOR THE SURREAL DOCUMENTATION

Surcomplex Spectral Theory

Singular Values, Infinitesimal Rank,
and Multiscale Stability

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Abstract

Finite matrices over the surcomplex numbers admit much of the familiar spectral theory, but its justification is algebraic rather than compactness based, and its interpretation changes when singular values occupy different infinitesimal scales. We develop Hermitian and normal diagonalization, positive square roots, singular value and polar decompositions, variational principles, least squares, low-rank approximation, conditioning, and spectral-subspace perturbation over a real closed field and its complexification.

The central synthesis identifies the valuations of singular values with successive differences of the minimum valuations of minors. A positive Cauchy–Binet identity then recovers both these scales and the leading singular coefficients from the characteristic coefficients of the Gram matrix. This gives an intrinsic rank filtration, an exact root-free scale algorithm, and precision certificates valid for arbitrary-rank Hahn value groups. Worked examples separate exact rank from standard-part rank, small perturbations from gap-relative perturbations, and algebraic inverses from Hahn-summable inverse series. Nonnormal pseudospectra, regularization, and a finite tensor application complete the picture.

Divisibility of the value group is then removed. A finite splitting tree, licensed by strong Hahn summability rather than by any convergence argument, produces Hermitian and normal diagonalizations and singular-value decompositions whose every exponent lies in the subgroup generated by the input exponents. Against this, positive matrix roots genuinely cost new scales, and the cost is exactly the determinantal one: for positive semidefinite A of rank r and integer $q \geq 2$, the principal q th root generates the Hahn field on $\Gamma + \sum_{k=1}^r \mathbb{Z}\Delta_k(A)/q$, a finite abelian totally ramified extension of degree the order of the subgroup the classes Δ_k span in $\Gamma/q\Gamma$; it is trivial exactly when every Δ_k lies in $q\Gamma$. At $q = 2$ this is also the exact criterion for a Gram factorization $A = C^*C$ over the base field. The single scalar $\text{tr}(A^{1/q})$ is always a primitive generator of that whole extension, by a positivity mechanism rather than by genericity.

The same question for a polynomial is answered by its Newton-sum Hankel matrix. For a monic squarefree real-rooted p over $F_\Gamma = \mathbb{R}((t^\Gamma))$, with leading principal Hankel minors $h_1, \dots, h_n$ and $h_0 = 1$, the splitting field is $F_\Gamma(\sqrt{h_1}, \dots, \sqrt{h_n}) = F_{\Gamma + \sum_k \mathbb{Z}v(h_k)/2}$, and the number of irreducible factors is $\#\{k : v(h_k) - v(h_{k-1}) \in 2\Gamma\}$. The full multiplicity profile of these classes is the character multiplicity of the root permutation representation and is stable under a stated precision. Applied to definite Hermitian pencils, it gives the exact field of the generalized eigenvalues, and size-optimal pencils hide half-scales behind eigenvalues of valuation zero.

Status and scope. A self-contained research exposition relative to standard surreal normal-form and Hahn-field foundations, assembled from three separately delivered manuscripts (see [Section 1.4](#)). Classical results are identified as such; no priority claim or solution of a named open problem is made. Older power-series spectral theorems, Hahn support lemmas, Kummer theory, trace forms and subdiscriminants are acknowledged as predecessors. The arbitrary-rank determinantal and primitive-trace package of [Sections 10](#) and [11](#) and the Hankel square-class package of [Section 12](#) are offered as candidate original contributions whose priority is not certified. All matrix dimensions are ordinary finite integers. General proofs are not proof-assistant verified as a complete package. The repository’s separate formalization ledger records current checked prerequisites and source-claim coverage. The accompanying exact symbolic checks exercise finite examples, not arbitrary Hahn supports or the general theorems.

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1 The documentation gap and the mathematical agenda

1.1 What was inspected

This article was chosen after inspecting the documentation catalogue and relevant inventories in the repository [VladimirReshetnikov/Surreal](#), at the fixed revision

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The catalogue lists fifteen packages in surreal mathematics, surcomplex analysis and geometry, and foundations and computation [1]. The detailed polynomial-algebra inventory and its opening source sections were especially relevant [2]; the foundations inventory was also inspected [3]. This is a targeted coverage audit, not a claim to have checked every line of all archived source manuscripts.

Matrix theory is not wholly absent. The polynomial report uses an algebraic Schur inequality and a differentiating compression in its treatment of Schoenberg’s inequality, and it uses signatures of Hermite forms to count real roots. Finite residue pairings and multiplication matrices also appear in the surrounding algebra. These are substantial ingredients, but they are organized around polynomial roots and residues, not around matrices as the objects of study. The inspected inventories do not offer a systematic singular-value, conditioning, or scale-resolved rank theory.

Existing strand	What it supplies	What this chapter adds
Polynomial algebra	Root geometry, Schur-type arguments, signatures, Newton profiles	Unitary geometry, SVD, positive Gram matrices, determinantal singular scales
Analytic geometry	Finite algebras and deformation questions	Gap-sensitive spectral projections and Schur-complement effective matrices
Foundations	Set-sized Hahn workspaces; limits on transfer	Explicit finite algebraic proofs without compactness or a class-sized choice principle
Computer algebra	Representation and precision distinctions	Root-free scale elimination and certificates for resolvable rank

Table 1: The gap is an organizing theory, not the absence of every matrix lemma.

1.2 Why this is more than replacing scalars by surreals

Over the ordinary reals, a positive singular value is often tacitly regarded as a non-negligible number. Over the surreals, a matrix can have singular values at scales 1 , ω^{-2} , and $\omega^{-\omega}$. It is then simultaneously invertible, singular after taking standard parts, and indistinguishable from a lower-rank matrix at many prescribed precisions. An infinitesimal perturbation can rotate an eigenspace through an ordinary angle when its spectral gap is infinitesimal too. Conversely, an infinite entry need not imply an unstable direction if every singular value has the same scale.

The issue is therefore not just whether a decomposition exists. We need to know what its invariants measure, which conclusions survive reduction to an ordinary complex matrix, and which operations can be certified without computing an entire Hahn expansion. The key bridge is

$$\min_{|I|=|J|=k} v(\det A_{I,J}) = \sum_{j=1}^k v(\sigma_j(A)), \quad 1 \leq k \leq \text{rank } A. \quad (1.1)$$

Here the singular values are ordered by their actual positive magnitudes, whereas the valuations

are ordered in the opposite sense. This identity turns analytic-looking size information into finite determinantal algebra.

1.3 Attribution and reading route

The spectral theorem over real closed fields is classical, not a novel surreal phenomenon; a recent explicit use is Wang [7]. The variational and perturbation principles have their classical matrix-analysis counterparts, for example in Tao’s lecture notes [8], the original low-rank approximation paper of Eckart and Young [9], and the Davis–Kahan literature [11, 12]. The pseudoinverse is the classical Moore–Penrose construction [10].

The contribution here is an integrated account with proofs in the exact field setting, the determinantal and Gram-coefficient description of surreal scales, and explicit boundaries for reduction, computation, and summability. We do not claim that these consequences of existing algebra first occur here. Readers already comfortable with real-closed-field linear algebra can start at [Section 7](#); readers interested primarily in computation can proceed from there to [Sections 14](#) and [15](#).

1.4 Provenance: three manuscripts

This article is assembled from three separately delivered manuscripts. The manuscripts themselves are not shipped with the report; their delivered code and data are, under `code/` and `data/`. The first two are both dated 21 September 2026. Every section except [Sections 9](#) to [12](#) comes from the first, delivered as `01-determinantal-singular-value-scales.tex`, which worked over a real closed field and, for the scale-sensitive results, in a *divisible* Hahn workspace. [Sections 9](#) to [11](#) come from the second, delivered as `02-determinantal-ramification-descent.tex`, which removes that divisibility hypothesis and measures exactly what it was buying.

The second manuscript cites the first by name and declines to reclaim its content: the SVD, the determinantal singular scales of [Theorem 7.5](#) and the positive Cauchy–Binet identity of [Theorem 8.1](#) are stated here once only, in the first manuscript’s form. What the second adds is the removal of divisibility from the decompositions, and then an exact field-of-definition and degree calculation for positive matrix roots, in which the very invariant Δ_k that measures singular scales turns out to control a finite radical extension.

The third manuscript, “Hankel Square-Class Profiles over Surreal Hahn Fields: Exact Factor Counts, Splitting Fields, and Definite Pencils” (archive `surreal_hankel_ramification`, dated 22 September 2026), supplies [Section 12](#). It cites this report and does not reclaim its descent, ramification or primitive-trace theorems. Its reproved inputs — positive square classes, the Pythagorean property of Hahn fields, finite monomial extensions and orthogonal diagonalization of real symmetric matrices over F_Γ — are already [Lemmas 9.8](#), [10.1](#) and [10.2](#) and [Theorem 9.12](#), and are cited there rather than reprinted. What it adds is the classification of intermediate half-scale fields, the Hankel splitting-field and factor-count package, its application to definite pencils, and a contour-free proof of real principal axes by coprime Hensel lifting, kept as a marked second route ([Remark 12.5](#)). The merge had to choose in four places. The material is placed here rather than in the polynomial-algebra report, because that report works in divisible workspaces, where $\Gamma/2\Gamma = 0$ and every invariant of [Section 12](#) vanishes. The source’s notation α_k , β_k , d_k , and its letter q for a quadratic form and for a second polynomial, are renamed to avoid this report’s γ_k , Δ_k and root order q ([Remark 12.1](#)). Two short observations linking the new section to the older ones are added and marked as such (the paragraph after [Corollary 12.10](#), and [Remark 12.11](#)). And the source’s theorem numbers change: its Theorems 5.1 and 6.2 are [Theorems 12.8](#) and [12.13](#) here.

The three manuscripts audited the repository at different pinned revisions,

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the first, the second and the third respectively. All three declared their audits targeted rather than line-by-line, with absence from a search result never used as proof of absence from the collection. At its pin the third found the spectral descent, ramification and primitive-trace results of this report and the Hermite signature criterion of the polynomial-algebra report, and no matching statement of its central package; at that pin the words “Hankel” and “subdiscriminant” occurred nowhere under docs/. The collection now also contains docs/surcomplex/prony-reconstruction-at-surreal-scales/, placed in the same batch, which uses Hankel moment matrices for a different question and shares no theorem with [Section 12](#).

The second manuscript’s half has genuine predecessors, which it concedes rather than displaces. (Those of the third manuscript, including Keller and Ochsenius’s 2008 multiquadratic constructions and the classical theory of subdiscriminants, are conceded in [Section 12.10](#).) Adkins proved a spectral theorem for normal matrices over hermitian discrete valuation rings, including one-variable formal and convergent power series [13]; Keller and Ochsenius proved orthogonal diagonalization over iterated real Laurent-series fields and a particular infinite-rank Hahn field by recursive block splitting [14]. So neither “a spectral theorem beyond $\mathbb{R}$ ” nor “no fractional exponents in a Laurent-series eigenbasis” is claimed as new. Parusiński and Rond prove multiparameter formal spectral and singular-value results under normal-crossing discriminant hypotheses [18], and the 2026 preprint of Dai, Liang, Lu and Zhi gives a splitting-and-projector criterion over multivariable formal power-series rings [19]. A Hahn field is a different ring: a totally ordered exponent group permits monomial division that a multivariable power-series ring need not permit, and we do not claim to remove the obstructions in that latter category ([Warning 9.24](#)). The support foundations are standard Hahn–Neumann and word-order ideas [20, 21], reproved here for self-containment, and the finite extension calculation is an elementary monomial form of Kummer theory.

1.5 Which spectrum this is

The collection uses the word *spectrum* for several inequivalent things, so this article fixes its own sense once. Here $\text{spec } H$ is the algebraic *eigenvalue list* of a finite matrix over K , together with the valuations of those eigenvalues and of the singular values. It is not the differential *phase spectrum* of docs/surcomplex/differential-equations/, which is a multiset of purely infinite surreal parts of primitives of eigenvalues, and whose members are not numbers of the same kind as ours and are not finite angles. It is not the *prime spectrum* of a ring, the sense in which docs/surcomplex/analytic-geometry/ discusses maximal and prime ideals. And the visible-rank levels $r_{<\theta}, r_{\leq\theta}$ of [Section 13](#), although they form a list of scales, are not a spectrum in any of these senses.

2 Fields, workspaces, and conventions

2.1 The algebraic setting

Until [Section 7](#), let F be a set-sized real closed ordered field and put $K = F(i)$, where $i^2 = -1$. We use the standard equivalent characterization that F is ordered and $F(i)$ is algebraically closed. Conjugation is $\overline{x + iy} = x - iy$, and

$$|z| = \sqrt{z\bar{z}} = \sqrt{x^2 + y^2} \in F_{\geq 0}.$$

Every positive element of F has a unique positive square root. No Archimedean property or completeness assumption is imposed.

For a matrix A , write $A^* = \bar{A}^\top$. A square matrix is *Hermitian* when $A^* = A$, *normal* when $A^*A = AA^*$, and *unitary* when $A^*A = AA^* = I$. A matrix with real entries is treated inside F , with transpose in place of conjugate transpose. The integers m, n, k used as dimensions and ranks are always ordinary finite integers. Ambient row and column sizes are positive; zero-dimensional invariant subspaces and empty products are allowed. Unless explicitly stated otherwise, vector spaces and subspaces are over K .

We use ∞ only as an added valuation symbol, with $v(0) = \infty$. It is not a surreal number. Expressions such as $\infty - \infty$ will never be used.

2.2 Hahn workspaces inside the surcomplex numbers

For the scale-sensitive results, choose a set-sized additive subgroup $\Gamma \subseteq \mathbf{No}$, with its inherited order, and set

$$F_\Gamma = \mathbb{R}((t^\Gamma)), \quad K_\Gamma = \mathbb{C}((t^\Gamma)), \quad t^\gamma := \omega^{-\gamma}. \quad (2.1)$$

The last expression uses Conway’s monomial map. It is not a convention about the ordinary real exponential. An element of K_Γ is a formal sum

$$a = \sum_{\gamma \in S} a_\gamma t^\gamma, \quad a_\gamma \in \mathbb{C},$$

where $S \subseteq \Gamma$ is a well-ordered set in increasing exponent order. Zero coefficients may be omitted. The field operations are Hahn addition and convolution. On F_Γ , the sign of a nonzero series is the sign of its coefficient at the least exponent.

No divisibility, finite-rank, countability, or Archimedean assumption is imposed on Γ .

We use two foundational facts. Conway normal forms identify these series with surcomplex numbers, and the Hahn field with divisible value group and algebraically closed coefficient field of characteristic zero is algebraically closed. Gonshor’s Chapter 5 gives the normal-form and support foundations [5]; Poonen’s Corollary 4 gives the relevant algebraic-closedness statement for Mal’cev–Neumann fields [6]. Thus *when Γ is divisible*, $K_\Gamma = F_\Gamma(i)$ is algebraically closed and F_Γ is real closed, so every result proved over a real closed F applies verbatim inside the workspace. These standard foundational theorems are inputs, not reproved here.

Remark 2.1 (What still needs divisibility). For a general Γ the field F_Γ need not be real closed: if $\gamma \notin 2\Gamma$ then the positive element t^γ has no square root in F_Γ , because the valuation of a square lies in 2Γ . Divisibility is nevertheless needed only where a root of an *arbitrary* positive element, or an algebraic closure, is actually used: the Schur triangularization of a non-normal matrix in [Proposition 4.2](#), the positive semidefinite square root of an arbitrary positive semidefinite matrix and the Cholesky factorization in [Proposition 4.3](#), and the reduction of the generalized Hermitian eigenproblem built on them, with its G -orthonormal eigenbasis — all of which can be carried out in the divisible hull $\Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$. [Theorem 10.4](#) computes the exact extension for principal roots of positive semidefinite matrices. [Theorem 12.22](#) computes the exact field of the generalized *eigenvalues* of a definite pencil over an arbitrary Γ , from the Hankel subdiscriminants of its characteristic polynomial, so the eigenvalues no longer need the divisible hull; a normalized eigenbasis can still need further square roots ([Warning 12.23](#)). Neither computes a Schur splitting field for an arbitrary matrix. The pseudospectrum below is a set of scalars $z \in K_\Gamma$; its perturbation characterization uses the SVD of $zI - A$ and remains valid without requiring every eigenvalue of A in the base field. Everything else in this article, including the Hermitian and normal spectral theorems, the SVD, the polar decomposition, every norm and variational principle, the determinantal scales of [Theorem 7.5](#), the positive Cauchy–Binet identity of [Theorem 8.1](#), the rank filtration, the scale algorithm, the perturbation certificates, and the matrix series of [Theorem 19.1](#),

holds over an arbitrary set-sized ordered abelian Γ , by the support-certified constructions of [Section 9](#). The Hankel results of [Section 12](#) are stated over an arbitrary Γ from the start.

Proposition 2.2 (Finite-data localization). *Every finite collection of surcomplex matrices has all its entries in one workspace (2.1). All the decompositions established below can be performed in that workspace.*

Proof. Take the union of the exponent supports in the real and imaginary normal forms of every entry. There are finitely many entries and every support is a set, so their union is a set. Its rational linear span is a divisible set-sized subgroup of $\mathbf{No}$. Reversing exponents to follow the convention $t^\gamma = \omega^{-\gamma}$ puts all entries in its Hahn field. Algebraic closedness supplies all roots needed in the spectral proofs; positive square roots and finite field operations remain in the same workspace. Every basis extension used below has finite length. $\square$

The same argument works for a set of scalar inputs, but we only need finite matrices here. No quantification over a class-sized set of all bases is required. A displayed theorem about matrices over $\mathbf{No}[i]$ means that the finite data are localized and the theorem is applied there. Its explicit algebraic witnesses are still witnesses in any larger workspace.

[Proposition 2.2](#) deliberately passes to a rational span, which is the smallest *divisible* workspace containing the data. There is a smaller and more informative choice.

Proposition 2.3 (Exact-support localization). *Let A be a finite surcomplex matrix. Take the union of the exponent supports of the real and imaginary Conway normal forms of all its entries, negate the exponents to follow the convention $t^\gamma = \omega^{-\gamma}$, and let $H \subseteq \mathbf{No}$ be the additive subgroup they generate. Then H is a set, every entry of A lies in K_H , and H is contained in every subgroup of $\mathbf{No}$ whose Hahn field contains all the entries.*

Proof. A subgroup generated by a set consists of the finite integer combinations of its generators, so it is a set. The remaining assertions are immediate from the definition of the support of a normal form. $\square$

The group H need not be divisible, and that is the point: the results of [Section 9](#) take place inside K_H itself, while [Sections 10](#) and [11](#) measure exactly how far a positive matrix root has to leave it. No proper-class Hahn sum is used anywhere in this article.

2.3 Order, valuation, and topology are separate structures

In a Hahn workspace the ordered modulus and the valuation are not the same thing. For $\rho > 0$, both t^ρ and $2t^\rho$ have valuation ρ , although the latter is outside the ordered disk $|z| \leq t^\rho$. We will use ordered inequalities for optimization and conditioning, and valuation inequalities for relative scale and support precision.

The principal proofs in this article use no infinite limiting process. A unit sphere is not assumed compact, a supremum is not assumed to exist, and no spectral integral is invoked. In [Section 19](#), where an infinite matrix series is used, its Hahn support is checked explicitly. Internal valuation convergence in a chosen Hahn field and the fine order/modulus topology inherited from the whole surreal class must not be conflated.

3 Hermitian geometry without compactness

For $x, y \in K^n$, define

$$\langle x, y \rangle = x^* y = \sum_{j=1}^n \bar{x}_j y_j, \quad \|x\| = \sqrt{x^* x}.$$

This convention is conjugate-linear in the first variable. Since F is ordered, a sum of squares is zero only if every term is zero. Consequently $x^*x > 0$ for $x \neq 0$.

Lemma 3.1 (Cauchy–Schwarz and orthogonal decomposition). *For $x, y \in K^n$,*

$$|x^*y| \leq \|x\| \|y\|.$$

If $x \neq 0$, the vector y decomposes uniquely as

$$y = x \frac{x^*y}{x^*x} + y_\perp, \quad x^*y_\perp = 0.$$

Every finite-dimensional subspace admits an orthonormal basis and has an orthogonal complement.

Proof. The asserted decomposition follows by substitution. Its squared norm is

$$y^*y = \frac{|x^*y|^2}{x^*x} + y_\perp^*y_\perp \geq \frac{|x^*y|^2}{x^*x},$$

which proves Cauchy–Schwarz; the case $x = 0$ is immediate. Gram–Schmidt now works on any finite linearly independent list: subtract the finitely many orthogonal components already constructed and divide the remaining nonzero vector by its positive norm. Extend a basis of the subspace to a basis of K^n and continue this process. The remaining orthonormal vectors span its orthogonal complement. $\square$

This proof still works when a norm is infinitesimal. Dividing by that norm may produce infinite coordinates at an intermediate step, but it is a legal field operation; the final unit vector has all coordinate moduli at most one. It is invalid to delete a nonzero vector merely because its standard part vanishes.

The triangle inequality follows from Cauchy–Schwarz:

$$\|x + y\|^2 \leq \|x\|^2 + 2\|x\| \|y\| + \|y\|^2.$$

Finite orthogonal projections therefore exist by an explicit basis formula, not by an infinite-dimensional projection theorem. If $u_1, \dots, u_k$ is an orthonormal basis of W , then

$$P_W = \sum_{j=1}^k u_j u_j^*, \quad P_W^2 = P_W = P_W^*, \quad K^n = W \oplus W^\perp. \quad (3.1)$$

The projector is independent of the chosen orthonormal basis because it acts as the identity on W and as zero on $W^\perp$.

4 Spectral, positive, and singular-value decompositions

4.1 The finite spectral theorem

Theorem 4.1 (Hermitian spectral theorem). *For every Hermitian $H \in K^{n \times n}$, there are a unitary U and $\lambda_1, \dots, \lambda_n \in F$ such that*

$$H = U \operatorname{diag}(\lambda_1, \dots, \lambda_n) U^*.$$

For real symmetric H , U can be chosen orthogonal with entries in F .

Proof. The characteristic polynomial splits in K , so there is an eigenpair $Hx = \lambda x$ with $x \neq 0$. The scalar x^*Hx is fixed by conjugation, hence lies in F , and

$$\lambda = \frac{x^*Hx}{x^*x} \in F.$$

Normalize x . If $y \perp x$, then $x^*Hy = (Hx)^*y = \lambda x^*y = 0$, so $x^\perp$ is invariant under H . The restriction to that complement is Hermitian. Induction on n completes the construction. For a real symmetric matrix, once $\lambda \in F$ is known, singularity of $H - \lambda I$ over F supplies a nonzero kernel vector in F^n . The same induction can therefore be carried out over F . $\square$

Proposition 4.2 (Schur form and normal matrices). *Every $A \in K^{n \times n}$ is unitarily similar to an upper triangular matrix. It is unitarily diagonalizable if and only if it is normal.*

Proof. Choose and normalize an eigenvector and extend it to an orthonormal basis. The first column below the diagonal then vanishes. Apply induction to the remaining square block to obtain upper triangular Schur form T . If T is normal, comparison of the $(1, 1)$ entries of TT^* and T^*T gives

$$|t_{11}|^2 + \sum_{j>1} |t_{1j}|^2 = |t_{11}|^2.$$

Positivity forces the rest of the first row to vanish. Induction on the remaining normal block makes T diagonal. Conversely a diagonal matrix is normal, and unitary conjugation preserves normality. $\square$

For distinct eigenvalues λ of a Hermitian matrix, its spectral projectors are finite polynomials:

$$P_\lambda = \prod_{\substack{\mu \in \text{spec } H \\ \mu \neq \lambda}} \frac{H - \mu I}{\lambda - \mu}, \quad H = \sum_{\lambda \in \text{spec } H} \lambda P_\lambda. \quad (4.1)$$

Evaluation on the diagonal form proves the formula. The eigenspace projector is canonical; a basis within a repeated eigenspace is not.

4.2 Positivity, square roots, and congruence

A Hermitian matrix is positive semidefinite, written $H \succeq 0$, when $x^*Hx \geq 0$ for every x , and positive definite, written $H \succ 0$, when this is strict for $x \neq 0$. These are equivalent respectively to all eigenvalues being nonnegative or positive. In particular, positive definite does not mean bounded below by a positive *ordinary real* multiple of I .

Proposition 4.3 (Positive square root and inertia). *A positive semidefinite matrix has a unique positive semidefinite square root. The numbers of positive, negative, and zero eigenvalues of a Hermitian matrix are unchanged by an invertible congruence $H \mapsto S^*HS$.*

Proof. Diagonalize H and replace each eigenvalue by its nonnegative square root. For uniqueness, a positive semidefinite R with $R^2 = H$ commutes with H , so it preserves each eigenspace of H . On the eigenspace with eigenvalue λ , diagonalization of R gives only nonnegative roots of $r^2 = \lambda$, hence $R = \sqrt{\lambda}I$ there.

If p is the number of positive eigenvalues of H , its positive eigenspace is a p -dimensional subspace on which the quadratic form is positive definite. Any subspace of dimension greater than p meets the span of the nonpositive eigenvectors nontrivially, so cannot have that property. Thus p is an intrinsic maximum dimension, preserved by invertible congruence. Apply the same argument to $-H$ for the negative count; subtract both counts from n for the nullity. $\square$

The usual Cholesky factorization also survives. If $H = \begin{pmatrix} a & b^* \\ b & C \end{pmatrix} \succ 0$, then $a > 0$ and $C - bb^*/a \succ 0$: evaluate the form at $(-a^{-1}b^*y, y)$. Inductively factor that Schur complement as LL^* . Then

$$H = \begin{pmatrix} \sqrt{a} & 0 \\ b/\sqrt{a} & L \end{pmatrix} \begin{pmatrix} \sqrt{a} & b^*/\sqrt{a} \\ 0 & L^* \end{pmatrix}.$$

This is a finite algorithm, not a limiting process.

4.3 Singular values and the polar decomposition

Theorem 4.4 (Singular value decomposition). *Let $A \in K^{m \times n}$ and $p = \min(m, n)$. There are unitaries U, V and a rectangular diagonal matrix Σ such that*

$$A = U\Sigma V^*, \quad \sigma_1 \geq \cdots \geq \sigma_r > 0, \quad \sigma_{r+1} = \cdots = \sigma_p = 0,$$

where $r = \text{rank } A$ and the diagonal entries of Σ are the σ_j . The positive σ_j are uniquely determined with multiplicity.

Proof. The matrix A^*A is positive semidefinite. Choose an orthonormal eigenbasis $v_1, \dots, v_n$ and write its positive eigenvalues as $\sigma_1^2, \dots, \sigma_r^2$. For $j \leq r$, put $u_j = Av_j/\sigma_j$. Then

$$u_j^* u_k = (\sigma_j \sigma_k)^{-1} v_j^* A^* A v_k = \delta_{jk}.$$

When the eigenvalue is zero, $\|Av_j\|^2 = 0$ implies $Av_j = 0$. Extend the u_j to an orthonormal basis of K^m . These bases give the claimed factorization, with precisely r nonzero diagonal entries. The characteristic polynomial of A^*A determines the squared singular values, and the positive square root determines the singular values themselves. $\square$

When convenient we extend the list by zeros up to length n , so that σ_j^2 lists all eigenvalues of A^*A . In particular, formulas using input subspaces of dimension k remain meaningful for $k > p$.

Set $|A| = (A^*A)^{1/2}$. The SVD gives the polar factorization

$$A = W|A|, \quad W = \sum_{j=1}^r u_j v_j^*. \quad (4.2)$$

Here W is the unique map that is an isometry from $(\ker A)^\perp$ to $\text{im } A$, is zero on $\ker A$, and satisfies (4.2). On that complement $|A|$ is invertible, so the uniqueness follows directly. For a square invertible matrix, W is unitary. For a singular square matrix, unitary extensions of W exist but are generally not unique. Confusing the canonical partial isometry with one of those extensions loses information.

4.4 A generalized Hermitian eigenproblem

If $G \succ 0$ and $H = H^*$, the equation $Hx = \lambda Gx$ reduces to the ordinary Hermitian problem for

$$C = G^{-1/2} H G^{-1/2}, \quad y = G^{1/2} x.$$

Consequently every generalized eigenvalue is in F , and there is a basis x_j with $x_j^* G x_k = \delta_{jk}$. This is useful when the metric itself contains infinitesimal weights. It does not require a uniform ordinary-real lower bound on G . It does, however, use a square root of an arbitrary positive definite matrix, and so is one of the statements listed in [Remark 2.1](#) as needing a divisible workspace. Over a nondivisible Hahn workspace the generalized eigenvalues themselves are handled exactly by [Theorem 12.22](#): they generate the Hahn field on $\Gamma + \sum_k \mathbb{Z}v(h_k)/2$, where

the h_k are the Hankel subdiscriminants of the squarefree part of $\det(XG - H)/\det G$. Only the G -orthonormal normalization can need further square roots ([Warning 12.23](#)).

Every proof in this section used real closedness of F , either through the splitting of a characteristic polynomial or through a square root of an arbitrary positive element. [Section 9](#) proves the Hermitian and normal spectral theorems and the singular-value decomposition again, by a completely different route, over a Hahn workspace whose value group is *not* assumed divisible, and with control on which exponents can occur. Neither proof subsumes the other: the argument here is shorter and applies to any real closed field, while the argument there applies to any set-sized ordered abelian value group and returns support information that the algebraic argument cannot see.

5 Norms and variational principles with attained extrema

5.1 Operator and Frobenius norms

Define

$$\|A\|_{\text{op}} = \sigma_1(A), \quad \|A\|_{\text{F}} = \sqrt{\text{tr}(A^*A)} = \sqrt{\sum_{i,j} |a_{ij}|^2},$$

with operator norm zero for the zero matrix. The SVD proves, rather than assumes, that

$$\|A\|_{\text{op}} = \max_{\|x\|=1} \|Ax\|. \quad (5.1)$$

Indeed, writing $x = Vy$ gives $\|Ax\|^2 = \sum_j \sigma_j^2 |y_j|^2 \leq \sigma_1^2$, with equality at a leading right singular vector. Thus the symbol “max” is legitimate even though the sphere was never declared compact.

Both norms are invariant under compatible unitary multiplication. They satisfy triangle inequalities, and

$$\max_{i,j} |a_{ij}| \leq \|A\|_{\text{op}} \leq \|A\|_{\text{F}} \leq \sqrt{mn} \max_{i,j} |a_{ij}|, \quad (5.2)$$

$$\|AB\|_{\text{op}} \leq \|A\|_{\text{op}} \|B\|_{\text{op}}, \quad \|AB\|_{\text{F}} \leq \|A\|_{\text{op}} \|B\|_{\text{F}}. \quad (5.3)$$

For the first inequality in (5.2), apply A to a coordinate unit vector and then bound one coordinate by the vector norm. The middle inequality follows from $\sigma_1^2 \leq \sum_j \sigma_j^2$, and the last one from the finite sum defining the Frobenius norm. The operator product bound follows from (5.1); the Frobenius bound follows by applying it to every column of B and summing squares. The companion inequality $\|AB\|_{\text{F}} \leq \|A\|_{\text{F}} \|B\|_{\text{op}}$ follows by taking adjoints.

5.2 Courant–Fischer and Ky Fan principles

Theorem 5.1 (Variational principles). *Let $H = H^*$ have eigenvalues $\lambda_1 \geq \dots \geq \lambda_n$. For $1 \leq k \leq n$,*

$$\lambda_k = \max_{\dim W=k} \min_{\substack{x \in W \\ \|x\|=1}} x^* H x, \quad (5.4)$$

$$= \min_{\dim W=n-k+1} \max_{\substack{x \in W \\ \|x\|=1}} x^* H x. \quad (5.5)$$

Also, for $0 \leq k \leq n$,

$$\sum_{j=1}^k \lambda_j = \max_{X^* X = I_k} \text{tr}(X^* H X). \quad (5.6)$$

All inner and outer extrema in these statements are attained.

Proof. For a fixed subspace W with orthonormal basis matrix X , the compression X^*HX is Hermitian. Its largest and smallest eigenvalues are the extrema of the restricted Rayleigh quotient, by diagonalization. This proves inner attainment without compactness.

Every k -dimensional W meets the span of the eigenvectors for $\lambda_k, \dots, \lambda_n$ in a nonzero vector. After normalization that vector has Rayleigh quotient at most λ_k . The span of the first k eigenvectors attains the opposite bound, proving (5.4). The complementary dimension argument proves (5.5).

For (5.6), the case $k = 0$ has value zero on both sides. For $k \geq 1$, express the columns of X in an orthonormal eigenbasis of H . If a_j is the sum of their squared moduli in row j , then $0 \leq a_j \leq 1$ and $\sum_j a_j = k$. Hence

$$\operatorname{tr}(X^*HX) = \sum_j a_j \lambda_j \leq \sum_{j=1}^k \lambda_j.$$

To see the inequality even for signed λ_j , subtract λ_k from every λ_j . The terms before k have nonnegative coefficients and can only decrease on replacing 1 by a_j ; the terms after k are nonpositive. The constant part is unchanged because the weights sum to k . The first k eigenvectors attain equality. $\square$

Applying (5.4) to A^*A and taking nonnegative square roots gives the singular-value principle

$$\sigma_k(A) = \max_{\dim W=k} \min_{\substack{x \in W \\ \|x\|=1}} \|Ax\|. \quad (5.7)$$

The list is padded with zeros to length n when necessary. Applying (5.6) to $-H$ similarly gives the minimum trace as the sum of the smallest k eigenvalues. These formulas are algebraic versions of familiar classical results [8]; their finite-dimensional proofs explain why they remain valid over F .

5.3 Perturbing eigenvalues and singular values

Theorem 5.2 (Ordered Lipschitz bounds). *If H, E are Hermitian of the same size, and both eigenvalue lists are ordered decreasingly, then*

$$|\lambda_j(H + E) - \lambda_j(H)| \leq \|E\|_{\text{op}}.$$

For arbitrary rectangular matrices of the same size,

$$|\sigma_j(A + E) - \sigma_j(A)| \leq \|E\|_{\text{op}}. \quad (5.8)$$

Proof. For a unit vector x , Cauchy–Schwarz gives $|x^*Ex| \leq \|E\|_{\text{op}}$. Insert the two inequalities for the perturbed Rayleigh quotient into (5.4), obtaining the Hermitian result. For singular values, the triangle inequality gives $|\|(A + E)x\| - \|Ax\|| \leq \|E\|_{\text{op}}$ on the unit sphere. Take the attained minima and maxima in (5.7). $\square$

Theorem 5.3 (A Frobenius eigenvalue bound). *For Hermitian H, G with decreasing eigenvalue lists λ, μ ,*

$$\sum_{j=1}^n (\lambda_j - \mu_j)^2 \leq \|H - G\|_F^2.$$

Proof. Let u_i, v_j be orthonormal eigenbases and set $p_{ij} = |u_i^* v_j|^2$. The matrix p has nonnegative entries and every row and column sum is one. Put $c_i = \sum_j p_{ij} \mu_j$. For every k ,

$$\sum_{i=1}^k c_i \leq \sum_{j=1}^k \mu_j,$$

by the weight argument in [Theorem 5.1](#): the column weights $\sum_{i \leq k} p_{ij}$ lie in $[0, 1]$ and sum to k . Equality holds for $k = n$. Summation by parts, using $\lambda_i - \lambda_{i+1} \geq 0$, now gives

$$\operatorname{tr}(HG) = \sum_i \lambda_i c_i \leq \sum_i \lambda_i \mu_i.$$

Expand $\|H - G\|_F^2 = \operatorname{tr} H^2 + \operatorname{tr} G^2 - 2 \operatorname{tr}(HG)$ to conclude. $\square$

This proof does not use a decomposition of a doubly stochastic matrix into permutations, a measure on a unitary group, or a real-valued limiting argument. It is useful when all the eigenvalue movements, not just the largest one, matter.

6 Least squares, pseudoinverses, and best low-rank models

6.1 The Moore–Penrose inverse

Given an SVD $A = U\Sigma V^*$, let $\Sigma^\dagger$ be the transposed rectangular diagonal matrix whose nonzero entries are σ_j^{-1} and whose remaining entries are zero. Define

$$A^\dagger = V\Sigma^\dagger U^*.$$

Direct diagonal multiplication proves the Penrose identities

$$AA^\dagger A = A, \quad A^\dagger AA^\dagger = A^\dagger, \quad (AA^\dagger)^* = AA^\dagger, \quad (A^\dagger A)^* = A^\dagger A. \quad (6.1)$$

They characterize $A^\dagger$ uniquely, as in the classical construction [\[10\]](#). For completeness, if B satisfies these identities, AB is the orthogonal projector onto $\operatorname{im} A$: it is a Hermitian idempotent with that image. Similarly BA is the orthogonal projector onto $(\ker A)^\perp$, since $\ker(BA) = \ker A$. The equation $B = BAB$ then says that B is zero on $(\operatorname{im} A)^\perp$ and, on $\operatorname{im} A$, is the inverse of the restriction $A : (\ker A)^\perp \rightarrow \operatorname{im} A$. This uniquely determines it and also shows that the SVD definition is basis-independent.

Theorem 6.1 (Least squares with an attained minimum). *For $b \in K^m$, all minimizers of $\|Ax - b\|$ are*

$$x = A^\dagger b + z, \quad z \in \ker A.$$

The unique minimizer of least norm is $A^\dagger b$, and the minimum residual is $\|(I - AA^\dagger)b\|$.

Proof. Write $c = U^*b$ and $y = V^*x$. The squared residual is

$$\sum_{j=1}^r |\sigma_j y_j - c_j|^2 + \sum_{j=r+1}^m |c_j|^2.$$

It is minimized precisely by $y_j = c_j/\sigma_j$ for $j \leq r$, with remaining input coordinates free. These free coordinates describe $\ker A$. They add a nonnegative squared norm to the fixed solution, so setting them to zero uniquely minimizes the norm. The residual formula is the corresponding orthogonal projection formula. $\square$

Small singular values are not discarded in this theorem. Their reciprocals may be infinite, and a mathematically exact least-squares solution may therefore have infinite norm even when the data are finite. Regularization in [Section 20](#) changes the optimization problem rather than silently changing what “nonzero” means.

6.2 Low-rank approximation

Let

$$A_k = \sum_{j=1}^k \sigma_j u_j v_j^*, \quad 0 \leq k < p.$$

Here zero terms can be included when $k > r$.

Theorem 6.2 (Eckart–Young bounds over F). *Among matrices B of rank at most k ,*

$$\min \|A - B\|_{\text{op}} = \sigma_{k+1}(A), \quad (6.2)$$

$$\min \|A - B\|_{\text{F}}^2 = \sum_{j=k+1}^p \sigma_j(A)^2. \quad (6.3)$$

Both minima are attained by A_k .

Proof. The SVD computes the two errors of A_k exactly. For the lower operator bound, a matrix B of rank at most k has a kernel meeting the span of $v_1, \dots, v_{k+1}$ in a nonzero vector. Normalize that vector x . Then

$$\|(A - B)x\| = \|Ax\| \geq \sigma_{k+1}.$$

For the Frobenius bound, choose $n - k$ orthonormal vectors w_j in $\ker B$ and extend them to an orthonormal basis. Summing squared column norms in that basis gives

$$\|A - B\|_{\text{F}}^2 \geq \sum_{j=1}^{n-k} \|Aw_j\|^2.$$

The minimum-trace version of (5.6), applied to A^*A , bounds the right side below by the sum of its smallest $n - k$ eigenvalues. That is exactly (6.3), after padding the singular list with zeros. $\square$

The least-squares lower-rank problem originates in Eckart and Young [9]. Here its optimal error is allowed to be infinitesimal or infinite. The theorem asserts existence of an optimizer, not uniqueness: repeated singular values and the operator norm in particular can produce many optimizers. It also gives a direct operational meaning to every singular scale: σ_{k+1} is the exact error required to replace the matrix by a rank- k model.

7 Hahn valuation and singular-value scales

From this section onward, unless a result explicitly returns to arbitrary F , work in (2.1). For $a \neq 0$, put

$$v(a) = \min \text{supp } a \in \Gamma, \quad \text{lc}(a) = a_{v(a)}.$$

Let

$$\mathcal{O} = \{a : v(a) \geq 0\}, \quad \mathfrak{m} = \{a : v(a) > 0\}.$$

The residue map $\text{st} : \mathcal{O} \rightarrow \mathbb{C}$ extracts the coefficient at exponent zero. On real finite elements it is the usual standard part. The valuation of a matrix or vector means the minimum valuation of its entries; all-zero objects have valuation ∞ . We sometimes write $v_{\text{mat}}(A)$ to make this convention explicit.

Remark 7.1 (Logical order). This section and the next use the singular-value decomposition. For a divisible Γ that is Theorem 4.4. For an arbitrary set-sized ordered abelian Γ it is Theorem 9.18, proved in Section 9 by an argument that uses nothing from this section onward; likewise the positive square root taken inside Lemma 7.2 is supplied without real closedness by Lemma 9.8. So the scale theory below is available over an arbitrary Γ , and there is no circularity — only a forward reference, made so that the classical development can be read first.

7.1 Positivity prevents leading cancellation

Lemma 7.2 (Valuation of a Euclidean norm). *For any nonzero finite vector x and any nonzero matrix A ,*

$$v(\|x\|) = \min_j v(x_j), \quad v(\|A\|_{\text{op}}) = v(\|A\|_{\text{F}}) = \min_{i,j} v(a_{ij}). \quad (7.1)$$

Proof. Let $\alpha = \min_j v(x_j)$. The coefficient of $t^{2\alpha}$ in $\sum_j |x_j|^2$ is

$$\sum_{v(x_j)=\alpha} |\text{lc}(x_j)|^2 > 0.$$

Thus its valuation is 2α , and taking its positive square root gives valuation α . Apply the same argument to the entries of A for the Frobenius norm. Finally (5.2) traps the operator norm between a largest entry modulus and a fixed positive ordinary-real multiple of it. Both bounds have the same valuation, proving the remaining equality. $\square$

A finite sum of positive leading coefficients is crucial. An arbitrary sum of complex entries can cancel at its leading exponent; a squared Euclidean norm cannot.

Lemma 7.3 (Unitary matrices preserve the integral lattice). *If U is unitary, then $U, U^{-1} \in \mathcal{O}^{n \times n}$. Its reduction $\text{st } U$ is an ordinary complex unitary matrix, and $U\mathcal{O}^n = \mathcal{O}^n$.*

Proof. Every entry has modulus at most one because its column has squared norm one. Hence every entry has nonnegative valuation. The inverse is U^* and has the same property. Reducing $U^*U = I$ gives $(\text{st } U)^*(\text{st } U) = I$, and the two lattice inclusions follow from integrality of U and its inverse. $\square$

For the nonzero singular values of A , define

$$\gamma_j = v(\sigma_j), \quad \gamma_1 \leq \cdots \leq \gamma_r. \quad (7.2)$$

Their order is reversed because a larger positive Hahn number has a smaller or equal leading exponent. Different singular values may have the same valuation. The tuple (7.2) records only scales, not the positive leading coefficients or the rest of each expansion.

7.2 The determinantal characterization

Define the k th determinantal valuation by

$$\Delta_0(A) = 0, \quad \Delta_k(A) = \min_{\substack{I \subseteq \{1, \dots, m\}, |I|=k \\ J \subseteq \{1, \dots, n\}, |J|=k}} v(\det A_{I,J}), \quad 1 \leq k \leq p. \quad (7.3)$$

It is ∞ exactly when $k > r$.

Lemma 7.4 (Integral row and column invariance). *If $L \in \text{GL}_m(\mathcal{O})$ and $R \in \text{GL}_n(\mathcal{O})$, then $\Delta_k(LAR) = \Delta_k(A)$ for every k .*

Proof. By the finite Cauchy–Binet formula, every k -minor of LA is a sum of k -minors of A multiplied by k -minors of L . The latter coefficients are integral, so every summand has valuation at least $\Delta_k(A)$. The ultrametric inequality gives $\Delta_k(LA) \geq \Delta_k(A)$. Apply the same argument to L^{-1} for the reverse inequality. Right multiplication is identical, or follows by transposition. The argument also covers vanishing minors. $\square$

Theorem 7.5 (Singular scales are determinantal scales). *For a matrix A of rank r and $1 \leq k \leq r$,*

$$\Delta_k(A) = \gamma_1 + \cdots + \gamma_k, \quad \gamma_k = \Delta_k(A) - \Delta_{k-1}(A). \quad (7.4)$$

In particular the finite slopes of the determinantal profile are nondecreasing. The tuple $(\gamma_1, \dots, \gamma_r)$ is a complete invariant for equivalence under integral invertible row and column operations, with the matrix dimensions fixed.

Proof. In an SVD $A = U\Sigma V^*$ both unitary factors and their inverses are integral by [Lemma 7.3](#). Thus [Lemma 7.4](#) reduces the calculation to Σ . Its nonzero k -minors are products of k distinct nonzero singular values. Their minimum valuation is the sum of the k smallest γ_j , proving (7.4).

Write $\sigma_j = t^{\gamma_j} u_j$, where $u_j \in \mathcal{O}^\times$ is a positive unit. Multiplying diagonal rows by u_j^{-1} shows that A is integrally equivalent to the rectangular diagonal matrix $\text{diag}(t^{\gamma_1}, \dots, t^{\gamma_r}, 0, \dots)$. Thus two matrices with the same tuple are equivalent. Conversely, equivalence preserves Δ_k , hence the tuple and rank. $\square$

For integral matrices this is the singular-value interpretation of their valuation-ring invariant factors. The valuation ring need not be Noetherian or a discrete valuation ring. Only finitely many entries and minors are involved, so a minimum valuation among them exists. Unit equivalence remembers less than unitary equivalence: the latter preserves all singular values, not merely their valuations.

An exterior-power interpretation is also useful. Give $\bigwedge^k K^n$ the inner product for which wedges of coordinate basis vectors are orthonormal. The entries of $\bigwedge^k A$ are the k -minors of A . Wedges of the singular bases yield

$$\left\| \bigwedge^k A \right\|_{\text{op}} = \sigma_1 \cdots \sigma_k.$$

Applying [Lemma 7.2](#) recovers (7.4). Thus a determinantal scale is the scale of the maximal k -dimensional volume factor.

7.3 Why entrywise scales alone are not enough

Example 7.6 (A nearly dependent matrix). Let $q \in \Gamma_{>0}$ and

$$A_q = \begin{pmatrix} 1 & 1 \\ 1 & 1 + t^q \end{pmatrix}.$$

Every entry has valuation zero. Nevertheless, $\det A_q = t^q$, so $\Delta_1 = 0$, $\Delta_2 = q$, and the singular scales are $0, q$. In contrast, the matrix of all ones has the same entrywise valuations and rank one. Changing q changes the hidden singular scale without changing any of those entrywise valuations.

The matrix A_q is real symmetric and positive definite, since its leading principal minors are 1 and t^q . Its singular values are its positive eigenvalues:

$$\sigma_{\pm} = \frac{2 + t^q \pm \sqrt{4 + t^{2q}}}{2}.$$

Hence $\text{lc}(\sigma_+) = 2$ and $\sigma_- = (1/2)t^q + \text{terms of valuation greater than } q$. A minimum-weight assignment using only entry valuations could miss the cancellation in the determinant. Exact minors or an equivalent elimination calculation are needed.

8 Gram coefficients recover scales and leading amplitudes

The polynomial report already develops Newton profiles of scalar polynomials [2]. Positive Gram matrices give an especially clean matrix input to that machinery: their characteristic coefficients carry no leading cancellation from squared minors.

Theorem 8.1 (Positive Cauchy–Binet identity). *For $A \in K^{m \times n}$ write*

$$\det(I_n + sA^*A) = \sum_{k=0}^p c_k s^k, \quad c_0 = 1. \quad (8.1)$$

Then

$$c_k = \sum_{|I|=|J|=k} |\det A_{I,J}|^2 = e_k(\sigma_1^2, \dots, \sigma_r^2). \quad (8.2)$$

For $1 \leq k \leq r$,

$$c_k > 0, \quad v(c_k) = 2\Delta_k(A). \quad (8.3)$$

For $k > r$, $c_k = 0$.

Proof. Expansion of $\det(I + sB)$ by the columns in which an entry of sB is selected shows that its coefficient of s^k is the sum of the principal k -minors of B . For $B = A^*A$, Cauchy–Binet expands the principal minor indexed by J as

$$\det((A^*A)_{J,J}) = \sum_{|I|=k} \overline{\det A_{I,J}} \det A_{I,J}.$$

This proves the first equality in (8.2). Diagonalizing A^*A gives the second. If a k -minor is nonzero, the leading coefficient of the sum of its squared moduli and all other squared minors of minimum valuation is a positive real number. Therefore no leading cancellation is possible, and the valuation is twice the minimum minor valuation. $\square$

Thus the Gram determinant polynomial provides all the singular scales without finding an eigenvector. Its nonzero coefficient valuations are $2\Delta_k$, with successive differences $2\gamma_k$. This statement works in any ordered group Γ ; it does not require drawing a polygon in an ordinary real plane. The same positivity mechanism, applied to forest weights of a row Laplacian, gives the eigenvalue valuations of positive rate matrices in the companion report held in `docs/surreal/markov-generators-at-every-scale/` (`markov:rem:gram`). The same mechanism governs spanning-tree sums of weighted incidence Laplacians in `docs/surreal/matrix-scaling-at-surreal-scales/` (`scale:thm:tree`).

8.1 Resolving singular values at a common scale

Define

$$C_k = \text{st}(t^{-2\Delta_k} c_k) > 0 \quad (0 \leq k \leq r), \quad C_0 = 1.$$

Let a maximal equal-scale block be

$$\gamma_{p_0+1} = \dots = \gamma_{p_0+d} = \gamma,$$

where the preceding scales, if any, are strictly smaller and the following ones, if any, are strictly larger. Write $a_j = \text{lc}(\sigma_j) > 0$.

Theorem 8.2 (Residual polynomial for a singular-scale block). *For $0 \leq \ell \leq d$,*

$$\frac{C_{p_0+\ell}}{C_{p_0}} = e_\ell(a_{p_0+1}^2, \dots, a_{p_0+d}^2). \quad (8.4)$$

Consequently the ordinary real polynomial

$$R_\gamma(X) = X^d + \sum_{\ell=1}^d (-1)^\ell \frac{C_{p_0+\ell}}{C_{p_0}} X^{d-\ell} \quad (8.5)$$

has roots precisely the squared leading coefficients in that block, with multiplicity.

Proof. In the elementary symmetric sum for $c_{p_0+\ell}$, a product of minimum valuation must contain all the p_0 singular values of smaller valuation, exactly ℓ singular values from the block, and none from later blocks. Indeed, omitting a strictly smaller scale while using a larger one increases the sum of valuations; the increasing ordering shows that only choices across the equal boundary can tie the minimum. Therefore its leading coefficient is

$$\left(\prod_{j=1}^{p_0} a_j^2 \right) e_\ell(a_{p_0+1}^2, \dots, a_{p_0+d}^2).$$

For $\ell = 0$, the prefactor is exactly C_{p_0} . Division proves (8.4), and the elementary symmetric factorization proves (8.5). $\square$

This does not give all higher coefficients of the singular values. It gives their valuations and first nonzero coefficients; repeated roots of R_γ may require another scale refinement. It is already stronger than knowing only the Newton slopes.

For [Example 7.6](#),

$$c_1 = 4 + 2t^q + t^{2q}, \quad c_2 = t^{2q},$$

so $C_1 = 4$ and $C_2 = 1$. The two one-element blocks give leading squares 4 and $1/4$, hence leading singular coefficients 2 and $1/2$. For a same-scale example,

$$A = \text{diag}(2t^\gamma, 3t^\gamma), \quad C_1 = 13, \quad C_2 = 36,$$

and $R_\gamma(X) = X^2 - 13X + 36 = (X - 4)(X - 9)$. The valuation alone cannot distinguish the factors 2 and 3, but the positive Gram coefficients do.

8.2 The Gram case and its parity constraint

One consequence of [Theorem 7.5](#) will be used repeatedly below and is worth recording separately. It is a corollary, not a new identity: it is what (8.3) says once the determinantal scales of A^*A are themselves read off by (7.4).

Corollary 8.3 (Even determinantal scales of Gram matrices). *For $A \in K_\Gamma^{m \times n}$ and $1 \leq k \leq \text{rank}(A)$,*

$$\Delta_k(A^*A) = 2\Delta_k(A). \quad (8.6)$$

Proof. The nonzero eigenvalues of A^*A are the σ_j^2 , of valuations $2\gamma_j$ in the same nondecreasing order. Apply (7.4) to A^*A and to A . $\square$

Equivalently, the positive Cauchy–Binet identity prevents leading cancellation in the sums of squared minors, which is why (8.6) has no error term. The parity visible in it is an arithmetic property of the group Γ , not a statement about an integer-valued valuation; [Corollary 10.5](#) turns it into an exact criterion for factoring a positive matrix as a Gram matrix.

9 Support-certified descent without divisibility

Every decomposition in [Section 4](#) was obtained from real closedness of F , which [Section 2](#) supplies inside a Hahn workspace only when Γ is divisible. Passing to the divisible hull is harmless for existence and fatal for arithmetic: it makes every positive square root free, and so erases the distinction between a matrix whose spectral data need no new scales and one whose positive root genuinely does. This section keeps the exact exponent group. Throughout it, Γ is an arbitrary set-sized ordered abelian group and

$$F_\Gamma = \mathbb{R}((t^\Gamma)), \quad K_\Gamma = \mathbb{C}((t^\Gamma)) = F_\Gamma(i),$$

with no divisibility, finite-rank, countability or Archimedean assumption. Conjugation fixes every t^γ , and, as before, a positive exponent means a small monomial.

9.1 Strong summation is not a limit convention

Definition 9.1 (Strongly summable family). A set-indexed family $(f_i)_{i \in I}$ of Hahn series is *strongly summable* if $\bigcup_i \text{supp}(f_i)$ is well ordered and, for each exponent γ , only finitely many i have a nonzero coefficient at γ . Its sum is the coefficientwise finite sum.

This licenses arbitrary regrouping of a jointly summable family. It does not declare a sequence of partial sums to converge, and the distinction is not academic at higher rank.

Example 9.2 (A rank-two warning). Let $\Gamma = \mathbb{Z}^2$ with the lexicographic order, and put $\delta = (0, 1)$ and $\Omega = (1, 0)$, so that $0 < n\delta < \Omega$ for every positive integer n . The Hahn identity

$$\sum_{n \geq 0} t^{n\delta} = (1 - t^\delta)^{-1}$$

is valid, but its partial sums do not converge to the right-hand side in the valuation topology: after the terms indexed $0, \dots, N$, the tail has valuation $(N + 1)\delta < \Omega$ for every N . A remainder bound $v(\text{remainder}_N) \geq N\delta$ is therefore not by itself a convergence proof in arbitrary rank, and it is never used as one here. This is the same separation of algebraic identity, strong summability and workspace-relative topology drawn in [Section 19](#), exhibited in a value group where even the first two do not bring the third with them.

9.2 A self-contained support engine

[Section 19](#) cites the positive-support lemma to Gonshor. Because finiteness *across all word lengths*, not merely at a fixed degree, is what the splitting construction needs, we prove it. The combinatorial method is the minimal-bad-sequence argument associated with Higman's lemma [\[21\]](#); the resulting support statement is the Hahn–Neumann lemma [\[20\]](#).

Lemma 9.3 (Ordered words). *Let S be a well-ordered set, and order finite words in S by subsequence embedding with coordinatewise increase. Every infinite sequence of words has a pair $i < j$ such that the i th word embeds in the j th.*

Proof. Suppose a bad sequence exists. Choose it successively so that at each position the word has the least possible length among words allowing a bad continuation. No chosen word is empty. Write $w_i = u_i a_i$ with last letter a_i . A sequence in a well order has an infinite nondecreasing subsequence; choose $a_{i_0} \leq a_{i_1} \leq \dots$.

Then $w_0, \dots, w_{i_0-1}, u_{i_0}, u_{i_1}, \dots$ is still bad. A comparison from its initial segment into some u_{i_j} would compare the corresponding original words; and a comparison $u_{i_j} \preceq u_{i_k}$, together with $a_{i_j} \leq a_{i_k}$, would compare w_{i_j} with w_{i_k} . Both are impossible. This contradicts minimality of the length at position i_0 . $\square$

Lemma 9.4 (Positive support monoids). *Let $S \subseteq \Gamma_{>0}$ be well ordered. The additive monoid $\langle S \rangle_{\mathbb{N}}$ generated by S is well ordered, and each $\gamma \in \Gamma$ is the sum of only finitely many finite ordered words with letters in S .*

Proof. If the monoid admitted an infinite strictly decreasing sequence, choose a word representing each term. Lemma 9.3 gives an embedding of an earlier word into a later one; positivity of the unused letters and increase of the matched letters make the earlier sum at most the later one, a contradiction.

If infinitely many distinct words had sum γ , apply the same lemma to those words. Equality of sums forbids any unused positive letter and any strict increase of a matched letter, so the two words have the same length and the same letters, contrary to being distinct. $\square$

Lemma 9.5 (Hahn evaluation). *Let $z_1, \dots, z_d \in \mathfrak{m}$ and $S = \bigcup_{j=1}^d \text{supp}(z_j)$. Every formal series $f \in \mathbb{C}[[X_1, \dots, X_d]]$ has a well-defined strongly summed evaluation $f(z_1, \dots, z_d) \in K_{\Gamma}$, with support in $M = \langle S \rangle_{\mathbb{N}}$, and in $M \setminus \{0\}$ when $f(0) = 0$. Evaluation preserves sums, products, conjugation with real formal variables, and formal composition whose inner series have zero constant coefficient.*

Proof. A finite union of well-ordered subsets of a linear order is well ordered. Every contribution to a coefficient of $f(z)$ determines an ordered word of exponents in S together with a choice of one of the d variables at each position. By Lemma 9.4 there are, for a fixed exponent, only finitely many such words and labels, so every coefficient sum is finite and the support lies in M . For a product or a composition, expanding either side gives the same labelled words with the same finite contributions at each exponent, so regrouping is legal and the formal identities survive evaluation. Conjugation preserves supports, and the same coefficientwise argument applies. $\square$

Matrix formal series reduce to this lemma by taking finitely many scalar entries as the variables; the matrix arguments need not commute, since the scalar coefficient bookkeeping keeps all ordered matrix products apart. No bound on the growth of the ordinary complex Taylor coefficients is needed, because a coefficient at a fixed Hahn exponent is still a finite sum. This is exactly the mechanism behind Theorem 19.1, stated once in the generality the splitting construction requires.

Corollary 9.6 (Exponent-preserving inverses and unit roots). *If $h \in \mathfrak{m}$, the geometric and binomial series define $(1+h)^{-1}$ and $(1+h)^a$ for every rational a , and all their exponents lie in the group generated by $\text{supp}(h)$. Every nonzero Hahn series has an inverse without enlarging its exponent group.*

Proof. Apply the formal geometric and binomial identities and Lemma 9.5. For a general series factor $f = ct^{\alpha}(1+h)$; its inverse is $c^{-1}t^{-\alpha}(1+h)^{-1}$, and both α and the exponents of h lie in the original support subgroup. $\square$

Lemma 9.7 (Formal maps are valuation nonexpanding). *Let $f \in \mathbb{C}[[X_1, \dots, X_d]]$ and $x, y \in \mathfrak{m}^d$. Then*

$$v(f(x) - f(y)) \geq \min_j v(x_j - y_j),$$

and the same entrywise assertion holds for a matrix of formal series.

Proof. There are formal series $g_j \in \mathbb{C}[[X, Y]]$ with $f(X) - f(Y) = \sum_j (X_j - Y_j)g_j(X, Y)$: for a monomial this is a telescoping product identity, and collecting by total degree gives the formal identity. Evaluate with Lemma 9.5; each $g_j(x, y)$ is integral. $\square$

Lemma 9.8 (Norms exist without real closedness). *For every nonzero $x \in K_\Gamma^n$ we have $x^*x > 0$, and x^*x has a unique positive square root in F_Γ . The same holds for any nonzero finite sum of squared moduli of entries, including $\|A\|_F^2 = \text{tr}(A^*A)$ for $A \neq 0$. A zero sum has the unique nonnegative square root zero.*

Proof. Let $\alpha = v(x)$ and write $x = t^\alpha y$ with y integral and at least one entry of nonzero residue. The constant coefficient of y^*y is $\sum_j |\text{st}(y_j)|^2 > 0$, so $x^*x = ct^{2\alpha}(1+h)$ with $c > 0$ real and $h \in \mathfrak{m} \cap F_\Gamma$. The binomial series $(1+h)^{1/2}$ is a Hahn series by Lemma 9.5, so $\sqrt{c}t^\alpha(1+h)^{1/2} \in F_\Gamma$ is the required root, and uniqueness follows from $(u-v)(u+v) = 0$ in an ordered field. $\square$

Here positivity ensures that the leading squared terms cannot cancel. Real closedness is not assumed: if $\gamma \notin 2\Gamma$, the positive element t^γ has no square root in F_Γ and therefore cannot be a nonzero sum of squares. That single observation is why an SVD can stay in the base field while an arbitrary positive matrix square root need not, and it is quantified exactly in Section 10. Note also that Lemma 9.8 supplies the root whose existence Lemma 7.2 takes for granted, so the valuation computation there is valid over an arbitrary Γ ; Lemma 7.3 likewise used only integrality and is unaffected.

9.3 Separated-block lifting

Let

$$D = \text{diag}(c_1 I_{n_1}, \dots, c_s I_{n_s}), \quad c_j \in \mathbb{R}, \quad c_j \neq c_\ell \ (j \neq \ell), \quad (9.1)$$

and write Q_j for the corresponding constant coordinate projections. Suppose $E = E^* \in \mathfrak{m}^{n \times n}$ and $B = D + E$. Let S be the union of the supports of the entries of E and put $M = \langle S \rangle_{\mathbb{N}}$.

Theorem 9.9 (Separated-block lifting). *There are Hermitian projections $P_1, \dots, P_s$ and a unitary W over K_Γ such that*

$$P_j P_\ell = 0 \ (j \neq \ell), \quad \sum_j P_j = I, \quad B P_j = P_j B, \quad P_j \equiv Q_j \pmod{\mathfrak{m}}, \quad (9.2)$$

$$W \equiv I \pmod{\mathfrak{m}}, \quad P_j W = W Q_j, \quad W^* B W \text{ is block diagonal relative to } (Q_j). \quad (9.3)$$

Every entry of $P_j - Q_j$ and of $W - I$ is supported in $M \setminus \{0\}$. The construction uses universal finite-variable formal series in the entries of E , with ordinary constant coefficients.

Proof. Put $R_0(\zeta) = (\zeta I - D)^{-1}$ and define

$$P_j(E) = \sum_{k \geq 0} \text{Res}_{\zeta=c_j} \left[R_0(\zeta) (E R_0(\zeta))^k \right]. \quad (9.4)$$

For each k the residue is a homogeneous polynomial of degree k in the entries of E whose coefficients are ordinary complex numbers, obtained as residues of rational functions with poles among the fixed real c_j . Hence Lemma 9.5 makes (9.4) a strongly summed Hahn series with the stated support; the term $k = 0$ is Q_j .

The projector identities are justified without presupposing a spectral theorem over K_Γ . Parametrize Hermitian E by n^2 independent *ordinary real* coordinates. For sufficiently small such coordinates, disjoint complex circles around the c_j separate the eigenvalues of the ordinary complex Hermitian matrix $D + E$: if λ is farther than $\|E\|$ from every c_j , then $I - (\lambda I - D)^{-1}E$ is invertible by its ordinary geometric series, so $\lambda I - D - E$ is invertible. On those circles the ordinary resolvent has its Neumann expansion, and the contour integrals are the orthogonal spectral projectors, as one sees by diagonalizing the ordinary Hermitian matrix. Their Taylor series at $E = 0$ are exactly (9.4). Therefore $P_j^* = P_j$, $P_j^2 = P_j$, $P_j P_\ell = 0$, $\sum_j P_j = I$ and $[B, P_j] = 0$ hold as finite-variable formal identities, every Taylor coefficient of the difference being

zero, and Hahn evaluation transports them to our E . This is a proof of universal coefficient identities by ordinary complex analysis, *not* contour integration in a surreal topology.

Now set

$$X = \sum_j P_j Q_j, \quad G = X^* X = \sum_j Q_j P_j Q_j, \quad W = X G^{-1/2}. \quad (9.5)$$

Here $X \in I + \mathfrak{m}^{n \times n}$ and $G = I + H$ with $H \in \mathfrak{m}^{n \times n}$, and

$$G^{-1/2} = \sum_{k \geq 0} \binom{-1/2}{k} H^k \quad (9.6)$$

is strongly summable by [Lemma 9.5](#). It is Hermitian, commutes with G and squares to G^{-1} , so $W^* W = I$; a square matrix with a left inverse is invertible, so also $W W^* = I$. Orthogonality of the projectors gives $P_j X = X Q_j$, while G , and hence $G^{-1/2}$, commutes with every Q_j ; thus $P_j W = W Q_j$. Since B commutes with each P_j , its conjugate $W^* B W$ commutes with every Q_j , which is block diagonality. All nonconstant terms are formal series without constant term in the entries of E , so their supports lie in $M \setminus \{0\}$. $\square$

Remark 9.10 (What is canonical and what is not). For a fixed residue decomposition D and projections Q_j , the formulas (9.4) and (9.5) specify a particular lift. They do not choose a preferred ordinary eigenbasis inside a repeated eigenspace of a later leading coefficient; those finite-dimensional choices are recorded in the splitting tree below. This is the same distinction as the one after (4.1): the eigenspace projector is canonical, a basis within it is not.

Writing $E_{j\ell} = Q_j E Q_\ell$ and taking the $k = 1$ residues,

$$W = I + K + O_{\text{formal}}(E^2), \quad K_{j\ell} = \frac{E_{j\ell}}{c_\ell - c_j} \ (j \neq \ell), \quad K_{jj} = 0, \quad (9.7)$$

with K skew-Hermitian; $X = I + K + O(E^2)$ gives $X^* X = I + O(E^2)$. The first-order off-diagonal term of $W^*(D + E)W$ is $E + [D, K]$, which vanishes between different blocks, and the diagonal block to second degree is

$$c_j I + E_{jj} + \sum_{\ell \neq j} \frac{E_{j\ell} E_{\ell j}}{c_j - c_\ell} + O_{\text{formal}}(E^3). \quad (9.8)$$

These O_{formal} symbols mean support-certified sums of monomials of the indicated total degrees, not a topological limiting convention. The second-order coupling correction in (9.8) is the formal-series counterpart of the Schur-complement correction (18.2), and [Example 18.2](#) is the same phenomenon at $n = 2$.

Proposition 9.11 (Local valuation estimate). *Apply the construction with the same D, Q_j to Hermitian $E, E' \in \mathfrak{m}^{n \times n}$. Then*

$$v(P_j(E) - P_j(E')) \geq v(E - E'), \quad v(W(E) - W(E')) \geq v(E - E').$$

If $A = \mu I + t^\alpha V(D + E)V^$ with V constant unitary, $A' = A'^*$, and $v(A' - A) > \alpha$, then the corresponding cluster projectors satisfy*

$$v(\Pi_j(A') - \Pi_j(A)) \geq v(A' - A) - \alpha. \quad (9.9)$$

Proof. The first pair of inequalities is [Lemma 9.7](#) applied to the universal formal entries. For the second, keep the same μ, α, V and put $E' = E + t^{-\alpha} V^*(A' - A)V$; its entries are infinitesimal, and [Lemma 7.3](#) gives $v(E' - E) = v(A' - A) - \alpha$. $\square$

The estimate is sharp already for a two-by-two constant diagonal matrix perturbed off the diagonal, since (9.7) has a nonzero linear term. Its role is to give a support-compatible *valuation* form of separated-cluster stability. It does not supersede [Theorem 17.1](#) and [Corollary 17.2](#), which measure an ordered Frobenius displacement against an ordered gap; the two are different statements, and [Example 17.3](#) shows that an infinitesimal valuation hypothesis alone cannot control an ordered subspace rotation when the gap is infinitesimal too.

9.4 Finite splitting trees and exact spectral descent

Theorem 9.12 (Hermitian Hahn spectral theorem). *Let Γ be any set-sized ordered abelian group. For every Hermitian $A \in K_\Gamma^{n \times n}$ there are a unitary U over K_Γ and $\lambda_1, \dots, \lambda_n \in F_\Gamma$ with $U^*AU = \text{diag}(\lambda_1, \dots, \lambda_n)$. For real symmetric input U can be chosen real orthogonal. Every exponent occurring in U or in the λ_j lies in the subgroup generated by the exponents of the input entries.*

Proof. Induct on n , using no algebraic closedness or real closedness assumption. The case $n = 1$ is immediate. Set

$$\mu = \frac{1}{n} \text{tr}(A), \quad C = A - \mu I. \quad (9.10)$$

If $C = 0$ take $U = I$. Otherwise let $\alpha = v(C)$ and write

$$t^{-\alpha}C = C_0 + E_0, \quad C_0 \in \mathbb{C}^{n \times n}, \quad E_0 \in \mathfrak{m}^{n \times n}. \quad (9.11)$$

The constant matrix C_0 is nonzero, Hermitian and traceless, so it has at least two distinct ordinary real eigenvalues: by the ordinary spectral theorem a Hermitian matrix with one eigenvalue is scalar, and a traceless scalar matrix in characteristic zero is zero. Choose an ordinary constant unitary V with $V^*C_0V = D$ of the form (9.1), with $s \geq 2$, and apply [Theorem 9.9](#) to $D + V^*E_0V$. Then $W^*V^*AVW = \text{diag}(A_1, \dots, A_s)$ with each A_j Hermitian of strictly smaller size. By induction choose unitary U_j diagonalizing A_j ; then $U = VW \text{diag}(U_1, \dots, U_s)$ diagonalizes A , and the diagonal entries are real because the diagonal matrix is Hermitian.

For the support claim let H be the subgroup generated by the input supports. Both μ and C have supports in H , and $\alpha \in H$, so normalization only translates supports inside H ; constant changes of basis add no exponents; and [Theorem 9.9](#) uses sums of positive exponents already in H , so W and the A_j stay in K_H . The induction therefore runs entirely inside K_H . For real input use real constant eigenbases; every coefficient of the projector and polar constructions is then real. $\square$

This gives the Hahn version of [Theorem 4.1](#) without divisibility and with control of the exponent group. The earlier proof is shorter and applies to any real closed field; the present version survives when K_Γ is not algebraically closed, and it alone says where the exponents can be. For real symmetric input a third proof, by coprime Hensel lifting and polynomial idempotents with no contour argument, is recorded in [Remark 12.5](#).

Corollary 9.13 (The splitting tree is finite). *The construction in [Theorem 9.12](#) has at most $n - 1$ nontrivial splitting nodes. This bound is independent of the order type of the input supports and of the rank or cardinality of Γ .*

Proof. Every internal node has at least two children, and the dimension at a node is the sum of its children's positive dimensions, so there are at most n leaves. A finite rooted tree in which every internal node has at least two children has at most one fewer internal node than it has leaves. Termination follows from the strictly decreasing positive dimensions along each branch. $\square$

At each internal node the construction records the scalar series μ , the exact leading exponent α , the ordinary constant spectral data (V, D, Q_j) , and the well-ordered positive support S of the normalized perturbation, followed by the inclusion of the lifted projector and unitary supports in $\langle S \rangle_{\mathbb{N}}$ and then the certificates of the smaller blocks. A scalar leaf records its series and multiplicity.

This is a *finite tree of set-valued support data*, not in general a finite list of monomials: a single node can carry an infinite or uncountable well-ordered support. Nor is it automatically an effective algorithm from an arbitrary presentation of a surreal number. Exact zero tests, support access and extraction of a leading coefficient must be supplied by the representation — the same contract as in [Section 14](#), now for a harder task. The coefficient fields remain the full $\mathbb{R}$ and $\mathbb{C}$; preservation of a smaller coefficient field such as $\mathbb{Q}$ is not asserted.

Example 9.14 (A scalar tail beyond every finite truncation). In the rank-two group of [Example 9.2](#), put

$$c = \sum_{m \geq 1} t^{m\delta}, \quad A = cI + t^\Omega \operatorname{diag}(1, -1).$$

Every coefficient encountered at the scales $m\delta$ is scalar, so a procedure that inspects more and more of those coefficients does not reach the first nonscalar scale after any finite number of inspections. The exact trace subtraction gives $A - \frac{1}{2} \operatorname{tr}(A)I = t^\Omega \operatorname{diag}(1, -1)$ at once. This is why [\(9.10\)](#) is stated as an exact operation, and it is the spectral counterpart of the truncation failure in [Section 14](#).

Corollary 9.15 (Normal spectral descent). *Every normal matrix over K_Γ is unitarily diagonalizable over K_Γ , without an exponent-group extension.*

Proof. Write $A = H + iJ$ with $H = (A + A^*)/2$ and $J = (A - A^*)/(2i)$ Hermitian; normality is equivalent to $HJ = JH$. Diagonalize H by [Theorem 9.12](#). The relation $HJ = JH$ makes the entries of J between distinct exact eigenspaces of H vanish, so diagonalize the Hermitian restriction of J inside each block. The resulting unitary diagonalizes H and J , hence A , and every operation stayed in the original support group. $\square$

[Proposition 4.2](#) proves more about general matrices — a Schur form for every square matrix — but it needs an eigenvalue to exist, hence an algebraically closed K . [Corollary 9.15](#) covers only normal matrices and needs nothing of the sort. Neither contains the other.

Lemma 9.16 (A finite-dimensional commutant fact). *If A is normal over K_Γ and $BA = AB$, then $BA^* = A^*B$.*

Proof. Use [Corollary 9.15](#) to make A diagonal. The entry b_{jk} must vanish when $\lambda_j \neq \lambda_k$, and these are exactly the pairs with $\overline{\lambda_j} \neq \overline{\lambda_k}$; so B also commutes with the diagonal conjugate of A . $\square$

Theorem 9.17 (Finite-witness simultaneous diagonalization). *Let $\mathcal{F} \subseteq K_\Gamma^{n \times n}$ be a set of pairwise commuting normal matrices. There is a simultaneous unitary diagonalizer, and it can be constructed from at most $n - 1$ selected members of $\mathcal{F}$; consequently its support may be confined to the subgroup generated by those finitely many members' supports. After selection, one ordinary complex linear combination of the selected matrices can be chosen whose eigenspaces are exactly the final joint eigenspaces.*

Proof. Begin with the whole space as one block. If every member acts as a scalar there, stop. Otherwise choose a nonscalar member and decompose the block into its distinct eigenspaces by [Corollary 9.15](#). Every other member and its adjoint preserve those spaces by [Lemma 9.16](#), so their restrictions remain normal; repeat on any block on which some member is nonscalar. Each

selection strictly increases the number of joint blocks, which never exceeds n , so at most $n - 1$ selections are needed, and the successive diagonalizers are produced from the selected matrices and previously constructed bases only.

Let $B_1, \dots, B_m$ be the selected matrices and let their joint scalar tuples on the final blocks be $\lambda^{(a)} \in K_\Gamma^m$; these are pairwise distinct. For each pair $a \neq b$ the condition $\sum_j c_j (\lambda_j^{(a)} - \lambda_j^{(b)}) = 0$ defines a proper complex-linear subspace of $\mathbb{C}^m$, and a finite union of proper subspaces over the infinite field $\mathbb{C}$ does not fill $\mathbb{C}^m$. Choose (c_j) outside that union; then $\sum_j c_j B_j$ has distinct scalar values on the final blocks, and it is normal because the selected matrices commute with one another and with one another's adjoints. $\square$

For a family of scalar matrices no member need be selected. This is an existence theorem for a set-sized family, not a procedure for searching an arbitrary undecidable one.

9.5 Singular values stay in the original scale group

Theorem 9.18 (SVD over a nondivisible Hahn field). *Every $A \in K_\Gamma^{m \times n}$ admits $A = U\Sigma V^*$ with U, V unitary over K_Γ and Σ rectangular diagonal with entries $\sigma_1 \geq \dots \geq \sigma_r > 0$ in F_Γ and zeros elsewhere. The factors can be chosen with every exponent in the subgroup generated by the input supports.*

Proof. Diagonalize A^*A by [Theorem 9.12](#). If v_j is a unit eigenvector with eigenvalue λ_j , then $\lambda_j = (Av_j)^*(Av_j) \geq 0$. When $\lambda_j > 0$, [Lemma 9.8](#) gives $\sigma_j = \sqrt{\lambda_j} \in F_\Gamma$ with no extension of the group; put $u_j = Av_j/\sigma_j$, which are orthonormal since $u_j^*u_k = v_j^*A^*Av_k/(\sigma_j\sigma_k) = \delta_{jk}$. When $\lambda_j = 0$, [Lemma 9.8](#) forces $Av_j = 0$. Extend the nonzero u_j to an orthonormal basis by finite Gram–Schmidt, selecting standard basis vectors not yet in the span; every nonzero residual vector has its norm in F_Γ by [Lemma 9.8](#), so no step needs a square root of an arbitrary positive scalar. Reordering the finitely many positive singular values gives the stated form and preserves the support field. $\square$

Corollary 9.19 (Gram square roots and polar decomposition). *For every rectangular A over K_Γ , the Gram matrix A^*A has its principal positive semidefinite square root over K_Γ ; in particular $|A|$ and the polar factorization (4.2) are defined over K_Γ with no enlargement of the value group. For square invertible A , the polar decomposition $A = QP$ has Q unitary and P positive definite over K_Γ .*

Proof. With the notation of [Theorem 9.18](#) put $P = V \operatorname{diag}(\sigma_1, \dots, \sigma_r, 0, \dots, 0)V^*$, so $P^2 = A^*A$. In the invertible square case take $Q = AP^{-1}$, or $Q = UV^*$. $\square$

This does *not* say that every positive semidefinite matrix is a Gram matrix over a nondivisible base field. The scalar matrix (t^γ) with $\gamma \notin 2\Gamma$ is a counterexample, and [Corollary 10.5](#) gives the exact characterization. Note also that [Theorem 7.5](#) and [Theorem 8.1](#) are now available over an arbitrary Γ : their proofs used only the SVD, integrality of unitary factors ([Lemma 7.3](#)), Cauchy–Binet and positivity of a sum of squared minors, and [Theorem 9.18](#) supplies the first without divisibility.

9.6 The subgroup, and what it is not

Three sharp examples fix the boundaries of the descent statement.

Example 9.20 (New integer combinations, but no new fractions). For $\Gamma = \mathbb{Z}$ and $t = t^1$, take

$$A = \begin{pmatrix} t^3 & t^4 \\ t^4 & 0 \end{pmatrix}, \quad \lambda_\pm = \frac{t^3}{2}(1 \pm \sqrt{1 + 4t^2}), \quad (9.12)$$

so that $\lambda_+ = t^3 + t^5 - t^7 + 2t^9 - \dots$ and $\lambda_- = -t^5 + t^7 - 2t^9 + \dots$. With $w = (\sqrt{1 + 4t^2} - 1)/(2t) = t - t^3 + 2t^5 - \dots$, the orthogonal matrix $U = (1 + w^2)^{-1/2} \begin{pmatrix} 1 & -w \\ w & 1 \end{pmatrix}$ satisfies $U^*AU = \text{diag}(\lambda_+, \lambda_-)$, and all exponents are integers. But the exponent 5 is *not* in the nonnegative monoid generated by the raw input exponents 3 and 4. A theorem asserting that eigenvalue supports stay in the raw input monoid would therefore be false. The correct invariant is the input *subgroup*, supplemented by the positive monoids of the normalized perturbations at each node.

Example 9.21 (An eigenvalue beyond all ordinary perturbation orders). In the rank-two group of [Example 9.2](#) write $t = t^\delta$ and $u = t^\Omega$, and let $A = \begin{pmatrix} 0 & u \\ u & t \end{pmatrix}$. Since $v(u/t) = \Omega - \delta > 0$, the smaller eigenvalue is

$$\lambda_- = \frac{t}{2}(1 - \sqrt{1 + 4u^2/t^2}) = -u^2t^{-1} + u^4t^{-3} - 2u^6t^{-5} + \dots,$$

of valuation $2\Omega - \delta$, which exceeds every finite multiple of δ . The support stays in $\mathbb{Z}\delta + \mathbb{Z}\Omega$ and the expansion is certified by the positive generator $2(\Omega - \delta)$. A finite-order expansion measured only in powers of t never sees this eigenvalue; compare [Example 13.3](#), where no finite integer cutoff reaches the last singular direction.

Proposition 9.22 (Order-preserving transport). *Let $\phi : \Gamma \hookrightarrow \Lambda$ be an injective order-preserving group homomorphism. Then $\phi_*(\sum_\gamma a_\gamma t^\gamma) = \sum_\gamma a_\gamma s^{\phi(\gamma)}$ is an injective star-preserving field homomorphism, and it preserves strongly summable families, the support certificates of the splitting construction, and each chosen spectral or singular-value decomposition.*

Proof. An order embedding sends a well-ordered support to a well-ordered support, and it preserves and reflects equality of sums of exponents, so the finite coefficient formulas for strong sums and products are unchanged. It preserves leading coefficients and minima. Hence the trace subtraction, the normalization, the formal projectors, the polar intertwiner and all selected ordinary constant eigenbases transport through every node of the finite tree. $\square$

This lets one symbolic support construction be interpreted at different surreal scales with the same ordered-group relations. It does not claim that arbitrary numerical substitutions, in particular ones that identify or reverse exponents, preserve Hahn summability.

Warning 9.23 (Relative degrees can drop in a larger workspace). Under an embedding into a larger group, an exponent not divisible by q in the source may become divisible in the target: $\mathbb{Z} \hookrightarrow \mathbb{Q}$ makes every positive monomial's q th root available and kills all ramification. The spectral identities transport by [Proposition 9.22](#), but the relative ramification group of [Section 10](#) must be recomputed in the new base group. Degree is preserved under an isomorphism onto the image group, *not* under arbitrary enlargement of the base workspace.

Warning 9.24 (This is not a multivariable analytic eigenbasis theorem). Consider the real symmetric formal matrix

$$H(x, y) = \begin{pmatrix} x & y \\ y & -x \end{pmatrix}, \tag{9.13}$$

whose characteristic equation is $\lambda^2 = x^2 + y^2$. There is no solution $\lambda \in \mathbb{C}[[x, y]]$: its lowest homogeneous term would have to be $ax + by$ with $a^2 = b^2 = 1$ and $2ab = 0$. So a general eigenvalue or eigenbasis theorem in that formal power-series ring is false [\[18, 19\]](#). After the ordered Hahn specialization $x = t^\alpha$, $y = t^\beta$ with $\beta > \alpha$, by contrast,

$$\sqrt{x^2 + y^2} = t^\alpha \sum_{m \geq 0} \binom{1/2}{m} t^{2m(\beta - \alpha)}$$

is a legitimate Hahn series in the original group, because the ordering permits division by the dominant monomial. Its displayed support may correspond to negative powers of x in the original two-variable expression, so it gives neither an element of $\mathbb{C}[[x, y]]$, nor a common analytic germ through the crossing, nor a global analytic choice of eigenvectors. The descent theorems resolve the totally ordered scale problem, and only that one.

10 Determinantal ramification and the exact root field

Spectral data need no new scales. Positive matrix roots sometimes do, and the cost is exactly computable from the invariant already in hand. Every ordered abelian group is torsion free, so its divisible hull $\Gamma_{\mathbb{Q}} = \Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$ carries a unique order extending that of Γ ; we write γ/q only as an element of this hull unless its membership in a smaller group has been established. Throughout this section and the next, Γ is again arbitrary.

10.1 Scalar roots are monomials times base-field units

Lemma 10.1 (The scalar root formula). *Let $a \in F_{\Gamma}$ be positive, let $\gamma = v(a)$, and let $q \geq 2$ be an integer. There is a unique positive q th root in $F_{\Gamma_{\mathbb{Q}}}$, and it has the form*

$$a^{1/q} = t^{\gamma/q} u, \quad u \in F_{\Gamma}, \quad u > 0. \quad (10.1)$$

It belongs to F_{Γ} if and only if $\gamma \in q\Gamma$.

Proof. Factor $a = ct^{\gamma}(1+h)$ with $c > 0$ real and $h \in \mathfrak{m} \cap F_{\Gamma}$. The binomial series gives $u = c^{1/q}(1+h)^{1/q} \in F_{\Gamma}$ by [Corollary 9.6](#), with positive constant coefficient, proving existence and the displayed factorization. The map $x \mapsto x^q$ is strictly increasing on positive elements of any ordered field, so the positive root is unique. A root in F_{Γ} must have valuation $\gamma/q \in \Gamma$, and the formula proves the converse. $\square$

Lemma 10.2 (Finite-index Hahn extensions). *Let $\Gamma \subseteq \Lambda \subseteq \Gamma_{\mathbb{Q}}$ with Λ/Γ finite. Then*

$$[K_{\Lambda} : K_{\Gamma}] = [F_{\Lambda} : F_{\Gamma}] = |\Lambda/\Gamma|, \quad (10.2)$$

and a choice of coset representatives η_h gives a basis $(t^{\eta_h})_{h \in \Lambda/\Gamma}$ over the respective base field. If $q\Lambda \subseteq \Gamma$, the complex extension is Galois and

$$\text{Gal}(K_{\Lambda}/K_{\Gamma}) = \text{Hom}(\Lambda/\Gamma, \mu_q), \quad (10.3)$$

where $\mu_q \subset \mathbb{C}^{\times}$ is the group of q th roots of unity and a character χ acts by $t^{\eta} \mapsto \chi(\eta + \Gamma)t^{\eta}$. The extension is totally ramified: its residue field is still $\mathbb{C}$ and its ramification index is the full degree.

Proof. Split the support of a Hahn series into the finitely many cosets of Γ . Translating the part in coset h by $-\eta_h$ gives a well-ordered subset of Γ , so every series has a unique expression $f = \sum_h t^{\eta_h} f_h$ with $f_h \in K_{\Gamma}$, and the analogous real expression; uniqueness holds because supports in distinct cosets are disjoint. This proves the basis and the degree.

Multiplication by a character on each coset preserves all coefficientwise sums and products, so it defines an automorphism. A finite abelian group killed by q has exactly its own order many characters into μ_q , and these separate points, as one sees by decomposing it into cyclic factors. We have thus exhibited as many distinct automorphisms as the degree, so the extension is Galois with the stated group. The residue field is unchanged and the value group grows by the full index, which is total ramification. $\square$

The finite-index hypothesis is essential, not cosmetic. For an infinite group extension, the field generated by all the new monomials need not contain every Hahn series supported in the enlarged group. We invoke [Lemma 10.2](#) only for extensions generated by finitely many fractional exponents of bounded denominator.

10.2 Existence, uniqueness, and the exact extension

Theorem 10.3 (Principal-root field). *Let $q \geq 2$ be an integer and let $A \in K_\Gamma^{n \times n}$ be Hermitian positive semidefinite, with positive eigenvalues $\lambda_1, \dots, \lambda_r$ counted with multiplicity, and put $\gamma_j = v(\lambda_j)$ and $\Lambda = \Gamma + \sum_{j=1}^r \mathbb{Z}\gamma_j/q$. There is a unique positive semidefinite Hermitian B over $K_{\Gamma_\mathbb{Q}}$ with $B^q = A$. It lies in $K_\Lambda^{n \times n}$, and*

$$K_\Gamma(B) = K_\Lambda, \quad (10.4)$$

where $K_\Gamma(B)$ denotes the field obtained by adjoining all entries of B .

Proof. Choose a unitary diagonalization over the base field, furnished by [Theorem 9.12](#): $A = U \operatorname{diag}(\lambda_1, \dots, \lambda_r, 0, \dots, 0)U^*$. The diagonal entries are nonnegative, being values of the quadratic form on unit eigenvectors. By [Lemma 10.1](#), $b_j = \lambda_j^{1/q} = t^{\gamma_j/q} u_j$ with $u_j \in F_\Gamma$ positive. Set

$$B = U \operatorname{diag}(b_1, \dots, b_r, 0, \dots, 0)U^*, \quad (10.5)$$

which is positive semidefinite, satisfies $B^q = A$, and has entries in K_Λ .

For uniqueness let $C = C^* \succeq 0$ satisfy $C^q = A$ in the enlarged field. Then C and A commute, so [Theorem 9.17](#) gives a simultaneous Hermitian diagonalization there. On an eigenspace of A with value λ , every eigenvalue of C is nonnegative with q th power λ , and the ordered scalar uniqueness of [Lemma 10.1](#) forces it to be the chosen root, including 0 when $\lambda = 0$. Hence $C = B$.

Let $L = K_\Gamma(B)$. Since U lies in the base field, the entries of U^*BU show every $b_j \in L$; dividing by the base-field unit u_j gives $t^{\gamma_j/q} \in L$. These finitely many monomials, together with monomials already in the base field, generate all coset representatives of Λ/Γ by multiplication and inversion, and the quotient is finite, being generated by r elements killed by q . The basis in [Lemma 10.2](#) therefore gives $K_\Lambda \subseteq L$, and the reverse inclusion is (10.5). $\square$

This is the exact sense in which [Proposition 4.3](#) needs a divisible workspace. The positive semidefinite root exists and is unique, but over a general Γ it lives in K_Λ , and Λ can be strictly larger than Γ .

10.3 Replacing eigenvalues by minors

The eigenvalues in [Theorem 10.3](#) can be eliminated entirely. By (7.4) the determinantal valuations and the eigenvalue valuations determine one another, so the field of definition is computable from finitely many minors and no spectral decomposition at all.

Theorem 10.4 (Determinantal ramification law). *Let $A \in K_\Gamma^{n \times n}$ be Hermitian positive semidefinite of rank r , let $\Delta_k = \Delta_k(A)$ be as in (7.3), and for an integer $q \geq 2$ put*

$$\Gamma_{A,q} = \Gamma + \sum_{k=1}^r \mathbb{Z} \frac{\Delta_k}{q} \subseteq \Gamma_\mathbb{Q}, \quad R_q(A) = \langle \Delta_1 + q\Gamma, \dots, \Delta_r + q\Gamma \rangle \subseteq \Gamma/q\Gamma. \quad (10.6)$$

Then the principal positive semidefinite root satisfies

$$K_\Gamma(A^{1/q}) = K_{\Gamma_{A,q}}, \quad (10.7)$$

$$[K_\Gamma(A^{1/q}) : K_\Gamma] = |R_q(A)| \leq q^r. \quad (10.8)$$

In particular

$$A^{1/q} \in K_\Gamma^{n \times n} \iff \Delta_k(A) \in q\Gamma \quad (1 \leq k \leq r). \quad (10.9)$$

The extension is finite, abelian and totally ramified, with Galois group isomorphic to $\operatorname{Hom}(R_q(A), \mu_q)$.

Proof. Order the γ_j nondecreasingly. By (7.4), $\Delta_k = \gamma_1 + \cdots + \gamma_k$, so

$$\Gamma + \sum_j \mathbb{Z} \frac{\gamma_j}{q} = \Gamma + \sum_k \mathbb{Z} \frac{\Delta_k}{q} : \quad (10.10)$$

each Δ_k/q is a sum of the γ_j/q , and each $\gamma_j/q = (\Delta_j - \Delta_{j-1})/q$. The map $\Gamma/q\Gamma \rightarrow (q^{-1}\Gamma)/\Gamma$ sending $\gamma + q\Gamma$ to $\gamma/q + \Gamma$ is an isomorphism, and it identifies $R_q(A)$ with $\Gamma_{A,q}/\Gamma$. Combining Theorem 10.3 with Lemma 10.2 gives (10.7), (10.8) and the Galois assertion, and the extension is trivial exactly when every generating class vanishes, which is (10.9). $\square$

Two things deserve emphasis. First, (10.8) is not merely a bound: it computes the exact degree from finitely many minor valuations, without computing an eigenbasis or the eigenvalues. Second, the same invariant Δ_k that Theorem 7.5 uses to record singular scales here controls a finite radical extension; the determinantal profile is doing two different jobs, and (10.9) is the second one in its sharpest form.

Corollary 10.5 (Exact Gram-factorization criterion). *A Hermitian positive semidefinite A over K_Γ can be written $A = C^*C$ with C over K_Γ if and only if $\Delta_k(A) \in 2\Gamma$ for $1 \leq k \leq \text{rank}(A)$. The condition permits C to be square, and it is necessary for every rectangular Gram representation.*

Proof. Necessity is Corollary 8.3: if $A = C^*C$ then $\Delta_k(A) = 2\Delta_k(C) \in 2\Gamma$. For sufficiency, Theorem 10.4 with $q = 2$ supplies the principal square root B over K_Γ ; take $C = B$. $\square$

Corollary 10.6 (Rational powers). *Let A be positive definite, $p \in \mathbb{Z}$, $q \geq 2$, $\gcd(p, q) = 1$. Then the principal power $A^{p/q}$ generates the same extension as $A^{1/q}$. For positive semidefinite A the same holds when $p > 0$.*

Proof. The nonzero diagonal spectral entries of $A^{p/q}$ are b_j^p . If $ap + bq = 1$, then $b_j = (b_j^p)^a \lambda_j^b$, so adjoining these entries is field-theoretically equivalent to adjoining the b_j . Zero eigenvalues are omitted, and negative powers require invertibility. $\square$

10.4 Prime and composite denominators

When q is prime, $\Gamma/q\Gamma$ is a vector space over $\mathbf{F}_q$ and

$$[K_\Gamma(A^{1/q}) : K_\Gamma] = q^{\dim_{\mathbf{F}_q} \text{span}\{\Delta_k + q\Gamma : 1 \leq k \leq r\}}.$$

For composite q the exact invariant is the finite subgroup, not the rank of a vector space: a single class of order 2 modulo 4Γ contributes a quadratic extension, not a quartic one.

If $\Gamma = \mathbb{Z}^d$ with any fixed compatible order and the Δ_k are integer vectors, put $L = q\mathbb{Z}^d + \sum_{k=1}^r \mathbb{Z}\Delta_k \subseteq \mathbb{Z}^d$. Then

$$|R_q(A)| = \frac{q^d}{|\mathbb{Z}^d : L|}, \quad (10.11)$$

and the index is computed by the Smith normal form of the integer matrix with columns $qI_d, \Delta_1, \dots, \Delta_r$. The root-field degree is therefore a finite integer-lattice computation once the minor valuations are known exactly, and no discreteness in the *order* of Γ is needed for it.

10.5 Three examples at the boundary

Example 10.7 (A positive matrix whose square root genuinely ramifies). Take

$$A = \begin{pmatrix} 1+t & 1 \\ 1 & 1 \end{pmatrix} \in \mathbb{C}((t))^{2 \times 2}, \quad (10.12)$$

positive definite because $x^*Ax = |x_1 + x_2|^2 + t|x_1|^2 > 0$ for $x \neq 0$. The minor minima are $\Delta_1 = 0$ and $\Delta_2 = v(t) = 1$, so the principal square root generates exactly $\mathbb{C}((t^{\frac{1}{2}\mathbb{Z}}))$, a quadratic extension of $\mathbb{C}((t^{\mathbb{Z}}))$. Explicitly

$$B = \frac{A + \sqrt{t}I}{\sqrt{2 + t + 2\sqrt{t}}}, \quad (10.13)$$

whose denominator has a positive constant leading term and so introduces no exponent beyond $\frac{1}{2}\mathbb{Z}$; Cayley–Hamilton gives $B^2 = A$, and positivity of numerator and denominator gives $B \succeq 0$. Moreover $T = \text{tr}(B) = \sqrt{2 + t + 2\sqrt{t}}$ and $\sqrt{t} = (T^2 - 2 - t)/2$, which makes the primitive-trace phenomenon of [Section 11](#) directly visible.

Example 10.8 (A Gram witness can use a larger group than its product). With $t = t^\delta$, $u = t^\Omega$ in the rank-two group of [Example 9.2](#), put

$$C = \begin{pmatrix} t & 1 \\ 0 & u \end{pmatrix}, \quad A = C^*C = \begin{pmatrix} t^2 & t \\ t & 1 + u^2 \end{pmatrix}.$$

Relative to $\Gamma = \mathbb{Z}\delta + \mathbb{Z}\Omega$ the minor valuations are $\Delta_1 = 0$ and $\Delta_2 = 2\delta + 2\Omega$, so the square root needs no extension while the fourth root generates one of degree 2. But the exact input support group of A alone is $H = \mathbb{Z}\delta + 2\mathbb{Z}\Omega$, and relative to H the value $2\delta + 2\Omega$ is not in $2H$, so $A^{1/2}$ generates a quadratic extension of K_H and recovers the missing scale u . This does not contradict [Corollary 9.19](#): the displayed Gram witness C already uses that scale. It is why every statement here names both the input matrix and the base group, and why [Proposition 2.3](#) is worth distinguishing from [Proposition 2.2](#).

Example 10.9 (The degree bound is attained in every rank). Let $\Gamma = \mathbb{Z}^r$ with the lexicographic order and standard basis $e_1, \dots, e_r$, each positive, and let $A = \text{diag}(t^{e_1}, \dots, t^{e_r})$. The nonzero eigenvalue valuations generate all of $\Gamma/q\Gamma$, and their partial sums, which are the determinantal values after ordering, generate the same subgroup. Hence $[K_\Gamma(A^{1/q}) : K_\Gamma] = q^r$. This rules out any degree bound depending only on q and not on the rank.

11 The trace is a primitive generator

The extension of [Theorem 10.4](#) is generated by the n^2 entries of $A^{1/q}$. It is generated by one of their linear functionals.

11.1 A positive-coset lemma

Lemma 11.1 (Positive radical sums retain every generator). *Let $a_1, \dots, a_r \in F_\Gamma$ be positive, let $q \geq 2$, and write $b_j = a_j^{1/q} = t^{\gamma_j/q} u_j$ as in [\(10.1\)](#). Put $\Lambda = \Gamma + \sum_j \mathbb{Z}\gamma_j/q$. Then for any strictly positive $w_j \in F_\Gamma$,*

$$K_\Gamma\left(\sum_{j=1}^r w_j b_j\right) = K_\Lambda. \quad (11.1)$$

Proof. Let $G = \Lambda/\Gamma$ and $h_j = \gamma_j/q + \Gamma$; these classes generate G . Choose a representative η_h for each class occurring among the h_j . The sum $T = \sum_j w_j b_j$ has the coset decomposition

$$T = \sum_h t^{\eta_h} s_h, \quad s_h = \sum_{j:h_j=h} w_j t^{\gamma_j/q - \eta_h} u_j \in F_\Gamma,$$

and every summand in s_h is positive in the ordered field F_Γ , so $s_h > 0$ and in particular $s_h \neq 0$ for each occurring class.

By [Lemma 10.2](#) every automorphism of K_Λ/K_Γ is a character $\chi : G \rightarrow \mu_q$. If such an automorphism fixes T , then

$$0 = \chi(T) - T = \sum_h (\chi(h) - 1) t^{\eta_h} s_h.$$

Distinct cosets have disjoint supports, so every coefficient $\chi(h) - 1$ vanishes; since the occurring h_j generate G , the character is trivial. Thus T has trivial stabilizer in the Galois group, and the fundamental correspondence gives $[K_\Gamma(T) : K_\Gamma] = |G|$, which is the asserted equality. $\square$

Theorem 11.2 (Primitive trace). *Under the hypotheses of [Theorem 10.4](#),*

$$K_\Gamma(\text{tr}(A^{1/q})) = K_\Gamma(A^{1/q}) = K_{\Gamma_{A,q}}. \quad (11.2)$$

More generally, if $\lambda_1, \dots, \lambda_r$ are the positive eigenvalues of A , then every sum $\sum_j a_j \lambda_j^{1/q}$ with $a_j \in F_\Gamma$ and $a_j > 0$ is a primitive generator of the same field.

Proof. For the principal root (10.5), trace invariance gives $\text{tr}(B) = \sum_{j=1}^r \lambda_j^{1/q}$. Apply [Lemma 11.1](#) with $w_j = 1$, and use [Theorem 10.3](#) together with (10.10). The weighted statement is the same lemma with arbitrary strictly positive w_j . $\square$

The proof identifies a mechanism rather than invoking an abstract primitive-element theorem. In a general finite extension most linear combinations may be primitive while special choices fail; here the *specific* unweighted trace always succeeds, because positivity makes its coset support a generating set. Strictly positive coefficients are the whole content of the hypothesis, as [Warning 11.5](#) shows.

Corollary 11.3 (The corresponding real field). *With $T = \text{tr}(A^{1/q})$ and $\Lambda = \Gamma_{A,q}$, we have $F_\Gamma(T) = F_\Lambda$.*

Proof. Both fields are ordered and $T \in F_\Lambda$. Adjoining i has degree 2 over each of these real fields, so the tower formula applied to $K_\Gamma(T) = K_\Lambda$ gives $[F_\Gamma(T) : F_\Gamma] = [K_\Lambda : K_\Gamma] = |\Lambda/\Gamma|$, which is $[F_\Lambda : F_\Gamma]$ by [Lemma 10.2](#). $\square$

11.2 An explicit irreducible orbit polynomial

Let $b_j = \lambda_j^{1/q}$, $G = \Gamma_{A,q}/\Gamma$ and $h_j = v(b_j) + \Gamma$. The minimal polynomial of the trace over K_Γ is

$$\mathcal{P}_{A,q}(X) = \prod_{\chi \in \text{Hom}(G, \mu_q)} \left(X - \sum_{j=1}^r \chi(h_j) b_j \right). \quad (11.3)$$

Its coefficients lie in K_Γ because the Galois group permutes the factors, and the factors are distinct by [Lemma 11.1](#), so the polynomial is irreducible of degree $|R_q(A)|$. It is a formula inside a finite extension, not necessarily a fast coefficient-extraction algorithm.

Example 11.4 (A biquadratic field encoded in one positive trace). Let $\Gamma = \mathbb{Z}\delta \oplus \mathbb{Z}\Omega$ with $0 < n\delta < \Omega$ for every positive integer n , and let $A = \text{diag}(t^\delta, t^\Omega)$ with $q = 2$. The minor valuations are $\Delta_1 = \delta$ and $\Delta_2 = \delta + \Omega$, and their classes span a subgroup of order 4 in $\Gamma/2\Gamma$. So $B = A^{1/2}$ generates a biquadratic extension, and so does $T = t^{\delta/2} + t^{\Omega/2}$. Writing $t = t^\delta$ and $u = t^\Omega$, the minimal polynomial is

$$X^4 - 2(t+u)X^2 + (t-u)^2, \quad (11.4)$$

whose degree 4 is certified by the two independent exponent classes and not by an unproved irreducibility guess about the displayed polynomial.

Warning 11.5 (Positivity cannot be dropped). For $A = t^\delta I_2$ with $\delta \notin 2\Gamma$, the matrix $C = \text{diag}(t^{\delta/2}, -t^{\delta/2})$ is a Hermitian square root generating a quadratic extension, and $\text{tr}(C) = 0$. It is not the positive root. [Theorem 11.2](#) concerns the principal positive root and strictly positive weights, never an arbitrary branch or a signed sum.

11.3 Several positive roots at once

Corollary 11.6 (Joint ramification and a joint primitive element). *Let $A_1, \dots, A_m$ be positive semidefinite Hermitian matrices over K_Γ , of possibly different sizes, and let $q_\ell \geq 2$. Put*

$$\Lambda = \Gamma + \sum_{\ell=1}^m \sum_{k=1}^{\text{rank}(A_\ell)} \mathbb{Z} \frac{\Delta_k(A_\ell)}{q_\ell}.$$

Then the field generated by all entries of all A_ℓ^{1/q_ℓ} is K_Λ , of degree $|\Lambda/\Gamma|$, and the single scalar $T = \sum_\ell \text{tr}(A_\ell^{1/q_\ell})$ is a primitive generator. No commutation assumption on the A_ℓ is needed.

Proof. The compositum of the individual monomial extensions contains the monomials generating Λ , and the quotient is finite and killed by $Q = \text{lcm}(q_1, \dots, q_m)$, so [Lemma 10.2](#) identifies the compositum with K_Λ . Decompose the finitely many positive scalar spectral roots occurring in the traces into cosets of Γ ; every occurring coefficient sum is positive and the occurring classes generate Λ/Γ , so the character argument of [Lemma 11.1](#), now with μ_Q , proves primitivity. $\square$

11.4 Back inside the surcomplex numbers

Corollary 11.7 (Exact-support surcomplex spectral calculus). *Let A be a finite surcomplex matrix and let H be its exact input support group, as in [Proposition 2.3](#). If A is Hermitian, it has real surreal eigenvalues and a unitary eigenbasis whose normal-form exponents stay in H ; the same support assertion holds for normal diagonalization and for the SVD of an arbitrary finite surcomplex matrix. If A is in addition positive semidefinite, then for every integer $q \geq 2$*

$$K_H(A^{1/q}) = K_{H + \sum_{k=1}^{\text{rank } A} \mathbb{Z} \Delta_k(A)/q} = K_H(\text{tr}(A^{1/q})),$$

and the degree over K_H is the order of the subgroup generated by the classes $\Delta_k(A) + qH$, $1 \leq k \leq \text{rank } A$, in H/qH .

Proof. Apply [Theorems 9.12, 9.18, 10.4](#) and [11.2](#) and [Corollary 9.15](#) inside the set-sized field K_H , then transport by the normal-form embedding of [Proposition 2.3](#). The positive branch agrees with the ambient surreal positive root by ordered-field uniqueness together with the uniqueness proof in [Theorem 10.3](#). $\square$

The ambient surreal exponent group is divisible, so the full class field has room for all of these fractional scales; the content of the corollary is not that positive roots fail to exist in $\mathbf{No}[i]$. It is their exact *relative cost* over the smallest Hahn workspace containing the data — a cost that becomes invisible the moment one replaces H by its divisible hull, which is precisely what [Proposition 2.2](#) does.

12 Hankel square-class profiles: exact splitting fields of real-rooted polynomials and definite pencils

[Sections 10](#) and [11](#) measured the exact cost of a positive *matrix* root from the minima Δ_k of all minors. This section, which comes from the third source manuscript ([Section 1.4](#)), asks the corresponding question for a real-rooted *polynomial*:

for a real-rooted polynomial in a fixed Hahn workspace, which finite data recover the exact splitting field and the number of irreducible factors, without first constructing the roots?

The answer uses *all* leading principal minors of the Newton-sum Hankel matrix, and for the factor count it uses their consecutive valuation differences, not just their span. The same computation gives the exact field of the generalized eigenvalues of a definite Hermitian pencil. For the eigenvalues, that removes the divisible workspace which the generalized eigenproblem of Section 4 used; Remark 2.1 records what still needs it.

Throughout this section Γ is an arbitrary set-sized ordered abelian group, of arbitrary rank and *not* assumed divisible, with $F_\Gamma = \mathbb{R}((t^\Gamma))$, $K_\Gamma = \mathbb{C}((t^\Gamma)) = F_\Gamma(i)$ and divisible hull $\Gamma_{\mathbb{Q}}$. Write $\mathcal{O}_F = \mathcal{O} \cap F_\Gamma$ for the real valuation ring. The Hahn field $F_{\Gamma_{\mathbb{Q}}}$ is real closed (Section 2), so it is a convenient ambient field for roots. It need not be an algebraic extension of F_Γ as a whole, and only the explicitly displayed finite subextensions are called finite or algebraic. The constant field $\mathbb{R}$ is essential; see Warning 12.31.

Three examples show what the answer has to see. Over $\mathbb{R}((t^{\mathbb{Z}}))$ the polynomial $X^2 - t$ needs $t^{1/2}$, whereas $X^2 - (1 + t)$ already splits. More subtly, $(X - 1)^2 - t$ has two roots of valuation zero but still needs $t^{1/2}$, so the valuations of the roots do not determine the splitting field. A full discriminant is not enough either: Example 12.28 has a square discriminant and nevertheless generates a biquadratic field.

- Remark 12.1* (Conventions for this section). (a) In this section $h_k = h_k(p)$ always denotes the k th leading principal minor of the Newton-sum Hankel matrix of a polynomial p (Definition 12.6), a *Hankel subdiscriminant*. It is not the coset h_j used in the proof of Lemma 11.1 and in (11.3), nor the infinitesimal h of Lemma 10.1. Other reports of the collection use the letter h_k for unrelated quantities (the Markov-generator and Prony reports among them); none of those meanings is intended here. The source manuscript's names $\alpha_k = v(h_k)$, $\beta_k = v(h_k) - v(h_{k-1})$ and $d_k = h_k/h_{k-1}$ are not used. We write π_k for the pivot h_k/h_{k-1} and ξ_k for its class modulo 2Γ , keeping these apart from the singular scales γ_k and the determinantal minima Δ_k of Theorem 7.5.
- (b) The *Hankel square-class profile* m_p is a multiplicity function on $\Gamma/2\Gamma$. It is not the determinantal profile (Δ_k) of Theorem 7.5, and not the Newton profile of the polynomial-algebra report.
- (c) The span $D_p \subseteq \Gamma/2\Gamma$ below is built from the *leading principal* minors of one Hankel matrix, whereas Δ_k is a minimum over *all* $k \times k$ minors. So D_p is an analogue, for $q = 2$, of the group $R_2(A)$ of (10.6), but for a different kind of minor, and the two must not be conflated; Remark 12.11 records the only relation stated here.

12.1 Inputs already proved, and one new ingredient

The source manuscript reproves its spectral and Kummer-theoretic inputs, to isolate exactly the field properties on which its equalities depend. Those already in this report are cited rather than reprinted.

- *Positive square classes.* For $a > 0$ in F_Γ , a is a square in F_Γ if and only if $v(a) \in 2\Gamma$. More precisely $\sqrt{a} = t^{v(a)/2}u$ with $u \in F_\Gamma$ positive, so its support lies in the single coset $v(a)/2 + \Gamma$. This is Lemma 10.1 at $q = 2$.
- *The Pythagorean property.* Every nonzero finite sum of squares in F_Γ is positive and is a square in F_Γ . This is Lemma 9.8 for real vectors. It holds for every Hahn field F_Λ , and it is used below for intermediate fields. It does *not* say that every positive element is a square: t^γ is a counterexample whenever $\gamma \notin 2\Gamma$.

- *Finite monomial extensions.* Degrees and coset bases are [Lemma 10.2](#).
- *Hahn principal axes.* Every real symmetric matrix over F_Γ is diagonalized by an orthogonal matrix over F_Γ , so its eigenvalues lie in F_Γ with no enlargement of the exponent group. This is the real case of [Theorem 9.12](#). The source's own proof is different, and it is kept below as a second route ([Remark 12.5](#)).

What is new in this subsection is the real Galois group of a half-scale extension and the classification of its intermediate fields.

Lemma 12.2 (Intermediate fields of half-scale extensions). *Let $\Gamma \subseteq \Lambda \subseteq \Gamma_{\mathbb{Q}}$ with Λ/Γ finite and $2\Lambda \subseteq \Gamma$. Then F_Λ/F_Γ is Galois, with group $\text{Hom}(\Lambda/\Gamma, \{\pm 1\})$; a character χ acts on every series by multiplying its coefficient at the exponent λ by $\chi(\lambda + \Gamma)$. Every intermediate field of F_Λ/F_Γ is $F_{\Lambda'}$ for a group $\Gamma \subseteq \Lambda' \subseteq \Lambda$. The same assertions hold with F replaced by K .*

Proof. The degree $|\Lambda/\Gamma|$ and the coset basis are [Lemma 10.2](#). With $q = 2$ the characters of that lemma take values in $\mu_2 = \{\pm 1\} \subset \mathbb{R}$, so they preserve real series. Its argument applies verbatim. Multiplicativity follows from the character law and the finite convolution defining each product coefficient. The finite elementary abelian group Λ/Γ has exactly $|\Lambda/\Gamma|$ such characters, and they separate its elements. So we have as many automorphisms of F_Λ over F_Γ as the degree, and the extension is Galois.

For a subgroup J of the Galois group, an invariant series can have support only in

$$\Lambda' = \{\lambda \in \Lambda : \chi(\lambda + \Gamma) = 1 \text{ for every } \chi \in J\},$$

because a nonzero coefficient in any other coset is changed by some character. Conversely every series supported in Λ' is fixed. So the fixed field of J is $F_{\Lambda'}$, and Λ' is a group between Γ and Λ . The fundamental theorem of finite Galois theory now gives all intermediate fields. Complex coefficients do not change the argument. $\square$

The sign-changing automorphisms are generally not order preserving: they may send the positive $t^{\gamma/2}$ to its negative. There is no conflict with the uniqueness of a positive square root in a *fixed* ordered field. Field automorphisms need not preserve that chosen order when the base field is not real closed.

Corollary 12.3 (Adjoining finitely many positive square roots). *For $a_1, \dots, a_m > 0$ in F_Γ , set $\Lambda = \Gamma + \sum_j \mathbb{Z}v(a_j)/2$. Then*

$$F_\Gamma(\sqrt{a_1}, \dots, \sqrt{a_m}) = F_\Lambda.$$

Its degree is 2^r , where r is the dimension over $\mathbf{F}_2$ of the span of the classes $v(a_j) + 2\Gamma$ in $\Gamma/2\Gamma$.

Proof. By [Lemma 10.1](#), each $\sqrt{a_j}$ is $t^{v(a_j)/2}$ times a nonzero element of F_Γ . So the same field is generated by these monomials. They generate all the coset summands of [Lemma 10.2](#), and so the field is F_Λ . Multiplication by 2 identifies Λ/Γ with the stated span in $\Gamma/2\Gamma$. $\square$

Over K_Γ the same equality is the case $A = \text{diag}(a_1, \dots, a_m)$, $q = 2$ of [Theorem 10.4](#).

Lemma 12.4 (Coprime Hensel lifting over F_Γ). *Let $f \in \mathcal{O}_F[X]$ be monic. If its reduction factors as $\bar{f} = \bar{g}\bar{h}$ with monic coprime polynomials $\bar{g}, \bar{h} \in \mathbb{R}[X]$, then $f = gh$ with monic $g, h \in \mathcal{O}_F[X]$ lifting those factors. The lifts are coprime over F_Γ .*

Proof. Choose the real-coefficient lifts of $\bar{g}, \bar{h}$, and regard the finitely many coefficients of $f - \bar{g}\bar{h}$ as formal parameters. The linearized multiplication map on corrections is

$$(u, w) \mapsto \bar{h}u + \bar{g}w, \quad \deg u < \deg \bar{g}, \quad \deg w < \deg \bar{h}.$$

It is invertible over $\mathbb{R}$. A vector in its kernel would give $\bar{g} \mid u$ and hence $u = w = 0$, and source and target have the same dimension. Solving successively by total parameter degree gives formal power series, without constant term, for all corrections of the factor coefficients. Evaluation at the actual positive-valuation errors is licensed by [Lemma 9.5](#), so the formal multiplication identity becomes $f = gh$ in $\mathcal{O}_F[X]$. The resultant of g and h reduces to the nonzero real resultant of the two residue factors. So it is nonzero, and the lifts are coprime. $\square$

The polynomial-algebra report proves a support-controlled version of this lemma over K_Γ , with several factors, by recursion on a well-ordered support monoid rather than by universal formal series (`polynomial:thm:hensel` there); it too needs no divisibility.

Remark 12.5 (A contour-free second route to real Hahn principal axes). The source proves the real case of [Theorem 9.12](#) without the contour argument used in [Theorem 9.9](#). Induct on the size n . Scalar matrices cause no difficulty. Otherwise, for a real symmetric A , set

$$A_0 = A - \frac{\operatorname{tr} A}{n} I \neq 0, \quad \gamma = v(A_0), \quad T = t^{-\gamma} A_0.$$

The residue $\bar{T}$ is a nonzero traceless real symmetric matrix. So it has at least two distinct real eigenvalues, exactly as in the proof of [Theorem 9.12](#). Partition them into two nonempty groups. The characteristic polynomial of T then has a coprime residue factorization, and [Lemma 12.4](#) lifts it. The polynomial Chinese remainder theorem supplies complementary idempotents e_1, e_2 modulo that characteristic polynomial; their coefficients are integral, because the resultant of the lifted factors is a unit. Substitution gives $E_j = e_j(T)$ with

$$E_j^2 = E_j, \quad E_1 + E_2 = I, \quad E_1 E_2 = 0, \quad E_j^\top = E_j.$$

Both are nonzero, since their residue projectors are nonzero. Their images are invariant under T , hence under A , and orthogonal, because $\langle E_1 x, E_2 y \rangle = x^\top E_1 E_2 y = 0$. Any basis of either image can be orthonormalized over F_Γ : at each Gram–Schmidt step the squared norm is a nonzero finite sum of squares, whose positive square root lies in F_Γ by [Lemma 9.8](#). In the combined orthonormal basis A has two smaller symmetric diagonal blocks, and induction finishes the proof.

This proof uses at most $n - 1$ nontrivial block splittings, as in [Corollary 9.13](#). It does not depend on the order type of a support, on the rank of Γ , or on the false assertion that the multiples of one positive valuation must be cofinal in Γ . The finite splitting count is not an algorithmic complexity bound for arbitrary representations of the input Hahn series. For real symmetric input, this is the “formal Hensel block-lifting” route that [Section 21](#) names for a formalization. It treats only real symmetric input, and it does not reproduce the support certificate $\langle S \rangle_\mathbb{N}$ of [Theorem 9.9](#). The statement cited below is [Theorem 9.12](#).

12.2 Newton sums, the Hankel trace form and the profile

Write

$$p(X) = X^n + c_1 X^{n-1} + \cdots + c_n \in F_\Gamma[X].$$

Its Newton sums are computed by the finite recurrences

$$s_0 = n, \tag{12.1}$$

$$s_j + c_1 s_{j-1} + \cdots + c_{j-1} s_1 + j c_j = 0 \quad (1 \leq j \leq n), \tag{12.2}$$

$$s_j + c_1 s_{j-1} + \cdots + c_n s_{j-n} = 0 \quad (j > n). \tag{12.3}$$

Equivalently, if M is the matrix of multiplication by X in the basis $1, X, \dots, X^{n-1}$ of $F_\Gamma[X]/(p)$, then $s_j = \operatorname{tr}(M^j) = \operatorname{Tr}_{F_\Gamma[X]/(p)/F_\Gamma}(X^j)$. Only $s_0, \dots, s_{2n-2}$ are needed below. Every one of these sums, and every determinant built from them, is a polynomial with integer coefficients in $c_1, \dots, c_n$.

Definition 12.6 (Hankel subdiscriminants and the square-class profile). Set

$$H_p = (s_{i+j})_{0 \leq i,j < n}, \quad h_k = h_k(p) = \det(s_{i+j})_{0 \leq i,j < k}, \quad h_0 = 1.$$

When H_p is positive definite, put, for $1 \leq k \leq n$,

$$\pi_k = \frac{h_k}{h_{k-1}}, \quad \xi_k = v(\pi_k) + 2\Gamma = v(h_k) - v(h_{k-1}) + 2\Gamma \in \Gamma/2\Gamma,$$

and for $\xi \in \Gamma/2\Gamma$

$$m_p(\xi) = \#\{k : \xi_k = \xi\}.$$

The function $m_p : \Gamma/2\Gamma \rightarrow \mathbb{Z}_{\geq 0}$ is the *Hankel square-class profile* of p . It has finite support and total mass n . Its span is

$$D_p = \text{span}_{\mathbf{F}_2} \{\xi_1, \dots, \xi_n\} = \text{span}_{\mathbf{F}_2} \{v(h_k) + 2\Gamma : 1 \leq k \leq n\}, \quad r_p = \dim_{\mathbf{F}_2} D_p,$$

and we put

$$\Lambda_p = \Gamma + \sum_{k=1}^n \mathbb{Z} \frac{v(h_k)}{2} \subseteq \Gamma_{\mathbb{Q}}.$$

The two descriptions of D_p agree, by taking consecutive differences or cumulative sums. That does not identify their multiplicities: for the profile, the differences ξ_k , not the cumulative classes, are essential. Since $h_1 = s_0 = n$, always $\xi_1 = 0$.

Proposition 12.7 (Classical trace-form and subdiscriminant identities). *For monic p of degree n , H_p represents the bilinear trace form*

$$(u, w) \longmapsto \text{Tr}_{F_{\Gamma}[X]/(p) / F_{\Gamma}}(uw).$$

If $\lambda_1, \dots, \lambda_n$ are its roots in an algebraic closure, listed with multiplicity, then

$$h_k = \sum_{\substack{I \subseteq \{1, \dots, n\} \\ |I|=k}} \prod_{\substack{i < j \\ i, j \in I}} (\lambda_j - \lambda_i)^2. \quad (12.4)$$

In particular $h_n = \text{Disc}(p)$. Furthermore the following are equivalent: p is squarefree with all roots real in $F_{\Gamma_{\mathbb{Q}}}$; H_p is positive definite; and $h_k > 0$ for $1 \leq k \leq n$.

Proof. The monomial entries of the trace form are s_{i+j} by definition. Let V be the Vandermonde matrix with entries $V_{\ell,j} = \lambda_{\ell}^j$ for $0 \leq j < n$. The Newton identities give $H_p = V^{\top}V$. Cauchy–Binet applied to the first k columns of V gives (12.4); the determinants of its row submatrices are the Vandermonde products shown there. This is the standard subdiscriminant identity [16]. With Roy’s descending index convention our h_k is $\text{SubDisc}_{n-k}(p)$.

For distinct real roots V is invertible, and $V^{\top}V$ is positive definite. Conversely, positive definiteness implies $h_n \neq 0$, so p is squarefree. Over the real closed field $F_{\Gamma_{\mathbb{Q}}}$ its quotient algebra is a product of real one-dimensional factors and conjugate-pair factors. A conjugate-pair factor has the real trace quadratic form $\text{Tr}((u + iv)^2) = 2u^2 - 2v^2$, which is indefinite. So positivity excludes every nonreal pair. (This is also the case of the form T_1 in the Hermite signature criterion of the polynomial-algebra report, `polynomial:thm:hermite`, applied over $F_{\Gamma_{\mathbb{Q}}}$.)

Finally, elimination of the first variable in a positive quadratic form, repeated inductively, gives the usual no-pivot $LDL^{\top}$ factorization, with diagonal entries h_k/h_{k-1} . This proves the equivalence with positivity of all leading determinants. It also shows that positivity over F_{Γ} persists over the real closure. These are the classical Hermite and Sylvester criteria in this finite algebraic setting. $\square$

This is the nondegenerate trace form of a squarefree algebra. It is not the discriminant-free residue pairing used elsewhere in the collection, in the finite-deformations and polynomial-algebra reports. At a repeated root the present trace form degenerates, so squarefree reduction is a substantive step, not a typographical choice. In the notation of the polynomial-algebra report, H_p is the matrix of the Hermite form T_1 in the monomial basis (`polynomial:thm:hermite` there). That theorem works over a real closed field, where $\Gamma/2\Gamma = 0$ and every invariant of this section is trivial.

Read as a Hankel matrix of unit-weight power moments, the identity $H_p = V^\top V$ is also the classical moment identity behind Prony's method. The report `docs/surcomplex/prony-reconstruction-at-surreal-scales/` is built on weighted versions of it. It shares only this classical Hankel/moment identity with the present section, and no theorem: it studies the metric stability of perturbed weighted moments and uses neither square classes nor leading principal minors.

12.3 The exact splitting field

Theorem 12.8 (Subdiscriminants give the entire splitting field). *Let $p \in F_\Gamma[X]$ be monic, squarefree, real-rooted, and of degree $n \geq 1$, and let L_p be its splitting field inside $F_{\Gamma\mathbb{Q}}$. Then*

$$L_p = F_\Gamma(\sqrt{h_1}, \dots, \sqrt{h_n}) = F_{\Gamma + \sum_k \mathbb{Z} v(h_k)/2} = F_{\Lambda_p}. \quad (12.5)$$

It has degree 2^{r_p} and Galois group $\text{Hom}(D_p, \{\pm 1\})$. The splitting field over K_Γ is exactly K_{Λ_p} , with the same degree and Galois group.

Proof. The upper field. The trace matrix has a no-pivot factorization

$$H_p = L \text{diag}(\pi_1, \dots, \pi_n) L^\top, \quad \pi_k = h_k/h_{k-1} > 0,$$

with L lower unitriangular over F_Γ . Adjoining all $\sqrt{\pi_k}$ and adjoining all $\sqrt{h_k}$ give the same field, since $h_k = \prod_{j \leq k} \pi_j$. By [Corollary 12.3](#) this field is $E = F_{\Lambda_p}$.

Over E set $C = \text{diag}(\sqrt{\pi_1}, \dots, \sqrt{\pi_n}) L^\top$, so $H_p = C^\top C$. Multiplication by X is self-adjoint for the trace form:

$$M^\top H_p = H_p M,$$

since the two sides express the equality of $\text{Tr}((Xu)w)$ and $\text{Tr}(u(Xw))$. Hence $N = CMC^{-1}$ is symmetric. The real case of [Theorem 9.12](#), applied over the Hahn field E , places all eigenvalues of N in E . Its characteristic polynomial is p . So p splits over E , and its splitting field satisfies $L_p \subseteq E$.

Exactness of the upper field. The extension E/F_Γ is finite elementary abelian. By [Lemma 12.2](#) its intermediate field L_p is itself a full Hahn field F_Λ , and so it is Pythagorean ([Lemma 9.8](#)). The roots of p belong to L_p , and [\(12.4\)](#) writes each h_k as a finite sum of squares of elements of L_p . Each h_k is nonzero, so its positive square root lies in L_p . Thus

$$E = F_\Gamma(\sqrt{h_1}, \dots, \sqrt{h_n}) \subseteq L_p.$$

This proves equality, not merely a ramification bound. The degree and Galois group follow from [Corollary 12.3](#) and [Lemma 12.2](#).

Complexification. The polynomial already has all its roots in the real field L_p , and $L_p(i) = K_{\Lambda_p}$. Since L_p is ordered, it does not contain i , and its intersection with the quadratic extension $F_\Gamma(i)$ is F_Γ . The finite Galois extensions are therefore linearly disjoint, and adjoining i preserves the degree and the Galois action on the roots. This proves the complex statement. $\square$

Corollary 12.9 (A root-free splitting criterion). *Under the hypotheses of Theorem 12.8, p splits over F_Γ if and only if every $v(h_k)$ lies in 2Γ . This is also equivalent to splitting over K_Γ .*

Corollary 12.10 (A coefficient-defined primitive element). *Choose indices $k_1, \dots, k_r$ such that the classes $v(h_{k_j}) + 2\Gamma$ form a basis of D_p . If $r > 0$, then*

$$\theta_p = \sqrt{h_{k_1}} + \dots + \sqrt{h_{k_r}}$$

is a primitive element of the entire splitting field, over either F_Γ or K_Γ . For $r = 0$ the field is the base field.

Proof. Each summand is supported in the distinct coset $v(h_{k_j})/2 + \Gamma$. A sign character fixing the sum must fix each summand, since different cosets cannot cancel. It therefore fixes a basis of the character-dual group of the extension, and it is the identity. The stabilizer is trivial, so the degree of θ_p equals that of the whole extension. $\square$

The argument is the positive-coset argument of Lemma 11.1. That lemma needs no choice of basis: applied with $a_k = h_k$, $q = 2$ and all weights 1, and combined with the tower argument of Corollary 11.3, it gives the same conclusion for $\sum_{k=1}^n \sqrt{h_k}$. (This observation is made in the present merge; it is not a statement of the source manuscript.) The radical generators are chosen from the *coefficients* of p , through its subdiscriminants. They need not themselves be roots of p . So this construction must be distinguished from the primitive trace of Theorem 11.2 for principal matrix roots: it is not a spectral functional, and it does not answer the question on spectral functionals raised in Section 21.

Remark 12.11 (The Hankel span and the determinantal root group). The two invariants are related only through the Gram structure. Over the splitting field, $H_p = V^\top V$ is a Gram factorization with V real over $L_p = F_{\Lambda_p}$. So Corollary 10.5, applied with base group Λ_p , gives $\Delta_k(H_p) \in 2\Lambda_p$ for every k ; that is, every class $\Delta_k(H_p) + 2\Gamma$ lies in D_p , and $R_2(H_p) \subseteq D_p$. (This consistency check is made in the present merge and is not a statement of the source manuscript.) Only this inclusion is asserted. The profile m_p , which counts factors, is read from the leading principal minors, and nothing here relates it to the Δ_k .

12.4 The pivot profile counts irreducible factors

Theorem 12.8 uses the span of the valuation classes. The next theorem keeps their multiplicities. Its proof needs a small valuation-theoretic fact about positive quadratic forms, which is proved rather than assumed.

Lemma 12.12 (Square-class multiplicities are invariant under congruence). *Let ϕ be a positive definite quadratic form on a finite-dimensional F_Γ -space V . In any diagonal expression*

$$\phi(x) = a_1 x_1^2 + \dots + a_n x_n^2, \quad a_j > 0,$$

the number of indices with $v(a_j) + 2\Gamma = \xi$ depends only on ϕ and ξ , not on the diagonalization.

Proof. For $x \neq 0$ define $w_\phi(x) = \frac{1}{2}v(\phi(x)) \in \frac{1}{2}\Gamma$, and put $w_\phi(0) = \infty$. In the displayed diagonal basis, positivity of every summand yields

$$w_\phi(x) = \min_{x_j \neq 0} (v(x_j) + \frac{1}{2}v(a_j)). \quad (12.6)$$

In particular $w_\phi(x + y) \geq \min(w_\phi(x), w_\phi(y))$. For $\eta \in \frac{1}{2}\Gamma$ let

$$V_{\geq \eta} = \{x : w_\phi(x) \geq \eta\}, \quad V_{> \eta} = \{x : w_\phi(x) > \eta\}, \quad \text{gr}_\eta(V, \phi) = V_{\geq \eta} / V_{> \eta}.$$

This is a vector space over the constant field $\mathbb{R}$.

Put $\nu_j = v(a_j)/2$. By (12.6) a coordinate can contribute to the quotient at level η precisely when $\eta - \nu_j \in \Gamma$. For each such coordinate take the coefficient of $t^{\eta - \nu_j}$ in x_j . These coefficients define a surjective map

$$V_{\geq \eta} \longrightarrow \mathbb{R}^{\{j: \eta - \nu_j \in \Gamma\}}$$

whose kernel is exactly $V_{> \eta}$. For coordinates not indexed on the right, equality with level η is impossible, so their whole contribution is already strictly above that level. Hence

$$\dim_{\mathbb{R}} \text{gr}_{\eta}(V, \phi) = \#\{j : v(a_j) + 2\Gamma = 2\eta + 2\Gamma\}.$$

The left side is defined from ϕ alone. Taking any representative η of the desired half-coset proves the lemma. The proof makes no assumption about the existence of a next valuation above η . $\square$

For the trace form $\tau_p(u) = \text{Tr}(u^2)$ of $F_{\Gamma}[X]/(p)$, the LDL^T diagonalization has entries $\pi_k = h_k/h_{k-1}$. So Lemma 12.12 makes m_p an invariant of the positive trace form itself.

Theorem 12.13 (Factor-count and character-profile theorem). *Let p satisfy the hypotheses of Theorem 12.8, and factor it over F_{Γ} as $p = p_1 \cdots p_b$ into distinct monic irreducibles. For a root of p_j , let its field be F_{Λ_j} , and identify*

$$D_j = 2\Lambda_j/2\Gamma \subseteq \Gamma/2\Gamma.$$

Then, for every class ξ ,

$$m_p(\xi) = \sum_{j=1}^b \mathbf{1}_{D_j}(\xi). \quad (12.7)$$

In particular

$$b = m_p(0) = \#\{k : v(h_k) - v(h_{k-1}) \in 2\Gamma\}. \quad (12.8)$$

The factor degrees are $\deg p_j = 2^{\dim D_j}$, and $D_p = D_1 + \cdots + D_b$.

Proof. The splitting field F_{Λ_p}/F_{Γ} has abelian Galois group, so every intermediate root field is Galois over the base. By Lemma 12.2 it is a Hahn field F_{Λ_j} , with $2\Lambda_j \subseteq \Gamma$. Being normal, the root field already contains all roots of the irreducible polynomial p_j . Its degree is $|D_j| = 2^{\dim D_j}$.

The Chinese remainder isomorphism

$$F_{\Gamma}[X]/(p) \simeq \prod_{j=1}^b F_{\Lambda_j}$$

identifies τ_p with the orthogonal sum of the field trace forms $\tau_j(z) = \text{Tr}_{F_{\Lambda_j}/F_{\Gamma}}(z^2)$. For the j th field choose one monomial t^{η_x} for each coset $x \in \Lambda_j/\Gamma$. The trace of a monomial with exponent outside Γ is zero, since summing its sign characters cancels; the trace of t^{γ} with $\gamma \in \Gamma$ is $|D_j|t^{\gamma}$. Distinct cosets $x \neq x'$ have $\eta_x + \eta_{x'} \notin \Gamma$, since the quotient has exponent 2. So the monomial basis is orthogonal, and

$$\tau_j \simeq \bigoplus_{x \in \Lambda_j/\Gamma} \langle |D_j| t^{2\eta_x} \rangle. \quad (12.9)$$

All diagonal coefficients are positive, and their valuation classes are exactly the elements of D_j , each occurring once.

Now compare (12.9), summed over j , with the Hankel LDL^T diagonalization of the same form. Lemma 12.12 gives (12.7). Every subspace D_j contains zero, so the zero class has multiplicity b . The compositum of the root fields is the full splitting field, which also proves the asserted sum of subspaces. $\square$

Corollary 12.14 (Irreducibility without factorization). *A monic squarefree real-rooted p is irreducible over F_Γ if and only if exactly one of its pivot valuations $v(h_k/h_{k-1})$ lies in 2Γ . The same test decides irreducibility over K_Γ . If p is irreducible, then m_p is the indicator function of D_p , and $n = 2^{\dim D_p}$.*

Proof. The real assertion is [Theorem 12.13](#). By [Theorem 12.8](#), adjoining i does not change the Galois action on the roots. So the irreducible factors over the complexified base have the same degrees and the same number. For a single factor, (12.7) is exactly the indicator function of its subspace. $\square$

The power-of-two phenomenon has older finite-rank precedents [15]. What the formula adds is a direct count from Hankel pivots in an arbitrary-rank workspace.

Corollary 12.15 (The root permutation character). *Let $\mathcal{G}_p = \text{Gal}(L_p/F_\Gamma)$, identified with $\text{Hom}(D_p, \{\pm 1\})$ by [Theorem 12.8](#). In the permutation representation of $\mathcal{G}_p$ on the distinct roots, the character indexed by $\xi \in D_p$ occurs with multiplicity $m_p(\xi)$. Consequently, for each $g \in \mathcal{G}_p$,*

$$\#\{\lambda : p(\lambda) = 0, g\lambda = \lambda\} = \sum_{\xi \in D_p} m_p(\xi) g(\xi).$$

Proof. The roots of p_j form one orbit, isomorphic to the quotient of $\mathcal{G}_p$ by the subgroup fixing F_{Λ_j} . Its permutation representation is the regular representation of that quotient, whose linear characters are exactly those indexed by D_j , each once. Summing over the orbits and using (12.7) gives the result. The trace of a permutation matrix is its number of fixed points. $\square$

Warning 12.16 (Factors are not base-field roots). The quantity $m_p(0)$ counts irreducible *factors*, not linear factors. An irreducible quartic has $m_p(0) = 1$ but no root in the base field. In fact the whole profile need not determine the number of base-field roots; [Example 12.29](#) gives an explicit example.

Theorem 12.17 (Realizability of the complete profile). *A nonzero finitely supported function $m : \Gamma/2\Gamma \rightarrow \mathbb{Z}_{\geq 0}$ is the Hankel square-class profile of a monic squarefree real-rooted polynomial over F_Γ if and only if it can be written*

$$m = \mathbf{1}_{D_1} + \cdots + \mathbf{1}_{D_b}$$

for finitely many finite-dimensional subspaces $D_j \subseteq \Gamma/2\Gamma$, repetitions allowed. Whenever it can occur, it can occur with all roots positive and of valuation zero. The degree is $\sum_\xi m(\xi) = \sum_j 2^{\dim D_j}$.

Proof. Necessity is [Theorem 12.13](#). For sufficiency, choose for each D_j a basis represented by positive elements $\gamma_{j1}, \dots, \gamma_{je_j} \in \Gamma$, and in its monomial extension set

$$\theta_j = t^{\gamma_{j1}/2} + \cdots + t^{\gamma_{je_j}/2}.$$

Distinct sign choices give distinct conjugates, because the summands occupy distinct independent support cosets. So θ_j generates the whole extension of degree 2^{e_j} . For $D_j = 0$ put $\theta_j = 0$. Choose pairwise distinct positive ordinary real numbers ϱ_j , and let p_j be the minimal polynomial of $\varrho_j + \theta_j$. Its conjugates are all positive units of standard part ϱ_j . So different p_j are coprime, each is real-rooted and squarefree, and its root field has subspace D_j . The product of the p_j has the desired profile by [Theorem 12.13](#). $\square$

Consequently the irreducible factor-degree lists compatible with a profile are exactly the lists $(2^{\dim D_1}, \dots, 2^{\dim D_b})$ coming from such indicator decompositions. This is a finite combinatorial classification, not a promise of an efficient search through subspaces. In particular $m(\xi) \leq m(0)$ for all ξ , and $m(0) = 1$ forces m to be the indicator of one subspace. The decomposition need not be unique.

12.5 Base change, invariance and precision

Theorem 12.18 (Exact base-change formulas). *Let $\Gamma \subseteq \Gamma'$ be an inclusion of ordered abelian groups, let $j : \Gamma/2\Gamma \rightarrow \Gamma'/2\Gamma'$ be the induced map, and regard p as a polynomial over $F_{\Gamma'}$. Its profile is the pushforward*

$$m_{p,\Gamma'}(\xi') = \sum_{j(\xi)=\xi'} m_{p,\Gamma}(\xi).$$

Its splitting group has rank $\dim j(D_p)$, and its number of irreducible factors is

$$N_{\Gamma'}(p) = \sum_{\xi \in \ker j} m_p(\xi) = \#\{k : v(h_k/h_{k-1}) \in 2\Gamma'\}. \quad (12.10)$$

These statements also hold over $K_{\Gamma'}$.

Proof. The Newton sums, the determinants and the positive pivot elements are unchanged by scalar extension (Proposition 9.22), and their valuations are the images of the original valuations in Γ' . The displayed profile formula follows directly from the definition. The polynomial remains squarefree and real-rooted in the larger real closed ambient field. Apply Theorems 12.8 and 12.13 there. $\square$

For a finite intermediate field F_{Λ_T} of L_p , corresponding to $T \subseteq D_p$, the kernel on D_p is exactly T . Thus $N_{\Lambda_T}(p) = \sum_{\xi \in T} m_p(\xi)$. At $T = 0$ this is the factor count over the base; at $T = D_p$ it is $\sum_{\xi} m_p(\xi) = n$, as a splitting field requires. For a divisible Γ' every class vanishes, which explains both why p splits there and why using that enlarged group from the outset loses the original arithmetic. This is the polynomial form of Warning 9.23: degrees can drop under enlargement of the base group, and (12.10) says exactly how.

Proposition 12.19 (Additivity and affine invariance). *For monic coprime squarefree real-rooted p and $\tilde{p}$,*

$$m_{p\tilde{p}} = m_p + m_{\tilde{p}}, \quad D_{p\tilde{p}} = D_p + D_{\tilde{p}}.$$

For $a \in F_{\Gamma}^{\times}$, $c \in F_{\Gamma}$ and $p_{a,c}(X) = a^n p((X - c)/a)$, one has

$$h_k(p_{a,c}) = a^{k(k-1)} h_k(p), \quad m_{p_{a,c}} = m_p, \quad D_{p_{a,c}} = D_p.$$

Indeed every individual pivot class, in its original index order, is unchanged by the affine substitution.

Proof. For products, the trace algebra is a product and its trace form is an orthogonal sum; apply Lemma 12.12, or equivalently Theorem 12.13. For the affine assertion, the roots change from λ_i to $a\lambda_i + c$. Each summand in (12.4) has $\binom{k}{2}$ squared differences, so it is multiplied by $a^{k(k-1)}$. Consecutive determinant valuations therefore change by $2(k-1)v(a)$, which is zero modulo 2Γ . $\square$

Affine invariance is useful for examples: any bounded real root configuration may be translated into a positive finite monad without changing its square-class arithmetic.

Theorem 12.20 (Stability of the complete profile). *Let p be monic, squarefree and real-rooted of degree n , and let $\tilde{p}$ be any monic polynomial of the same degree over F_{Γ} . If*

$$v(h_k(\tilde{p}) - h_k(p)) > v(h_k(p)) \quad (1 \leq k \leq n), \quad (12.11)$$

then $\tilde{p}$ is also squarefree and real-rooted, and

$$m_{\tilde{p}} = m_p, \quad D_{\tilde{p}} = D_p, \quad L_{\tilde{p}} = L_p$$

as Hahn subfields of the fixed ambient field. The irreducible factor counts remain identical after every ordered Hahn base enlargement.

Proof. A perturbation of strictly larger valuation preserves both the valuation and the leading coefficient of a nonzero Hahn element. So every $h_k(\tilde{p})$ has the same valuation as $h_k(p)$ and is positive, and [Proposition 12.7](#) shows that $\tilde{p}$ is squarefree and real-rooted. Equality of all determinant valuations gives equality of all pivot classes, hence of the profiles and their spans. The exact splitting fields are the Hahn fields specified by the same span. Finally apply [Theorem 12.18](#). $\square$

Corollary 12.21 (A sufficient coefficient precision for integral inputs). *Assume that the coefficients of p lie in $\mathcal{O}_F$, and set $\eta = \max_{1 \leq k \leq n} v(h_k(p))$. If every coefficient difference between the monic degree- n polynomials p and $\tilde{p}$ has valuation strictly greater than η , the conclusions of [Theorem 12.20](#) hold.*

Proof. The determinants h_k are integer polynomials in the coefficients, so $h_k(p) \in \mathcal{O}_F$ and $\eta \geq 0$. The assumed differences make the coefficients of $\tilde{p}$ integral as well. Expand a difference of polynomial values by telescoping products. Each term contains at least one coefficient difference of valuation greater than η and otherwise only integral factors, so its valuation exceeds η . There are finitely many terms, so the same holds for $h_k(\tilde{p}) - h_k(p)$, which proves (12.11). $\square$

The certificate is sufficient, not optimal. It certifies field and factor-count invariants, not equality of roots or of all their series coefficients, and there is no topological convergence argument hidden in it. One must actually know the required precision in the ordered group; repeatedly increasing a smaller Archimedean scale may never reach η . Precision certificates in the collection sit at different levels and should be cross-referenced, not merged: this one on the Hankel determinants, [Theorem 15.3](#) on singular scales, the polynomial-algebra report's stability theorem on coefficients (`polynomial:thm:stability`), and the Prony report's on moments.

12.6 Definite pencils and sharp dimension bounds

Let $A = A^*$ and $B = B^* \succ 0$ be $n \times n$ matrices over K_Γ . Positivity means $z^* B z > 0$ for every nonzero vector, and conjugation acts coefficientwise, fixing every t^γ . (This is the generalized eigenproblem of [Section 4](#), with its pair (H, G) renamed (A, B) .) The generalized characteristic polynomial

$$p_{A,B}(X) = \frac{\det(XB - A)}{\det B} \quad (12.12)$$

is monic of degree n and has real coefficients, since conjugating each determinant gives the determinant of its transpose. In the real closed ambient field, B has a positive square root and $B^{-1/2} A B^{-1/2}$ is Hermitian; its eigenvalues are real by the finite-dimensional real-closed-field spectral argument of [Theorem 4.1](#). So (12.12) is real-rooted. These standard reductions are not new. The point below is the *minimal* field, not the existence of an ambient one.

Theorem 12.22 (Exact eigenvalue fields of definite pencils). *Let p be the monic squarefree part of (12.12), of degree d . The field generated over K_Γ by the generalized eigenvalues is K_{Λ_p} , where Λ_p is computed from the d Hankel subdiscriminants of p . Its degree is 2^{r_p} , and*

$$r_p \leq \min(\lfloor d/2 \rfloor, d - m_p(0)).$$

The number of irreducible factors of the generalized eigenvalue polynomial p is $m_p(0)$.

Proof. After [Theorems 12.8](#) and [12.13](#) and [Corollary 12.14](#) only the rank bounds remain. More generally, let a squarefree real-rooted polynomial of degree d have irreducible factor fields described by the subspaces $D_1, \dots, D_b$, with $e_j = \dim D_j$. Then

$$r_p = \dim(D_1 + \dots + D_b) \leq \sum_j e_j, \quad d = \sum_j 2^{e_j}, \quad b = m_p(0).$$

For $e \geq 1$ we have $2e \leq 2^e$, and for all $e \geq 0$ we have $e \leq 2^e - 1$. Summing proves $2r_p \leq d$ and $r_p \leq d - b$. Equivalently, the first bound is the elementary bound on the minimal faithful permutation degree of an elementary abelian 2-group. $\square$

Warning 12.23 (Eigenvalues versus normalized eigenvectors). [Theorem 12.22](#) identifies the eigenvalue field, not the least field containing a B -orthonormal eigenbasis. Even the one-dimensional pencil $A = B = t^\gamma$ has eigenvalue 1 in the base field, while a vector of B -norm 1 requires $t^{-\gamma/2}$ when $\gamma \notin 2\Gamma$. A non-normalized eigenbasis is available over the eigenvalue splitting field, because the split generalized operator is semisimple; normalizing that basis may require additional positive square roots.

Theorem 12.24 (A positive definite realization in dimension $2r$). *Let $D \subseteq \Gamma/2\Gamma$ have finite dimension $r \geq 1$. There is a real symmetric positive definite pencil of size $2r$ over F_Γ whose eigenvalue splitting field has ramification space exactly D . Every eigenvalue is positive, finite and of valuation zero. No definite pencil of smaller size can have this ramification space.*

Proof. Choose positive representatives $\gamma_1, \dots, \gamma_r \in \Gamma$ of a basis of D ; replacing a nonzero representative by its negative does not change its class modulo 2Γ . Put $a_j = t^{\gamma_j}$, so $0 < a_j < 1$. For each j take

$$B_j = \begin{pmatrix} 1 & 0 \\ 0 & a_j \end{pmatrix}, \quad A_j = \begin{pmatrix} 1 & a_j \\ a_j & a_j \end{pmatrix}. \quad (12.13)$$

The leading principal minors of A_j are 1 and $a_j(1 - a_j)$, which are positive, and B_j is plainly positive definite. Direct calculation gives

$$\frac{\det(XB_j - A_j)}{\det B_j} = (X - 1)^2 - a_j.$$

The eigenvalues are $1 \pm t^{\gamma_j/2}$, which are positive units. They are distinct within each pair and across different pairs, since the a_j have distinct nonzero square classes. Their collective field is $F_\Gamma(t^{\gamma_1/2}, \dots, t^{\gamma_r/2})$. Take the block direct sums of all A_j and all B_j .

Conversely, a smaller pencil has fewer than $2r$ distinct eigenvalues, contradicting the rank bound of [Theorem 12.22](#). $\square$

This realizes arbitrary independent half-scales while ordinary leading magnitudes see only the repeated value 1. It also separates two notions of minimal dimension: the field generated *collectively* by the spectrum needs only $2r$ dimensions, while one primitive eigenvalue needs 2^r .

Proposition 12.25 (One primitive eigenvalue costs 2^r dimensions). *For the field of [Theorem 12.24](#), an eigenvalue generating the entire extension of degree 2^r requires a pencil of size at least 2^r . There is a positive definite pencil of that size with all eigenvalues positive units and every eigenvalue primitive.*

Proof. An eigenvalue of an $n \times n$ pencil has minimal polynomial of degree at most n , which proves the lower bound. For existence, write $E = F_\Gamma(t^{\gamma_1/2}, \dots, t^{\gamma_r/2})$ and set

$$\lambda = 1 + t^{\gamma_1/2} + \dots + t^{\gamma_r/2}.$$

Its 2^r sign conjugates are distinct, because the summands lie in independent, distinct support cosets. All are positive units, since a finite sum of positive infinitesimals is infinitesimal. So λ is primitive.

Choose a monomial basis $e_1, \dots, e_{2^r}$ of E/F_Γ and define

$$B_{ij} = \text{Tr}_{E/F_\Gamma}(e_i e_j), \quad A_{ij} = \text{Tr}_{E/F_\Gamma}(\lambda e_i e_j).$$

These are real symmetric matrices over the base field. For $u \neq 0$ in E all conjugates of u are nonzero, so

$$\mathrm{Tr}(u^2) = \sum_g g(u)^2 > 0, \quad \mathrm{Tr}(\lambda u^2) = \sum_g g(\lambda)g(u)^2 > 0.$$

Hence A and B are positive definite. If M_λ is multiplication by λ , then $A = BM_\lambda$. The generalized characteristic polynomial is therefore the minimal polynomial of λ , and the eigenvalues are exactly the sign conjugates described above. $\square$

The sum-of-independent-radicals construction has a direct antecedent in Keller–Ochsenius [15]. Its role here is to calibrate the collective $2r$ bound against the single-root 2^r bound, not to claim that such primitive radicals are new.

12.7 Examples and limitations

Example 12.26 (A quadratic whose roots hide their ramification). For $p(X) = (X - 1)^2 - t^\gamma$,

$$H_p = \begin{pmatrix} 2 & 2 \\ 2 & 2 + 2t^\gamma \end{pmatrix}, \quad h_1 = 2, \quad h_2 = 4t^\gamma.$$

The pivot classes are 0 and $\gamma + 2\Gamma$. If $\gamma \notin 2\Gamma$, the polynomial is irreducible and its splitting field has degree 2, although both roots have valuation zero. This is also the single block (12.13). For comparison, $X^2 - (1 + t^\gamma)$ has $h_2 = 4(1 + t^\gamma)$, of valuation zero, so both pivot classes vanish and it has two factors over F_Γ : the positive binomial square root stays in the original group.

In the next three examples fix $\delta, \Omega \in \Gamma$ with $0 < \delta < \Omega$ whose classes $\bar{\delta}, \bar{\Omega}$ modulo 2Γ are independent, and put $a = t^\delta$, $b = t^\Omega$; bars denote classes modulo 2Γ . The group $\mathbb{Z}\delta \oplus \mathbb{Z}\Omega$ of Example 11.4 is one instance.

Example 12.27 (Two quadratic factors and two independent scales). For $p_{\text{pair}}(X) = (X^2 - a)(X^2 - b)$,

$$\begin{aligned} h_1 &= 4, & h_2 &= 8(a + b), \\ h_3 &= 8(a - b)^2(a + b), & h_4 &= 16ab(a - b)^4. \end{aligned}$$

Since $v(a + b) = v(a - b) = \delta$, the determinant valuations are

$$(v(h_1), v(h_2), v(h_3), v(h_4)) = (0, \delta, 3\delta, 5\delta + \Omega),$$

and the consecutive pivot classes are $(0, \bar{\delta}, 0, \bar{\Omega})$. The coefficient formula gives two irreducible factors and splitting degree 4, in agreement with the displayed factorization. The full discriminant alone sees only the class $\bar{\delta} + \bar{\Omega}$; it cannot specify the whole two-dimensional extension space.

Example 12.28 (An irreducible quartic with a square discriminant). Consider

$$p_{\text{prim}}(X) = X^4 - 2(a + b)X^2 + (a - b)^2,$$

with roots $\pm\sqrt{a} \pm \sqrt{b}$. With a, b renamed t, u it is the minimal polynomial (11.4) of the primitive trace in Example 11.4. Its Hankel determinants are

$$\begin{aligned} h_1 &= 4, & h_2 &= 16(a + b), \\ h_3 &= 256ab(a + b), & h_4 &= 4096a^2b^2(a - b)^2. \end{aligned}$$

The last is the square $[64ab(a - b)]^2$ in the base field. Nevertheless the determinant valuations are $0, \delta, 2\delta + \Omega, 4\delta + 2\Omega$, and the pivot classes are $(0, \bar{\delta}, \bar{\delta} + \bar{\Omega}, \bar{\Omega})$. Exactly one pivot class is zero,

so the polynomial is irreducible and its splitting field is the full biquadratic extension. This can be checked independently: the four sign conjugates of $\sqrt{a} + \sqrt{b}$ are distinct, so its minimal polynomial has degree 4, which is how [Example 11.4](#) certifies the degree. The determinant calculation is a root-free certificate of the same fact.

For an explicit surreal interpretation take $\Gamma = \mathbb{Z} + \mathbb{Z}\omega \subset \mathbf{No}$, $\delta = 1$, $\Omega = \omega$, so $a = \omega^{-1}$ and $b = \omega^{-\omega}$. The new exponent classes needed are $1/2$ and $\omega/2$. Translating X by 1 makes all four roots positive units and leaves the whole profile unchanged ([Proposition 12.19](#)).

Example 12.29 (The profile does not recover all factor degrees). Define two polynomials of degree six,

$$P(X) = p_{\text{prim}}(X)(X-2)(X-3), \quad Q(X) = (X^2 - a)(X^2 - b)(X^2 - ab).$$

Their roots are distinct: all radical roots are infinitesimal, the constants 2, 3 are different, and a, b, ab have different nonzero square classes. The factor degrees are 4, 1, 1 for P and 2, 2, 2 for Q . By [Theorem 12.13](#) both have the same profile,

$$m(0) = 3, \quad m(\bar{\delta}) = m(\bar{\Omega}) = m(\bar{\delta} + \bar{\Omega}) = 1.$$

Both have three irreducible factors and the same splitting field of degree four, and they have the same number of irreducible factors after every ordered Hahn base enlargement ([Theorem 12.18](#)). But P has two base-field roots and Q has none. So the profile must not be advertised as a complete reconstruction of the Galois set of roots or of the list of factor degrees. The underlying representation identity is elementary: for the group $(\mathbb{Z}/2)^2$, one regular orbit together with two fixed points has the same permutation character as the union of its three distinct orbits of index two. The profile detects that representation, not the finer orbit-by-orbit structure.

Example 12.30 (Shared scales need not give independent extensions). Over $\Gamma = \mathbb{Z}$ the polynomial $(X^2 - t)(X^2 - t^3)$ has two irreducible quadratic factors, but its splitting field is only $F_{\frac{1}{2}\mathbb{Z}}$, since $\sqrt{t^3} = t\sqrt{t}$. Its profile has two occurrences of 0 and two of the unique nonzero class. So the factor count is 2 and the splitting rank is 1. Counting visible radical expressions instead of independent square classes would wrongly predict degree 4.

Warning 12.31 (The hypotheses are needed). (a) *Real roots*. The polynomial $X^3 - t$ over $F_{\mathbb{Z}}$ is not real-rooted: it has one real root and a nonreal conjugate pair. Its Hankel matrix is not positive definite (already $h_2 = 0$), and its complex splitting field requires $t^{1/3}$. So no “only half-scales” principle for all polynomials follows from this section.

(b) *Squarefreeness*. A polynomial with repeated roots must first be replaced by its monic squarefree part. For $(X^2 - t)^3$ the unreduced trace matrix of size six is singular; treating its zero determinants as ordinary valuations, or dividing by them, is invalid. The correct squarefree polynomial of degree two has the quadratic profile computed above.

(c) *Real closed constants*. Over $\mathbb{Q}((t^{\Gamma}))$ positive constants need not be squares, and irreducible residue polynomials can create extensions not described by $\Gamma/2\Gamma$. The hypothesis of real closed constant coefficients cannot be silently omitted.

(d) *A field, not a multivariable ring*. A multivariable formal power-series ring is not a totally ordered Hahn field: the crossing matrix (9.13) of [Warning 9.24](#) has eigenvalues $\pm\sqrt{x^2 + y^2}$, which do not belong to $\mathbb{C}[[x, y]]$. The multiparameter results of [18, 19] have different hypotheses and are not consequences of the workspace formulas here.

12.8 The exact-operation contract

The formulas are finite, but arbitrary Hahn numbers do not automatically come with effective representations. The following procedure is an algorithm relative to explicit scalar and value-group operations.

Given a nonconstant polynomial over F_Γ , first make it monic and compute its monic squarefree part using $\gcd(p, p')$. For the reduced degree n , compute the Newton sums through s_{2n-2} , form H_p , and perform exact symmetric elimination. If the leading determinants are not all positive, the input does not satisfy the real-rooted squarefree hypothesis, and the factor-count formulas are not applied. For a successful input the pivots are all nonzero and positive, and the rest is small:

$$\begin{aligned} \pi_k &= h_k/h_{k-1}, & \xi_k &= v(\pi_k) + 2\Gamma, \\ m(\xi) &= \#\{k : \xi_k = \xi\}, & b &= m(0), & r &= \dim_{\mathbf{F}_2} \text{span}\{\xi_1, \dots, \xi_n\}. \end{aligned} \quad (12.14)$$

Return b , the profile, a basis of the span and the splitting degree 2^r . A basis chosen instead from the cumulative classes $v(h_k) + 2\Gamma$ gives the primitive-element certificate of [Corollary 12.10](#). No root extraction is needed to compute the invariants themselves.

The required scalar operations are exact field arithmetic, equality, order or sign, and the valuation of a nonzero element. The group representation must in addition decide the relevant linear relations modulo 2Γ . A valuation oracle alone does not decide membership in 2Γ ; this is [Warning 14.2](#) at $q = 2$. When $\Gamma \cong \mathbb{Z}^m$ is given with an explicit basis and an explicit compatible order, classes are coordinate vectors modulo 2, and ordinary binary Gaussian elimination suffices (compare [\(10.11\)](#)). For more general groups this is a separate representation requirement.

Constructing the Newton sums uses $O(n^2)$ field operations with the standard recurrences, and one $LDL^\top$ factorization uses $O(n^3)$; the determinants can be accumulated from its pivots rather than computed separately. These are arithmetic-operation counts. They are not bit-complexity bounds, and not bounds on the cost of an arbitrary Hahn-series oracle. Squarefree reduction, coefficient expression swell, valuation extraction and the cost of the quotient-group operations must be accounted for separately.

A finite prefix is not a universal exact certificate. A truncated input can conceal a term at a previously unseen scale: a name known only to have initial part $(X - 1)^2$ is compatible both with that repeated-root polynomial and with $(X - 1)^2 - t^\gamma$ beyond the inspected support, and the latter can have irreducible quadratic splitting at an odd class. Exact factorization information cannot be inferred from an uncertified finite prefix alone. [Theorem 12.20](#) is the positive replacement: once the leading terms of the relevant nonzero Hankel determinants are certified, the profile and all the stated field data are certified as well. This matches the representation distinctions of the computer-algebra report [\[4\]](#) and of [Section 14.3](#).

12.9 Back inside the surreal numbers

Let Γ be a set-sized additive subgroup of $\mathbf{No}$. The normal-form map $\sum_\gamma a_\gamma t^\gamma \mapsto \sum_\gamma a_\gamma \omega^{-\gamma}$ embeds F_Γ into $\mathbf{No}$, and coefficientwise complexification embeds K_Γ into $\mathbf{No}[i]$, as in [\(2.1\)](#) and [Proposition 2.3](#). The reversed sign in the exponent is essential: increasing support for t becomes decreasing support in the surreal normal-form convention, and the transported valuation of $\omega^{-\gamma}$ is γ . For finitely many surreal or surcomplex coefficients the union of their normal-form exponent supports is a set, and the subgroup it generates, after the sign conversion, is a legitimate choice of Γ ; there is no need to replace it by its divisible hull. [Theorems 12.8](#) and [12.13](#) then determine exactly which half-cosets a real-rooted polynomial needs, and count its irreducible factors over that workspace. The same applies to the real generalized characteristic polynomial of a definite surcomplex pencil.

Warning 12.32 (Relative fields, not an extension of the whole surreal field). The full surreal field is real closed, and it acquires no new real algebraic numbers from this construction. The extension being measured is $F_\Gamma \subseteq F_{\Lambda_p}$, both already embedded in $\mathbf{No}$. Similarly, $\mathbf{No}[i]$ already contains all the relevant complex algebraic roots. The theorems are about the exact exponent workspace required by finite data, not about a failure of algebraic closure of $\mathbf{No}[i]$; compare the remark after [Corollary 11.7](#).

Also, F_Γ is not in general the smallest ordinary field generated by the finitely many coefficients: it contains *all* real Hahn series on Γ . The splitting-field equality is relative to that explicitly stated base, not to a smaller rational-function or finitely generated coefficient field.

Arbitrary rank is a genuine part of the statements. The example $\Gamma = \mathbb{Z} + \mathbb{Z}\omega$ already contains an exponent larger than every integer multiple of the smaller positive scale, and a general Γ can have much larger rank, non-finitely generated subgroups and supports of long order type. Nevertheless every object in the coefficient certificate is finite: n determinants, n pivot classes, and a finite-dimensional binary span. The proofs never turn a Hahn sum into an ordinary convergent sequence, or assume that one infinitesimal parameter controls all smaller scales. Divisibility is not assumed even locally. An equation such as $2\eta = \gamma$ is solved in the divisible ambient group only to specify the precise finite extension; whether its solution already belongs to Γ remains visible throughout.

12.10 What this section claims, and what it does not

This section is not a new spectral theorem. Hahn spectral descent is [Section 9](#), with the predecessors conceded in [Section 1.4](#); spectral descent over suitable power-series fields is already in Adkins and Keller–Ochsenius [13, 14]. Trace forms and the Hermite and Sylvester criteria are classical. The Hankel and Vandermonde-square descriptions of subdiscriminants are Roy’s [16], and their invariant-theoretic and sums-of-squares aspects are developed by Domokos [17]. Keller and Ochsenius’s 2008 paper treats positive inner-product spaces over finite-rank iterated Laurent fields, gives a multiquadratic upper field, obtains power-of-two irreducible degrees for self-adjoint operators, and exhibits irreducible operators built from sums of independent square roots [15]. No priority is claimed for those mechanisms or examples. The Parusiński–Rond and Dai–Liang–Lu–Zhi results on several power-series parameters are relevant boundaries, not interchangeable versions of the field results here [18, 19].

The proposed addition is narrower: the *exact* field (12.5), the whole pivot multiplicity profile with its factor-count interpretation (12.8), and their arbitrary-rank, base-change and precision consequences. The size bounds and pencil examples make the package concrete; their elementary group theory and classical radical constructions are not advertised as independent discoveries. The source’s targeted search combined “subdiscriminants” with “splitting field” and “square classes”, and “Hankel” or “trace form” with “irreducible factors” and “valuation”. It did not locate (12.5) or (12.8), or their full arbitrary-rank profile package. That is evidence about this search, not a proof of absence from the literature. The formulas may have equivalents in the language of quadratic trace forms, ordered valued fields or permutation characters. The package is a candidate original contribution with proofs, pending independent mathematical review and a broader priority check; no named published conjecture is claimed solved.

The logical inputs are finite field theory, finite real spectral theory, and the standard Hahn support and real-closedness facts of [Sections 2](#) and [9](#). The arguments are mathematical proofs, but they have not been independently refereed or checked by a proof assistant. The finite checks of [Section B](#) do not establish the general Hahn support theorem, the classification of intermediate fields ([Lemma 12.2](#)) or the graded-form invariant ([Lemma 12.12](#)), and the repository’s existing formalizations are not asserted to verify this section. Its continuation questions are recorded in [Section 21](#); they are directions, not theorems.

13 Rank at a chosen scale

13.1 Exact rank and standard-part rank

Corollary 13.1 (Rank after reduction). *If $A \in \mathcal{O}^{m \times n}$ has nonzero singular scales $\gamma_1, \dots, \gamma_r$, then every $\gamma_j \geq 0$ and*

$$\text{rank}_{\mathbb{C}}(\text{st } A) = \#\{j : \gamma_j = 0\}.$$

Proof. By Lemma 7.2, $v(\sigma_1) = v(A) \geq 0$, so all singular values are integral. Reduce an SVD. Its unitary factors reduce to invertible ordinary unitary matrices, and its diagonal entries reduce to nonzero numbers exactly at valuation zero. $\square$

An invertible matrix can therefore have a singular standard part. Conversely, an invertible standard part certifies invertibility of an integral square matrix. The inverse need not be integral in the first situation, but is integral in the second: the latter is exactly the case in which all its singular scales are zero.

13.2 An intrinsic input filtration

Fix $\theta \in \Gamma$ and define an $\mathcal{O}$ -submodule of the input lattice by

$$M_{\theta}(A) = \{x \in \mathcal{O}^n : Ax \in t^{\theta} \mathcal{O}^m\}. \quad (13.1)$$

Its image $\text{st}(M_{\theta}(A))$ under coordinatewise reduction is a $\mathbb{C}$ -linear subspace of $\mathbb{C}^n$. Also define

$$M_{>\theta}(A) = \{x \in \mathcal{O}^n : v(Ax) > \theta\}.$$

These definitions work even when A or t^{θ} is nonintegral; only the input vectors are required to lie in the integral lattice.

Theorem 13.2 (Filtration dimensions). *For every $\theta \in \Gamma$,*

$$\dim_{\mathbb{C}} \text{st}(M_{\theta}(A)) = n - \#\{j : \gamma_j < \theta\}, \quad (13.2)$$

$$\dim_{\mathbb{C}} \text{st}(M_{>\theta}(A)) = n - \#\{j : \gamma_j \leq \theta\}. \quad (13.3)$$

Only the nonzero singular values enter the counts.

Proof. Use an SVD $A = U\Sigma V^*$. Both unitary factors preserve integral lattices and their reductions are invertible. Write $x = Vy$. For $j \leq r$, condition (13.1) becomes

$$v(y_j) \geq \max(0, \theta - \gamma_j).$$

The kernel coordinates $j > r$ are unrestricted in $\mathcal{O}$. The reduction of y_j must be zero exactly when $\gamma_j < \theta$; all other reduced coordinates can be chosen arbitrarily. This gives (13.2). For the strict condition, reduction of y_j vanishes exactly when $\gamma_j \leq \theta$. Multiplication by the invertible $\text{st } V$ preserves the dimension, proving (13.3). $\square$

The two formulas deliberately differ at the boundary. A direction of singular scale exactly θ remains visible when the output threshold is strict, and disappears if only outputs of strictly larger magnitude are counted. It is helpful to fix explicit notation:

$$r_{<\theta}(A) = \#\{j : \gamma_j < \theta\}, \quad r_{\leq\theta}(A) = \#\{j : \gamma_j \leq \theta\}.$$

These are not replacements for algebraic rank. They describe how much rank is visible at a specified valuation resolution. In the integral case, $r_{\leq 0}(A) = \text{rank}(\text{st } A)$.

Example 13.3 (Three genuinely different levels). Choose a workspace containing 1 and ω in its value group and let

$$A = \text{diag}(1, t^2, t^\omega), \quad t = \omega^{-1}.$$

Its exact rank is three and its standard-part rank is one. Its closed-scale rank $r_{\leq \theta}$ is one for $0 \leq \theta < 2$, two for $2 \leq \theta < \omega$, and three for $\theta \geq \omega$. No finite integer cutoff N reaches the last direction. This is not a statement about a slow numerical convergence rate; it is an ordered-group fact, $N < \omega$ for every ordinary N .

14 A root-free scale algorithm and its computational contract

14.1 Minimum-valuation elimination

The determinantal formula is finite, but enumerating all minors is expensive. A valuation-aware elimination computes the same scale tuple without singular vectors or algebraic root extraction.

Proposition 14.1 (Exact scale elimination). *Suppose exact field arithmetic, exact zero testing, extraction of valuations of nonzero elements, and comparison of those valuations are available. The following algorithm terminates after at most $\min(m, n)$ pivots and returns $\gamma_1, \dots, \gamma_r$ in nondecreasing order.*

In the active submatrix, choose a nonzero entry of minimum valuation and move it to the leading position by row and column swaps. Clear the remaining entries in its column by row operations, and then clear its row by column operations. Record its valuation and recurse on the trailing submatrix. Stop when the active submatrix is zero.

Proof. Write one active block as

$$\begin{pmatrix} a & b \\ c & D \end{pmatrix}, \quad v(a) = \alpha \leq v(\text{every entry}).$$

The row multipliers c_i/a and column multipliers b_j/a are integral. Their elementary matrices and inverses therefore lie in $\text{GL}(\mathcal{O})$. The two clearings leave the Schur complement

$$D - ca^{-1}b. \tag{14.1}$$

Every term in it has valuation at least α , because $v(c_i) + v(b_j) - v(a) \geq \alpha$. Cancellation can raise the valuation, but cannot lower it. Thus subsequent pivot valuations are nondecreasing. After finitely many steps the result is a diagonal matrix and a zero block. All transformations were integral and invertible, so they preserve the Δ_k . The diagonal pivot valuations must therefore be their successive differences, which are the singular scales by [Theorem 7.5](#). $\square$

The algorithm uses $O(mn \min(m, n))$ field operations, apart from pivot comparisons. This counts algebraic operations; it is not a bit-complexity bound for arbitrary Hahn-series representations. A single exact zero test may already be unavailable for a given symbolic representation. Moreover, the elimination matrices are generally not unitary: this algorithm returns the singular *scales*, not the singular values or the best low-rank approximation itself.

14.2 What must be provided by a symbolic scalar representation

The algorithm has four logically separate requirements: a representation closed under the field operations it performs, exact zero testing, exact leading-exponent access for nonzero outputs, and an effective order on those exponents. Rational functions over a finite rational or Gaussian-rational monomial field can meet these requirements. A name for an arbitrary surreal number

does not automatically meet any of them. This is the same kind of denotation-versus-computation distinction emphasized by the repository’s computer-algebra strand [1].

Cancellation is the principal practical issue. A representation that stores only the current leading term is insufficient: (14.1) may cancel it. Correct implementations must either compute further exact coefficients, attach a certified remainder bound, or report that the pivot scale is unresolved. Reporting a zero pivot merely because all currently stored coefficients vanish is unsound.

14.3 Two further contracts: spectral descent and ramification

The support-certified spectral construction of Section 9 is more demanding than the scale algorithm. It needs exact arithmetic and conjugation, exact zero tests, leading exponents and leading coefficients for nonzero expressions, ordinary constant Hermitian diagonalization, and a representation of strongly summable substitutions together with their support certificates. An exact representation need not provide any of these effectively merely because it can denote a surreal number. Example 9.14 is the concrete obstruction: a procedure that inspects more and more coefficients never reaches the first nonscalar scale, whereas the exact trace subtraction (9.10) reaches it in one step. A practical implementation should also keep two outputs apart. A matrix identity such as $U^*AU = D$ is an *algebraic* certificate; a support certificate is what licenses the Hahn expressions occurring in it. Finite truncation can test an identity to a declared order, but cannot certify exact zero, all coefficients, or well-ordering of an arbitrary support.

The determinantal ramification computation of Section 10 needs much less: exact valuations of finitely many minors, and a way to compute the finite subgroup generated by specified classes modulo $q\Gamma$. In a finite integer lattice (10.11) makes this a Smith normal form, hence a concrete algorithm. The Hankel certificates of Section 12 need the same kind of group data at $q = 2$, together with exact sign tests for the positivity of the Hankel minors; their contract is spelled out in Section 12.8.

Warning 14.2 (Membership in $q\Gamma$ is extra data). For an abstract ordered group Γ presented only through a black-box comparison of elements, membership of Δ_k in $q\Gamma$ is additional group-theoretic data and need not be decidable. Order comparison alone does not answer it. Consequently (10.9) — “ $A^{1/q}$ is defined over K_Γ if and only if every $\Delta_k(A)$ lies in $q\Gamma$ ” — is an exact mathematical criterion that a given representation may nevertheless be unable to evaluate, even when it can compute every Δ_k exactly.

This is the same denotation-versus-computation distinction as above, one tier further out, and it is exactly the question that the repository’s computer-algebra report formalizes as a hierarchy of representation capabilities [4]. That report’s own inventory already makes the matching distinction on the implementation side: a fixed rational exponent lattice permits division by q , whereas a fixed integer lattice need not contain the required fractional exponents. This is a condition on allowable exponents, not a complete representation of a root. Even $\sqrt{1+t}$ stays in the integer exponent group but is not a rational function of t : a rational square has even zero order at $t = -1$, whereas $1+t$ has order one. A rational-function kernel therefore needs an additional algebraic or series representation. Equation (10.9) is therefore a useful concrete instance of a decision that needs a capability tier the black-box order interface does not supply. Nothing here claims to solve that report’s representation or computability problems.

14.4 Why finite truncation cannot certify every exact rank

Assume $\Gamma \neq \{0\}$ and fix any precision threshold $\theta \in \Gamma$. Choose $\eta > \theta$. The matrices

$$\text{diag}(1, 0), \quad \text{diag}(1, t^\eta)$$

agree at all exponents at most θ but have different exact ranks. Thus finite-precision coefficient access alone cannot decide exact rank uniformly. An exact representation may still decide it, for example by computing a nonzero symbolic determinant, but truncation alone cannot. For $\Gamma = \{0\}$, truncation through exponent zero already gives the entire matrix, so this obstruction does not apply.

The right numerical-style output is therefore often a certified rank profile up to a declared cutoff, with the remaining directions unresolved. The perturbation theorem in [Section 15](#) specifies exactly what precision is sufficient for such a certificate. A finite exact rational kernel for testing the elimination and Gram identities is included in the archive; its scope is described in [Section B](#).

15 Condition numbers and perturbation certificates

15.1 Distance to singularity is a minimum, not merely an infimum

This subsection again holds over any real closed F and $K = F(i)$.

Theorem 15.1 (Distance to singularity). *For an invertible square A ,*

$$\|A^{-1}\|_{\text{op}} = \sigma_n(A)^{-1}, \quad \min_{A+E \text{ singular}} \|E\|_{\text{op}} = \sigma_n(A). \quad (15.1)$$

The latter minimum is attained by $E = -\sigma_n u_n v_n^$ for any matched least-singular vector pair.*

Proof. Invert the SVD to obtain the first formula. The stated perturbation deletes one diagonal entry of Σ and has norm σ_n . Conversely, if $A + E$ is singular, choose a unit vector x in its kernel. Then

$$\|E\|_{\text{op}} \geq \|Ex\| = \|Ax\| \geq \sigma_n.$$

□

Define the ordered spectral condition number by

$$\kappa(A) = \|A\|_{\text{op}} \|A^{-1}\|_{\text{op}} = \frac{\sigma_1}{\sigma_n} \geq 1.$$

It is an element of F , not necessarily an ordinary real. In a Hahn workspace,

$$v(\kappa(A)) = \gamma_1 - \gamma_n. \quad (15.2)$$

Consequently $\kappa(A)$ is infinite precisely when $\gamma_1 < \gamma_n$. Matrices whose nonzero singular values all have the same valuation have finite condition number, although its ordinary leading coefficient can still be large. Overall scale and relative conditioning are separate: $t^{-\omega}I$ has infinite norm but condition number one.

Theorem 15.2 (Inverse and linear-system perturbation). *Let A be invertible and put $q = \|A^{-1}\|_{\text{op}} \|E\|_{\text{op}}$. If $q < 1$, then $A + E$ is invertible,*

$$\|(A + E)^{-1}\|_{\text{op}} \leq \frac{\|A^{-1}\|_{\text{op}}}{1 - q}, \quad (15.3)$$

$$\|(A + E)^{-1} - A^{-1}\|_{\text{op}} \leq \frac{\|A^{-1}\|_{\text{op}}^2 \|E\|_{\text{op}}}{1 - q}. \quad (15.4)$$

If $Ax = b \neq 0$ and $(A + E)(x + \delta x) = b + f$, then

$$\frac{\|\delta x\|}{\|x\|} \leq \frac{\kappa(A)}{1 - q} \left(\frac{\|f\|}{\|b\|} + \frac{\|E\|_{\text{op}}}{\|A\|_{\text{op}}} \right). \quad (15.5)$$

Proof. For every vector x ,

$$\|(A + E)x\| \geq (\sigma_n(A) - \|E\|_{\text{op}}) \|x\|.$$

The coefficient is positive under the hypothesis. The map is injective, hence invertible in finite dimension, and this inequality gives (15.3). The finite resolvent identity

$$(A + E)^{-1} - A^{-1} = -(A + E)^{-1}EA^{-1}$$

gives (15.4). Also $\delta x = (A + E)^{-1}(f - Ex)$. Apply (15.3), divide by $\|x\|$, and use $\|b\| \leq \|A\|_{\text{op}} \|x\|$ to obtain (15.5). $\square$

No geometric series was used. This matters because the ordered inequality $q < 1$ is not by itself a Hahn-summability hypothesis.

15.2 A certified precision theorem for singular scales

Return to a Hahn workspace. When $v(A) = \eta$, every entry of A has valuation at least η ; this is the matrix precision convention below.

Theorem 15.3 (Preserving visible singular data). *Let $A, B = A + E$ have the same dimensions. If $v(E) > \gamma_k(A)$ for a nonzero singular value $\sigma_k(A)$, then for all $1 \leq j \leq k$,*

$$v(\sigma_j(B)) = \gamma_j(A), \quad \text{lc}(\sigma_j(B)) = \text{lc}(\sigma_j(A)).$$

More generally, if $v(E) > \theta$, then

$$r_{\leq \theta}(A) = r_{\leq \theta}(B), \quad r_{< \theta}(A) = r_{< \theta}(B).$$

The exact ranks need not be equal.

Proof. By Lemma 7.2, $v(\|E\|_{\text{op}}) = v(E)$ for $E \neq 0$. The ordered singular-value estimate (5.8) therefore implies

$$v(\sigma_j(B) - \sigma_j(A)) \geq v(E).$$

An error of strictly larger valuation cannot alter a nonzero leading term. This proves the first assertion. If a singular value has valuation greater than θ , or is zero, adding an error of valuation greater than θ keeps it above that threshold. Singular values at or below the threshold retain their leading terms. Apply this coordinatewise to the ordered singular lists for both asserted counts. The case $E = 0$ is immediate. $\square$

Strictness is essential. For $A = \text{diag}(1, t^\theta)$ with $\theta > 0$, a perturbation $E = \text{diag}(0, -t^\theta)$ has exactly valuation θ and destroys the second singular direction. Even under the strict hypothesis, a zero singular value can become a nonzero value of a still smaller magnitude; $\text{diag}(1, 0)$ and $\text{diag}(1, t^\eta)$ with $\eta > \theta$ show why exact rank is not certified. The theorem certifies the rank *visible at the stated scale*.

16 Nonnormal matrices and ordered pseudospectra

Eigenvalues alone do not control a general matrix. For example,

$$N_L = \begin{pmatrix} 0 & L \\ 0 & 0 \end{pmatrix}$$

has only the eigenvalue zero, while $\|N_L\|_{\text{op}} = |L|$ can be infinite. The distinction between eigenvalues and singular values is as important in the surcomplex field as it is over $\mathbb{C}$.

For a square matrix and an ordered tolerance $\varepsilon \in F_{\geq 0}$, define the closed ordered pseudospectrum by

$$\Lambda_\varepsilon(A) = \{z \in K : \sigma_n(zI - A) \leq \varepsilon\}. \quad (16.1)$$

Proposition 16.1 (Exact perturbation interpretation). *A scalar z belongs to $\Lambda_\varepsilon(A)$ if and only if there is E with $\|E\|_{\text{op}} \leq \varepsilon$ for which z is an eigenvalue of $A + E$. If $z \notin \text{spec } A$ and $\varepsilon > 0$, this is equivalent to*

$$\|(zI - A)^{-1}\|_{\text{op}} \geq \varepsilon^{-1}.$$

For normal A ,

$$\Lambda_\varepsilon(A) = \bigcup_{\lambda \in \text{spec } A} \{z : |z - \lambda| \leq \varepsilon\}. \quad (16.2)$$

Proof. Apply the distance-to-singularity argument to $zI - A$. If it is already singular, use $E = 0$. Otherwise [Theorem 15.1](#) gives an attained perturbation of norm $\sigma_n(zI - A)$ making it singular; changing its sign converts it to a perturbation of A . The lower bound follows by applying $zI - A$ to an eigenvector of $A + E$. The inverse characterization is [\(15.1\)](#). For normal A , unitary diagonalization gives singular values of $zI - A$ equal to the numbers $|z - \lambda_j|$, proving [\(16.2\)](#). $\square$

Example 16.2 (An infinitesimal perturbation with a larger spectral effect). In a workspace containing 1 in its exponent group, write $t = t^1$ and let

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 0 \\ t^2 & 0 \end{pmatrix}.$$

Then $\|E\|_{\text{op}} = t^2$, whereas $N + E$ has eigenvalues $\pm t$. Thus $t \in \Lambda_{t^2}(N)$ although its distance from the unperturbed spectrum is $t > t^2$. More generally, for nonzero infinitesimal z , $zI - N$ has $\Delta_1 = 0$ and $\Delta_2 = 2v(z)$, so its least singular value has valuation $2v(z)$. The nonlinear sensitivity is visible directly in the determinantal scale profile of the resolvent matrix.

Here ε is an actual ordered magnitude, not just a value-group element. A valuation threshold gives a coarser notion. In particular, replacing the ordered disks in [\(16.2\)](#) by valuation balls would change the assertion. No fine-continuous contour is needed in the pseudospectral argument.

17 Spectral subspaces and infinitesimal gaps

Individual eigenvectors are not canonical, especially at multiplicities. Orthogonal spectral projectors are the better perturbation objects. The following finite-dimensional Frobenius version of a Davis–Kahan bound is proved directly over F ; the classical context is [\[11, 12\]](#). No inverse trigonometric function is needed to measure the subspace displacement.

Theorem 17.1 (A separated-subspace bound). *Let $H = H^*$, $B = H + E = B^*$, and let P be an orthogonal projector onto an H -invariant subspace. Let Q be the orthogonal projector onto a B -invariant subspace of rank k . Suppose that every eigenvalue λ of $H|_{\text{im}(I-P)}$ and every eigenvalue μ of $B|_{\text{im } Q}$ satisfy*

$$|\lambda - \mu| \geq \delta > 0.$$

Then

$$\|(I - P)Q\|_F \leq \frac{\|E\|_F}{\delta}. \quad (17.1)$$

If $\text{rank } P = \text{rank } Q$, also

$$\|P - Q\|_F \leq \frac{\sqrt{2}\|E\|_F}{\delta}. \quad (17.2)$$

Proof. Choose orthonormal eigenbasis matrices Y for $\text{im}(I - P)$ and X for $\text{im} Q$, so $HY = Y\Lambda$ and $BX = XM$, with real diagonal Λ, M . Put $C = Y^*X$. Multiplying $HX + EX = XM$ on the left by Y^* gives the Sylvester equation

$$\Lambda C - CM = -Y^*EX.$$

Entrywise, $(\lambda_a - \mu_b)c_{ab} = -(Y^*EX)_{ab}$, hence

$$\|C\|_F^2 \leq \delta^{-2} \|Y^*EX\|_F^2 \leq \delta^{-2} \|E\|_F^2.$$

The last inequality follows because X and Y have orthonormal columns; extend them to unitary matrices and observe that taking a submatrix cannot increase the sum of squared moduli. Since $(I - P)Q = YCX^*$, $\|(I - P)Q\|_F = \|C\|_F$. This proves (17.1).

For equal ranks,

$$\|P - Q\|_F^2 = \text{tr } P + \text{tr } Q - 2\text{tr}(PQ) = 2\|(I - P)Q\|_F^2,$$

which gives (17.2). $\square$

Corollary 17.2 (Using an unperturbed cluster gap). *Let $H = H^*$ and $E = E^*$. Select a nonempty proper set of eigenvalue indices of H , separated from all complementary indices by at least $g > 0$. Let P project onto this cluster, and let Q select the same indices in the decreasing eigenvalue list of $H + E$. If $e = \|E\|_{\text{op}} < g/2$, then*

$$\|P - Q\|_F \leq \frac{\sqrt{2} \|E\|_F}{g - e}.$$

Proof. The eigenvalue estimate in Theorem 5.2 moves each selected eigenvalue by at most e . Its distance from an unselected eigenvalue of H is at least $g - e$. No eigenvalue can cross the external gap under $e < g/2$. The chosen subspaces have the same dimension, so apply Theorem 17.1 with $\delta = g - e$. $\square$

The Frobenius separation condition is all that was used in the proof. We are not asserting every operator-norm variant of the classical Davis–Kahan theorem under an arbitrary separation hypothesis. The Frobenius estimate is sufficient for an important surreal consequence: if $v(E) > v(g)$ in a Hahn workspace, then the projector displacement is infinitesimal. Indeed, $\|E\|_F/g$ then has positive valuation, so $e < g/2$ and the displayed bound has positive valuation.

Example 17.3 (Infinitesimal error, ordinary subspace rotation). In a workspace containing 1 in its exponent group, write $t = t^1$ and consider

$$H = \begin{pmatrix} 0 & 0 \\ 0 & t^2 \end{pmatrix}, \quad B = \frac{t^2}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

They have the same eigenvalues $0, t^2$. Their zero-eigenspace projectors are

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Exact multiplication gives

$$\|B - H\|_{\text{op}} = \frac{t^2}{\sqrt{2}}, \quad \|P - Q\|_F = 1, \quad \|P - Q\|_{\text{op}} = \frac{1}{\sqrt{2}}.$$

Thus the matrix error is infinitesimal but the subspace change is not. Its size is comparable to the gap t^2 , so the gap-relative smallness hypothesis fails. Discarding the common infinitesimal factor before asking about eigenvectors would conceal the entire phenomenon.

17.1 Singular subspaces and a warning about Gram squaring

Right singular subspaces are eigenspaces of A^*A . Under $A \mapsto A + E$,

$$(A + E)^*(A + E) - A^*A = A^*E + E^*A + E^*E,$$

so its operator norm is at most $2\|A\|_{\text{op}}\|E\|_{\text{op}} + \|E\|_{\text{op}}^2$. One can therefore apply [Theorem 17.1](#) to squared-singular-value gaps. This is valid but can be quantitatively wasteful: it changes gaps $\sigma_i - \sigma_j$ into $\sigma_i^2 - \sigma_j^2$. A direct singular-subspace or Hermitian-dilation argument can yield better bounds, but no sharp Wedin theorem is claimed here. The distinction is particularly relevant when both singular values are infinitesimal.

18 Degenerate clusters and Schur-complement corrections

A cluster of repeated or nearly repeated eigenvalues cannot generally be understood by retaining only the perturbation inside that cluster. Coupling to its complement can create a smaller, second-order splitting. The right starting point is an exact finite identity.

Proposition 18.1 (Exact effective matrix). *Suppose a Hermitian matrix is written in an orthogonal decomposition as*

$$H = \begin{pmatrix} \lambda_0 I + E_{11} & E_{12} \\ E_{21} & D + E_{22} \end{pmatrix}, \quad E_{21} = E_{12}^*.$$

For $z \in K$ such that $D + E_{22} - zI$ is invertible,

$$\det(H - zI) = \det(D + E_{22} - zI) \det S(z), \quad (18.1)$$

$$S(z) = (\lambda_0 - z)I + E_{11} - E_{12}(D + E_{22} - zI)^{-1}E_{21}. \quad (18.2)$$

If the ordered distance from z to every eigenvalue of $D + E_{22}$ is at least $\delta > 0$, then

$$\|E_{12}(D + E_{22} - zI)^{-1}E_{21}\|_{\text{op}} \leq \frac{\|E_{12}\|_{\text{op}}^2}{\delta}. \quad (18.3)$$

Proof. Multiply $H - zI$ on the left by the determinant-one block matrix

$$\begin{pmatrix} I & -E_{12}(D + E_{22} - zI)^{-1} \\ 0 & I \end{pmatrix}.$$

The resulting matrix is block lower triangular with diagonal blocks $S(z)$ and $D + E_{22} - zI$, proving (18.1). Diagonalization of the Hermitian complement shows that its shifted inverse has norm at most δ^{-1} . Apply the product norm bounds and $\|E_{21}\|_{\text{op}} = \|E_{12}\|_{\text{op}}$. $\square$

For real z outside the complementary spectrum, $S(z)$ is Hermitian. It is an eigenvalue-dependent effective matrix, not in general one fixed matrix whose eigenvalues already solve the original problem. Freezing z or dropping the inverse correction requires a separate error estimate. Formula (18.3) supplies one such estimate: the correction has the scale of the square of the coupling divided by the complementary gap. If that gap is itself infinitesimal, a seemingly second-order term can be as important as a first-order term.

Example 18.2 (A spectral shift invisible in the small diagonal block). In a workspace containing 1 in its exponent group, write $t = t^1$ and let

$$H_t = \begin{pmatrix} 0 & t \\ t & 1 \end{pmatrix}.$$

Its eigenvalues and low-eigenvalue projector are exactly

$$\lambda_{\pm} = \frac{1 \pm \sqrt{1 + 4t^2}}{2}, \quad (18.4)$$

$$P_- = \frac{\lambda_+ I - H_t}{\lambda_+ - \lambda_-}. \quad (18.5)$$

The small eigenvalue is

$$\lambda_- = -t^2 + t^4 - 2t^6 + 5t^8 + O_v(10). \quad (18.6)$$

Here $O_v(10)$ denotes a Hahn remainder of valuation at least 10; t is not a real variable tending to zero. The expansion follows either from the binomial series justified in [Section 19](#), or by substituting the displayed finite polynomial into $\lambda^2 - \lambda - t^2$ and checking the remainder order, then selecting the root with positive valuation.

The direct perturbation inside the first coordinate is zero. Nevertheless, (18.2) gives the scalar equation $-z - t^2/(1 - z) = 0$, whose leading solution is $z = -t^2$. An eigenvector may be taken as $(1, \lambda_-/t)^\top$, displaying an order- t mixing with a spectral shift of order t^2 .

The companion matrix

$$\tilde{H}_t = \begin{pmatrix} t^2 & t \\ t & 1 \end{pmatrix} = \begin{pmatrix} t \\ 1 \end{pmatrix} \begin{pmatrix} t & 1 \end{pmatrix}$$

is positive semidefinite of exact rank one. In its effective scalar at $z = 0$, the direct t^2 term and the coupling correction t^2 cancel exactly. The difference between a small eigenvalue and a zero eigenvalue can therefore depend on cancellation between terms of the same scale. Neither leading entry valuations nor a generic “second-order is negligible” rule resolves it.

19 Finite functional calculus versus infinite Hahn series

19.1 Finite spectral calculus needs no convergence

Let H be Hermitian with distinct eigenvalues in a finite set S . For any prescribed scalars $f(\lambda) \in K$, define

$$f(H) = \sum_{\lambda \in S} f(\lambda) P_\lambda. \quad (19.1)$$

Lagrange interpolation gives a polynomial $p \in K[X]$ of degree less than $|S|$ with $p(\lambda) = f(\lambda)$; thus $f(H) = p(H)$. The construction preserves finite sums and products and depends only on the values of f at the spectrum. For real-valued spectral data it gives a Hermitian matrix; for nonnegative data it gives a positive semidefinite matrix. It also yields

$$\|f(H)\|_{\text{op}} = \max_{\lambda \in S} |f(\lambda)|.$$

For $z \notin S$, the resolvent is

$$(zI - H)^{-1} = \sum_{\lambda \in S} \frac{P_\lambda}{z - \lambda}.$$

All of this is finite algebra. It does not assert that an arbitrary scalar function already has a chosen surreal extension. In particular, one cannot use (19.1) to define unspecified values such as a global surreal sine at an infinite argument by circular reasoning. One must first supply the values to be interpolated.

19.2 A genuine matrix Hahn-summability theorem

A family of Hahn series is *strongly summable* if the union of its supports is well ordered and every exponent occurs in only finitely many members (the notion of [Definition 9.1](#)). Its sum is then defined coefficientwise by finite sums. For matrices this means entrywise strong summability, equivalently the same condition on the finitely many entry families together.

We use the standard support lemma underlying Hahn-field arithmetic: if $S \subseteq \Gamma_{>0}$ is well ordered, its additive monoid

$$S^* = \{0\} \cup \{s_1 + \cdots + s_k : k \geq 1, s_j \in S\}$$

is well ordered, and every exponent has only finitely many representations as an ordered finite sum of members of S . This is the positive-support form of the Neumann lemma; see the support development in Gonshor, Chapter 5A [5], and [Lemma 9.4](#) for a proof from the ordered-word lemma. The lemma does not assert that $k\rho$ is cofinal in Γ for a positive ρ .

Theorem 19.1 (Matrix inverse and square-root series). *If $E \in K_\Gamma^{n \times n}$ has $v(E) > 0$, then*

$$\sum_{k \geq 0} E^k \tag{19.2}$$

is strongly Hahn-summable and equals $(I - E)^{-1}$. If in addition $E = E^$, then*

$$\sum_{k \geq 0} \binom{1/2}{k} E^k \tag{19.3}$$

is strongly Hahn-summable and equals the positive square root of $I + E$.

Proof. The finite union of the entry supports of E is a well-ordered subset $S \subseteq \Gamma_{>0}$. An entry of E^k is a finite sum of products of k entries, so its support is contained in sums of k elements of S . The support lemma gives a well-ordered union and finitely many contributing lengths and tuples at every exponent. For a fixed length there are only finitely many matrix-index paths. This proves strong summability of both series, and justifies the coefficientwise rearrangements below.

The finite formal identity $(1 - X) \sum_{k \geq 0} X^k = 1$ is valid coefficientwise after substituting E , giving a two-sided inverse. Likewise the formal binomial identity makes the square of (19.3) equal to $I + E$. Its sum R is Hermitian, because every power of E is, and $v(R - I) > 0$. The norm of $R - I$ is therefore infinitesimal by [Lemma 7.2](#); all eigenvalues of R are positive by [Theorem 5.2](#). Uniqueness of the positive square root from [Proposition 4.3](#) identifies R with $(I + E)^{1/2}$. $\square$

19.3 Two failures of a familiar shortcut

The ordered condition $\|E\|_{\text{op}} < 1$ does not imply strong Hahn-summability of (19.2). For $E = \frac{1}{2}I$, every nonzero entry of every power has exponent zero, so infinitely many members meet the same exponent. The Hahn summability criterion fails, although $(I - E)^{-1} = 2I$ exists algebraically. Ordinary analytic summation of real coefficients is a different operation.

Even when the Hahn series exists, it need not be a topological limit of its partial sums in the full surreal field. With $E = tI$ and $t = \omega^{-1}$,

$$(I - tI)^{-1} - \sum_{k=0}^N t^k I = \frac{t^{N+1}}{1-t} I.$$

For every ordinary N , this error is larger in modulus than t^ω . Thus the partial sums fail the fine-surreal convergence test at the single tolerance t^ω .

There is no contradiction with internal valuation convergence in $K_{\mathbb{Q}} = \mathbb{C}((t^{\mathbb{Q}}))$: the natural-number exponents are cofinal in $\mathbb{Q}$, so these same partial sums do converge in that field's own valuation topology. In the larger workspace with value group $\mathbb{Q} + \mathbb{Q}\omega$, they do not, because the threshold ω is now present. Algebraic identities, strong summability, and workspace-relative topology are three different claims; only the first two were needed for [Theorem 19.1](#).

20 Two applications: regularization and finite tensors

20.1 Regularization as an explicit scale filter

For $\alpha > 0$, consider the regularized least-squares objective

$$J_{\alpha}(x) = \|Ax - b\|^2 + \alpha^2 \|x\|^2.$$

The matrix $A^*A + \alpha^2 I$ is positive definite, regardless of the rank of A . Define

$$x_{\alpha} = (A^*A + \alpha^2 I)^{-1} A^*b. \quad (20.1)$$

Expanding and using its normal equation yields

$$J_{\alpha}(x) = J_{\alpha}(x_{\alpha}) + (x - x_{\alpha})^*(A^*A + \alpha^2 I)(x - x_{\alpha}).$$

Hence (20.1) is the unique minimizer over the whole field, with no compactness or coercivity argument required.

In singular coordinates,

$$x_{\alpha} = \sum_{j=1}^r \frac{\sigma_j}{\sigma_j^2 + \alpha^2} (u_j^* b) v_j, \quad Ax_{\alpha} = \sum_{j=1}^r \frac{\sigma_j^2}{\sigma_j^2 + \alpha^2} (u_j^* b) u_j. \quad (20.2)$$

The fitted-output factor has a particularly clear reduction. Put $\beta = v(\alpha)$ and $\gamma = v(\sigma_j)$, with positive leading coefficients $b_0 = \text{lc}(\alpha)$ and $a_0 = \text{lc}(\sigma_j)$. Positivity of the denominator gives

$$\text{st} \left(\frac{\sigma_j^2}{\sigma_j^2 + \alpha^2} \right) = \begin{cases} 1, & \gamma < \beta, \\ \frac{a_0^2}{a_0^2 + b_0^2}, & \gamma = \beta, \\ 0, & \gamma > \beta. \end{cases} \quad (20.3)$$

The quotient always lies in $[0, 1]$, so its standard part exists. This is a scale filter: larger singular directions pass, smaller ones disappear in the ordinary shadow, and the boundary scale retains a nontrivial leading amplitude. At that boundary, a valuation-only description is insufficient.

Proposition 20.1 (Ordinary data and infinitesimal regularization). *If A and b have ordinary complex entries and α is a positive infinitesimal in a Hahn workspace, then x_{α} is finite and*

$$\text{st}(x_{\alpha}) = A^{\dagger}b,$$

where the right side is the ordinary complex pseudoinverse solution.

Proof. Choose an ordinary complex SVD. Every nonzero singular value is a positive ordinary real number. In (20.2), the coefficient $\sigma_j/(\sigma_j^2 + \alpha^2)$ is finite and has standard part $1/\sigma_j$. There are finitely many terms, so reduction gives the pseudoinverse formula. $\square$

The ordinary-data hypothesis matters. For $A = (t)$, $b = 1$, and $\alpha = t^2$, the regularized solution is $t^{-1}/(1 + t^2)$, which is infinite and has no standard part. Nor does a universal choice of one infinitesimal α resolve every surreal scale: [Example 13.3](#) already exhibits a direction smaller than all finite powers of a chosen basic infinitesimal.

20.2 Schmidt decomposition and infinitesimal components

Give the finite tensor space $K^m \otimes_K K^n$ the inner product for which the coordinate tensor basis is orthonormal. A vector

$$\psi = \sum_{a,b} A_{ab} e_a \otimes f_b$$

is identified with its coefficient matrix A . The SVD immediately gives

$$\psi = \sum_{j=1}^r \sigma_j u_j \otimes \bar{v}_j, \quad \|\psi\|^2 = \sum_{j=1}^r \sigma_j^2. \quad (20.4)$$

Conjugation on the second vector is required by the convention $A = U\Sigma V^*$. The least number of simple tensors in a representation of ψ is its matrix rank: a sum of k simple tensors gives a matrix of rank at most k , and (20.4) attains $k = r$.

For a unit vector, the eigenvalues of the positive trace-one matrix AA^* are the σ_j^2 and zeros. Thus the finite algebraic parts of Schmidt rank and reduced-matrix spectral calculations make sense over K . For example,

$$\psi_t = \frac{e_0 \otimes f_0 + t e_1 \otimes f_1}{\sqrt{1+t^2}}$$

has exact tensor rank two, whereas its standard part is the rank-one tensor $e_0 \otimes f_0$. Its second squared Schmidt coefficient is the positive infinitesimal $t^2/(1+t^2)$, not zero. This illustrates precisely what is lost on taking an ordinary shadow.

This is a finite algebraic application, not a proposed physical interpretation of surreal probabilities. No infinite tensor product, measure-valued probability theory, entropy extension, or quantum dynamics is supplied by these calculations.

21 What the chapter establishes, and what remains separate

The finite-dimensional order theory is robust: orthogonality, spectral and singular-value decompositions, least squares, low-rank optimization, and the perturbation bounds proved above require real closedness, not Archimedean completeness. Their proofs exhibit the relevant witnesses, so questions about class-sized compactness never arise.

The scale theory contains additional information not visible in an ordinary complex matrix. Nonzero singular values have a finite ordered list of valuations. Determinantal minima recover that list, positive Gram coefficients recover its leading amplitudes in blocks, and the input filtration gives an intrinsic meaning to rank at a chosen resolution. The same data govern conditioning and perturbation certificates. The minimum-valuation elimination gives a finite algebraic route to those invariants when the scalar representation supplies the required exact operations.

Divisibility of the value group turns out to be separable from all of this. The decompositions survive its removal, with every exponent confined to the subgroup generated by the input exponents and with a finite splitting tree of at most $n - 1$ nontrivial nodes, a bound independent of the order type of the supports and of the rank or cardinality of Γ . What does not survive its removal is the free availability of positive roots, and the determinantal invariant measures the deficit exactly: the field of definition of $A^{1/q}$, its degree, the criterion for no extension at all, the matching criterion for a Gram factorization, and a single scalar generator of the whole extension. That the same Δ_k answers both the scale question and the arithmetic one is the point at which the two halves of this article meet.

Section 12 is the polynomial counterpart. For a monic squarefree real-rooted polynomial the leading principal minors of the Newton-sum Hankel matrix play the part of the Δ_k : their square

roots generate the exact splitting field, and their consecutive valuation classes, counted with multiplicity, give the number of irreducible factors over the base and over every ordered Hahn enlargement of it, with a finite precision certificate. The same computation gives the exact field of the generalized eigenvalues of a definite pencil, one of the items [Remark 2.1](#) used to list as needing divisibility. A normalized eigenbasis of the pencil can still need further roots, and the profile does not determine the list of factor degrees ([Warning 12.23](#) and [Example 12.29](#)).

Several nearby subjects remain distinct. There is no theorem here about infinite-dimensional Hilbert spaces over $\mathbf{No}[i]$, bounded-operator spectra, compact or trace-class operators, or spectral measures. A separate report now treats the first two of these, and operators that are compact on an ordinary Hilbert space, under its own hypotheses. No theorem of it is used here, and neither report treats trace-class operators or spectral measures. That report is [docs/surcomplex/infinite-dimensional-hahn-spectral-theory/](#). A support-controlled Hahn expansion of an eigenbasis *is* now available for a single matrix, over an arbitrary set-sized value group ([Theorems 9.12](#) and [9.18](#) and [Corollary 9.15](#)). There is still no global analytic choice of eigenvectors through collisions or across multiple parameters, and [Warning 9.24](#) shows that this is not a gap in the argument but a genuine failure in the multivariable formal power-series category: the crossing matrix [\(9.13\)](#) has no eigenvalue in $\mathbb{C}[[x, y]]$ at all. The descent theorems resolve the totally ordered scale problem and not that one. Nor do the exact decompositions automatically provide effective algorithms for arbitrary surreal inputs; [Section 14.3](#) lists what a representation must supply, and [Warning 14.2](#) records that even the ramification criterion can outrun a black-box order interface. Finally, the strong matrix series of [Theorem 19.1](#) is not a claim of fine-topological convergence, and neither is the splitting tree: [Example 9.2](#) exhibits a value group in which a valid Hahn identity has non-convergent partial sums, so a remainder bound is never used here as a convergence proof.

All field and extension statements here are set-sized. No proper-class Hahn sum, no proper-class vector space, and no class-sized workspace occurs anywhere; [Proposition 2.3](#) is what keeps the surcomplex statements inside a set.

Three further questions are concrete enough to name. One is to replace the finite tree of set-valued supports by finite effective support grammars for useful transseries subclasses. Another is to characterize smaller non-Hahn subfields closed under the separated-block construction, so that the full Hahn field can be replaced by a more economical domain. A third is to ask which other positive spectral functionals, beyond rational powers and their strictly positive sums, furnish similarly exact primitive generators. The coefficient-defined primitive element of [Corollary 12.10](#) is related, but it is not a spectral functional and does not answer this question. These are questions suggested by the results, not advertised as previously published open problems.

The third manuscript adds three continuation questions of the same status, directions for further work rather than theorems. The subspace-indicator classification of profiles ([Theorem 12.17](#)) suggests studying the complexity of recovering all compatible factor-degree lists, rather than only their number. Extracting the pivot parity information efficiently, without a full Hankel elimination, is an algorithmic question. And extending the field formula to residue fields that are not real closed would require keeping residue-field extensions and residue quadratic-form data, which the valuation-only certificate of [Section 12](#) deliberately avoids.

A formalization would have a natural order, which is worth recording because it is *not* a report of code that has been compiled: positive sum-of-squares valuation ([Lemma 9.8](#)) and integral unitary invariance ([Lemma 7.3](#)) first; then finite-index Hahn coset decompositions, the monomial character action, and the positive-coset primitive-element argument ([Lemmas 10.2](#) and [11.1](#)), all of which are set-sized and independent of the full surreal construction; then an interface for finite-variable formal series evaluated at Hahn infinitesimals, with the universal separated-block projectors and their polar intertwiner, where the ordinary complex contour justification in [Theorem 9.9](#) could be replaced by formal resolvent identities or a formal Hensel

block-lifting theorem; then dimension induction for the splitting tree; and finally the normal-form embeddings. For real symmetric input, such a block-lifting route is now written out: [Lemma 12.4](#) and [Remark 12.5](#) replace the contour argument by coprime Hensel lifting through universal formal series and polynomial idempotents. That route does not reproduce the support certificate of [Theorem 9.9](#) or treat complex Hermitian input. The Hankel package of [Section 12](#) would then need only finite field theory, the intermediate-field lemma [Lemma 12.2](#) and the graded-form invariant [Lemma 12.12](#). This is a dependency proposal, not a claim that the full package has been formalized. Current checked prerequisites and source-claim mappings are recorded separately in `docs/FORMALIZATION.md`.

For the repository, this report sits at `docs/surcomplex/spectral-theory/`. It follows the workspace conventions of the foundations and polynomial-algebra reports and can be read independently of the analytic coefficient-ring machinery. The most useful cross-reference from polynomial algebra is the bridge from its Newton profiles to [Theorems 8.1](#) and [8.2](#), and its Hermite trace form T_1 , which in the monomial basis is the Hankel matrix H_p of [Section 12](#); the most useful cross-references to computer algebra are the exact-operation contract in [Section 14](#) and its extensions in [Sections 12.8](#) and [14.3](#), where [Warning 14.2](#) names a decision that needs a capability tier the black-box order interface does not provide [4].

A A proof and hypothesis ledger

Result	Scalar hypotheses	Essential finite mechanism
Spectral theorem and SVD	Real closed F , $K = F(i)$	Algebraic roots, positivity, finite orthogonal complements
Variational and low-rank extrema	Same	Eigenbasis witnesses and dimension intersection, not compactness
Least squares and pseudoinverse	Same	Orthogonal coordinates and finite sums of nonnegative squares
Singular scales from minors	Hahn workspace; arbitrary Γ	Integral unitary factors and Cauchy–Binet
Leading singular amplitudes	Same, real/complex residue field	Positive sum of squared minor coefficients
Rank filtration	Same, chosen integral lattice	Lattice preservation by unitaries and coordinate reduction
Scale elimination	Exact field and valuation operations	Minimum pivot, integral operations, Schur complement
Inverse perturbation	Real closed F ; $q < 1$	A positive lower singular bound and a finite inverse identity
Visible-rank stability	Hahn workspace; strict precision threshold	Ordered singular-value Lipschitz bound plus leading-term preservation
Spectral-subspace bound	Real closed F ; separated selected/complementary spectra	Entrywise Sylvester equation and Frobenius norm
Infinite matrix series	Hahn workspace; $v(E) > 0$	Positive support lemma, not cofinality of integer multiples
Norm square roots without real closure	Real/complex Hahn coefficients; arbitrary Γ	Positive leading squared terms and binomial unit root

Result	Scalar hypotheses	Essential finite mechanism
Separated-block lift	Distinct ordinary residue eigenvalues; Hermitian positive-support error	Universal formal projector identities and polar intertwiner, evaluated by the support lemma
Hermitian spectral descent	Hermitian input over K_Γ ; arbitrary Γ	Trace subtraction, constant spectral theorem, block lift and induction on dimension
Normal and simultaneous descent	A set of commuting normal matrices over K_Γ	Hermitian descent, commutant argument and at most $n - 1$ selected refinements
SVD without divisibility	A rectangular matrix over K_Γ	Gram diagonalization, norm-square roots and finite Gram–Schmidt
Exact root field	Positive semidefinite Hermitian matrix; integer $q \geq 2$	Scalar binomial roots, spectral descent and finite-index coset basis
Primitive trace	Principal positive roots; strictly positive weights in F_Γ	Nonzero coset sums and trivial character stabilizer
Surcomplex interpretation	Set-supported normal forms; finite matrices	Exact input subgroup and the normal-form embedding
Real principal axes, second route	Real symmetric input over F_Γ ; arbitrary Γ	Coprime Hensel lifting by universal formal series, polynomial idempotents, norm roots
Intermediate half-scale fields	$\Gamma \subseteq \Lambda$, Λ/Γ finite, $2\Lambda \subseteq \Gamma$	Sign characters on support cosets and finite Galois theory
Hankel trace form	Monic p over F_Γ	Newton sums, $H_p = V^\top V$, Cauchy–Binet, no-pivot $LDL^\top$
Exact splitting field	Monic squarefree real-rooted p ; constant field $\mathbb{R}$; arbitrary Γ	Pivot roots make multiplication by X symmetric; descent; intermediate Hahn fields are Pythagorean
Factor count and profile	Same	Congruence invariance of square-class multiplicities; monomial trace forms of the root fields
Base change and precision	An ordered inclusion $\Gamma \subseteq \Gamma'$; strict precision on every h_k	Pushforward of classes; leading-term preservation
Definite pencils	$A = A^*$, $B = B^* \succ 0$ over K_Γ	Squarefree part of $\det(XB - A)/\det B$; elementary 2-group bounds

The order-theoretic arguments also apply directly to real symmetric matrices over F . They do not apply to an arbitrary algebraically closed field equipped with an arbitrary involution: positivity of x^*x is the indispensable additional structure. Conversely, the valuation-ring elimination argument itself requires no conjugation or positivity; those structures are what identify its invariant factors with singular values. This separation helps avoid attributing a positivity theorem to general valued-field linear algebra.

B Reproducible finite checks

The report directory contains `code/verify-examples.py`, an exact SymPy verification program, together with its recorded JSON and text reports. The valuation and elimination kernel uses rational functions over $\mathbb{Q}(i)$ in two formal positive monomials t, u , with the lexicographic valuation convention

$$v(t) = (0, 1), \quad v(u) = (1, 0).$$

These correspond to the exponents 1 and ω in the subgroup $\mathbb{Z} + \mathbb{Z}\omega$. A rational function is first canceled exactly; the valuation is the least lexicographic monomial exponent of its numerator minus that of its denominator. This finite rational-function field embeds in a divisible Hahn workspace, but the code is not a general Hahn-series engine. Separate finite identity checks also use exact square-root expressions for the two-by-two spectral examples.

The checks compare minimum-valuation elimination with exhaustive minors on small exact matrices, including matrices obtained by integral invertible row and column operations from prescribed diagonal scales. They also check the positive Gram coefficient identity, the residual polynomial examples, Penrose identities for a rank-deficient matrix, the exact projector and rotation examples, the second-order spectral expansion, the nilpotent pseudospectral example, and the regularization and tensor identities. All symbolic assertions must pass for the program to write a successful report; no floating-point sampling is used.

The bundled run passed 176 exact assertions across 18 matrix-profile cases (Python 3.13.5, SymPy 1.14.0). The recorded run is a reproducibility check of these finite identities and of the implementation on its stated test cases. It is not verification of the general mathematical proofs, of arbitrary ordered groups or Hahn supports, or of proper-class foundations. In particular, a finite test cannot prove the support lemma or certify a general surreal equality algorithm. The exact assertion count and software versions are recorded in `data/verification-report.json` rather than being used as a proxy for mathematical correctness.

The second check suite

The material of [Sections 9 to 11](#) arrived with its own exact SymPy program, also with no floating-point sampling. It exercises the resolvent-projector construction of (9.4) and its polar intertwiner (9.5) through formal degree six for a three-dimensional Hermitian example with a repeated leading eigenvalue; it tests the displayed scalar and two-by-two identities; it checks minor-valued profiles under rational unitary changes of basis, including rank-deficient profiles; and it tests finite quotient computations, subgroup enumeration against a maximal-minor lattice index, and character and stabilizer counts in finite integer lattices, the last of these being the computational face of (10.11) and (11.3).

The delivered run passed **807 exact scalar assertions** — 378 projector and polar-series coefficients, 144 finite ramification-group checks, 105 rational-matrix checks, 96 character checks, 55 block and low-degree terms, 29 displayed identities — across 48 finite quotient cases and six rational positive semidefinite profiles, under Python 3.13.5 and SymPy 1.14.0. It was rerun for this collection with an explicit output directory and reproduced PASS with exactly 807 assertions and the identical category split under Python 3.14.4 and SymPy 1.14.0.

The same caveats apply, and one more. The counts include individual zero coefficients, so they are not counts of independent theorems. The checks are finite identities and examples only: there is no exhaustive computation over arbitrary Hahn supports or arbitrary ordered groups, and the general support and field-theoretic results rest on the proofs in this article and not on the scripts. Both programs default their output directory to the shipped `data/`, so both must be run on a copy or with an explicit output path; the repository README records the safe invocations.

The third check suite

The material of [Section 12](#) arrived with `code/03-hankel-ramification-check_examples.py`, an exact SymPy program. The file `data/03-hankel-ramification-requirements.txt` pins SymPy 1.14.0. The program uses Python `assert` statements, so it must not be run with `-O`. For six polynomials — $X^2 - a$, $(X - 1)^2 - a$, p_{pair} , p_{prim} , and the two polynomials P and Q of degree six of [Example 12.29](#) — it builds the companion matrix, the Newton sums as traces of its powers, the Hankel matrix and its leading minors, and asserts $M^T H_p = H_p M$ and $h_n = \text{Disc}(p)$. It then computes valuations, pivot classes, the profile, the factor count and the splitting degree. Valuations are exponent pairs in the explicit group $\mathbb{Z} + \mathbb{Z}\omega$, with $v(a) = 1$ and $v(b) = \omega$, compared with the ω -coordinate first; it is not an unspecified black-box ordered group. The program asserts the factor counts and splitting degrees of the displayed examples, the equality of the two degree-six profiles, the characteristic polynomial and leading minor of the 2×2 pencil block ([12.13](#)), the sign-product identity for the primitive quartic, the affine covariance $h_k \mapsto 2^{k(k-1)} h_k$ on p_{pair} , the squarefree reduction of $(X^2 - a)^3$, and the discriminant $4096a^2b^2(a - b)^2$ of the primitive quartic. The factor counts and degrees it asserts are those the formulas of [Section 12](#) predict from the computed valuations; they are not independent factorizations.

The delivered run under SymPy 1.14.0 ended with “ALL EXACT SYMBOLIC CHECKS PASSED”; its output is recorded in `data/03-hankel-ramification-verification_results.json` and `data/03-hankel-ramification-verification_output.txt`. The program records no assertion count. It was rerun for this collection on a copy, under Python 3.14.4 and SymPy 1.14.0, and reproduced both files exactly, apart from line endings. It writes `verification_results.json` into its own directory, so in this layout it must also be run on a copy. It does not implement general infinite Hahn arithmetic, prove strong summability, certify the literature search or verify the theorems. In particular it does not check the general Hahn support theorem, the classification of intermediate fields ([Lemma 12.2](#)) or the graded-form invariant ([Lemma 12.12](#)).

C Notation at a glance

Symbol	Meaning
$F, K = F(i)$	A real closed ordered field and its algebraic complexification
F_Γ, K_Γ	Real and complex Hahn workspaces, $t^\gamma = \omega^{-\gamma}$
A^*	Conjugate transpose
$\sigma_1 \geq \dots \geq \sigma_r > 0$	Nonzero singular values, in decreasing ordered magnitude
$\ A\ _{\text{op}}, \ A\ _F$	Operator and Frobenius norms, valued in $F_{\geq 0}$
$v(a), \text{lc}(a)$	Least Hahn exponent and coefficient at that exponent
$v(A)$	Minimum valuation of the entries; $v(0) = \infty$
γ_j	$v(\sigma_j)$, in nondecreasing order
Δ_k	Minimum valuation of a k -minor, with $\Delta_0 = 0$
$\mathcal{O}, \mathfrak{m}, \text{st}$	Integral valuation ring, positive-valuation ideal, residue map
c_k	Coefficient of s^k in $\det(I + sA^*A)$
C_k	Positive leading coefficient of c_k , after monomial normalization
$r_{<\theta}, r_{\leq\theta}$	Numbers of nonzero singular scales below or at a cutoff
$A^\dagger$	Moore–Penrose inverse
$\kappa(A)$	Ordered spectral condition number of an invertible matrix
$O_v(\eta)$	A formal remainder with valuation at least the exponent η , not a real limit
$\Gamma_{\mathbb{Q}}$	The divisible hull $\Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$, with its unique compatible order

Symbol	Meaning
$\langle S \rangle_{\mathbb{N}}$	The additive monoid generated by a well-ordered $S \subseteq \Gamma_{>0}$
H	The exact input support subgroup of a matrix, not assumed divisible
$\Gamma_{A,q}$	The root group $\Gamma + \sum_{k=1}^{\text{rank } A} \mathbb{Z} \Delta_k(A)/q$
$R_q(A)$	The subgroup of $\Gamma/q\Gamma$ generated by $\Delta_k(A) + q\Gamma$, $1 \leq k \leq \text{rank } A$
μ_q	The group of q th roots of unity in $\mathbb{C}^\times$
$K(B)$	The field obtained from K by adjoining every entry of a finite matrix B
$O_{\text{formal}}(E^k)$	A support-certified sum of monomials of formal degree at least k
$\mathcal{O}_F$	The real valuation ring $\mathcal{O} \cap F_\Gamma$
s_j, H_p	Newton sums of a monic polynomial p , and its Hankel matrix $(s_{i+j})_{0 \leq i,j < n}$
$h_k = h_k(p)$	The k th leading principal minor of H_p (a Hankel subdiscriminant), $h_0 = 1$; not the coset h_j of Lemma 11.1
π_k, ξ_k	The pivot h_k/h_{k-1} and its class $v(\pi_k) + 2\Gamma$
m_p, D_p, r_p	The Hankel square-class profile on $\Gamma/2\Gamma$, its span, and $\dim_{\mathbf{F}_2} D_p$; D_p is not $R_2(A)$
Λ_p, L_p	$\Gamma + \sum_k \mathbb{Z} v(h_k)/2$, and the splitting field F_{Λ_p} of p
$p_{A,B}$	$\det(XB - A)/\det B$ for a definite pencil

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